2. Working

The angular position of the moving system is determined by the phase angle ϕ between the line current and the respective phase voltage. In other words, deflection θ is equal to ϕ, although, in practice, the instrument is calibrated to read the power factor directly.

3. Advantages

(*i*) Since no ***rotating*** magnetic field is produced, there is no tendency for the moving system to be dragged around continuously in one direction.

(*ii*) This instrument is not much affected by the type of variations of frequency, voltage and waveform as might be expected in an ordinary supply.

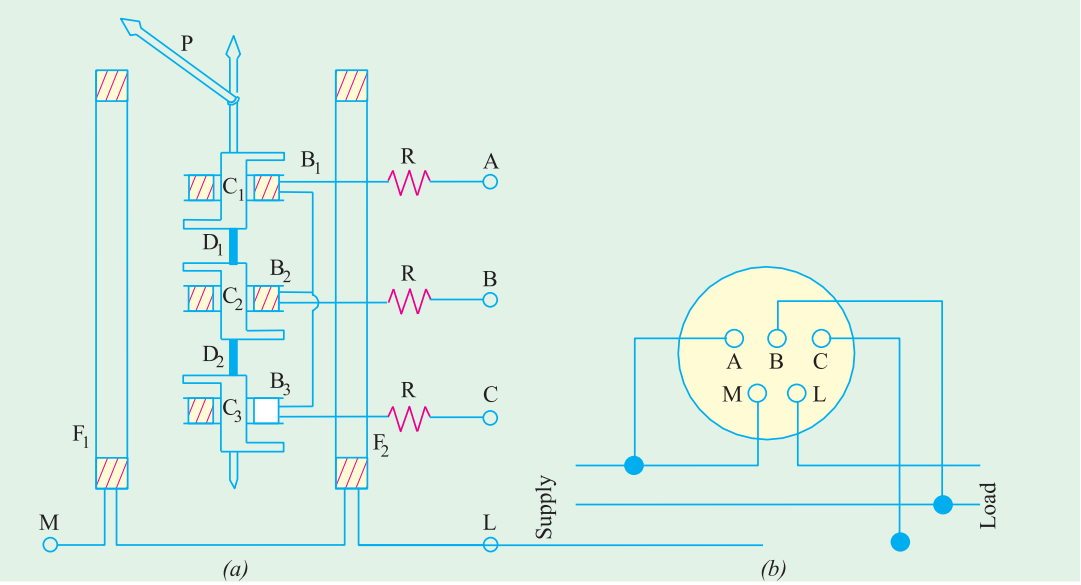


Fig. 10.77

10.60. D.C. Potentiometer



DC potentiometer

A potentiometer is used for measuring and comparing the e.m.fs. of different cells and for calibrating and standardizing voltmeters, ammeters etc. In its simplest from, it consists of a German silver or manganin wire usually one meter long and stretched between two terminals as shown in Fig. 10.78.

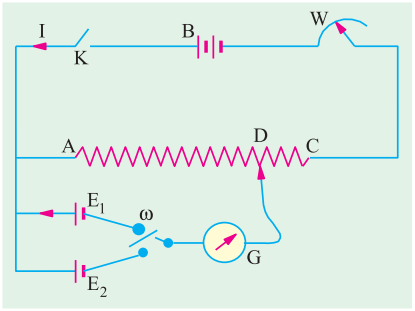


Fig. 10.78

This wire is connected in series with a suitable rheostat and battery *B* which sends a steady current through the resistance wire *AC*. As the wire is of uniform cross-section throughout, the fall in potential across it is uniform and the drop between any two points is proportional to the distance between them. As seen, the battery voltage is spread over the rheostat and the resistance wire *AC*. As we go along *AC*, there is a progressive fall of potential. If ρ is the resistance/cm of this wire, *L* its length, then for a current of *I* amperes, the fall of potential over the whole length of the wire is ρ *LI* volts.

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The two cells whose e.m.fs are to be compared are joined as shown in Fig. 10.78, always remembering that ***positive terminals of the cells and the battery must be joined together.*** The cells can be joined with the galvanometer in turn through a two-way key. The other end of the galvanometer is connected to a movable contact on AC. By this movable contact, a point like D is found when there is no current in and hence no deflection of G. Then, it means that the e.m.f. of the cell just balances the potential fall on AD due to the battery current passing through it.

Suppose that the balance or null point for first cell of e.m.f. E_1 occurs at a length L_1 as measured from point A. The $E_1 = \rho\ L_1 I$.

Similarly, if the balance point is at L_2 for th other cell, then $E_2 = \rho\ L_2 I$.

Dividing one equation by the other, we have $\dfrac{E_1}{E_2} = \dfrac{\rho L_1 I}{\rho L_2 I} = \dfrac{L_1}{L_2}$

If one of the cells is a standard cell, the e.m.f. of the other cell can be found.

10.61. Direct-reading Potentiometer

The simple potentiometer described above is used for educational purposes only. But in it commercial form, it is so calibrated that the readings of the potentiometer give the voltage directly, thereby eliminating tedious arithmetical calculations and so saving appreciable time.

Such a direct-reading potentiometer is shown in Fig. 10.79. The resistance R consists of 14 equal resistances joined in series, the resistance of each unit being equal to that of the whole slide were S (which is divided into 100 equal parts). The battery current is controlled by slide wire resistance W.

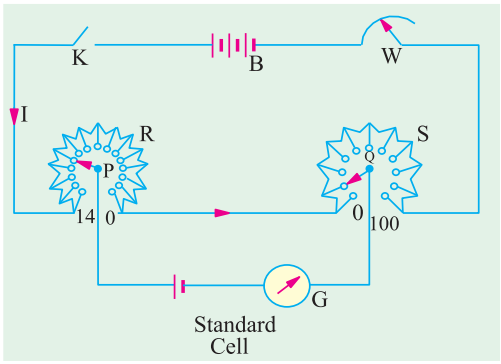


Fig. 10.79

10.62. Standardizing the Potentiometer

A standard cell *i.e.* Weston cadmium cell of e.m.f. 1.0183 V is connected to sliding contacts P and Q through a sensitive galvanometer G. First, P is put on stud No. 10 and Q on 18.3 division on S and then W is adjusted for zero deflection on G. In that case, potential difference between P and Q is equal to cell voltage *i.e.* 1.0183 V so that potential drop on each resistance of R is 1/10 = 0.1 V and every division of S represents 0.1/100 = 0.001 V. After standardizing this way, the ***position of W is not to be changed in any case*** otherwise the whole adjustment would go wrong. After this, the instrument becomes direct reading. Suppose in a subsequent experiment, for balance, P is moved to stud No. 7 and Q to 84 division, then voltage would be $= (7 \times 0.1) + (84 \times 0.001) = 0.784$ V.

It should be noted that since most potentiometers have fourteen steps on R, it is usually not possible to measure p.ds. exceeding 1.5 V. For measuring higher voltges, it is necessary to use a volt box.

10.63. Calibration of Ammeters

The ammeter to be calibrated is connected in series with a variable resistance and a standard resistance F, say, of 0.1 Ω across battery B_1 of ample current capacity as shown in Fig. 10.80. Obviously, the resistance of F should be such that with maximum current flowing through the ammeter A, the potential drop across F should not exceed 1.5 V. Some convenient current, say

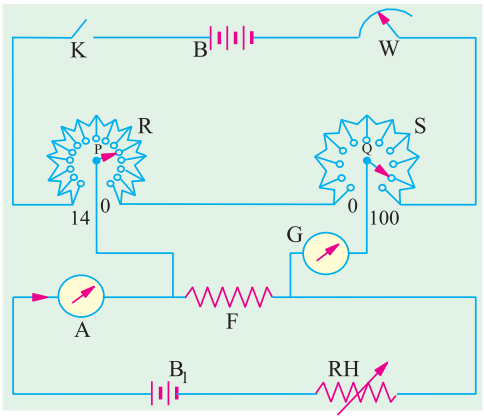


Fig. 10.80

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6 amperes (as indicated by *A*) is passed through the circuit by adjusting the rheostat *RH*.

The potential drop across *F* is applied between *P* and *Q* as shown. Next the sliding contacts *P* and *Q* are adjusted for zero deflection on *G*. Suppose *P* reads 5 and *Q* reads 86.7. Then it means that p.d. across *F* is 0.5867 V and since *F* is of 0.1 Ω, hence ***true*** value of current through *F* is 0.5867/0.1 = 5.867 amperes. Hence, the ammeter reads high by (6 – 5.867) = 0.133 A. The test is repeated for various values of current over the entire ranges of the ammeter.

10.64. Calibration of Voltmeters

As pointed out in Art. 10.62, a voltage higher than 1.5 cannot be measured by the potentiometer directly, the limit being set by the standard cell and the type of the potentiometer (since it has only 14 resistances on *R* as in Fig. 10.79). However, with the help of a volt-box which is nothing else but a voltage reducer, measurements of voltage up to 150 V or 300 V can be made, the upper limit of voltage depending on the design of the volt-box.

The diagram of connections for calibration of voltmeters is shown in Fig. 10.81. By calibration is meant the determination of the extent of error in the reading of the voltmeter throughout its range. A high value resistor *AB* is connected across the supply terminals of high voltage battery B_1 so that it acts as a voltage divider. The volt-box consists of a high resistance *CD* with tapings at accurately determined points like *E* and *F* etc. The resistance *CD* is usually 15,000 to 300 Ω. The two tappings *E* and *F* are such that the resistances of portions *CE* and *CF* are 1/100th and 1/10th the resistance *CD*. Obviously, whatever the potential drop across *CD*, the corresponding potential drop across *CE* is 1/100th and that across *CF*, 1/10th of that across *CD*.

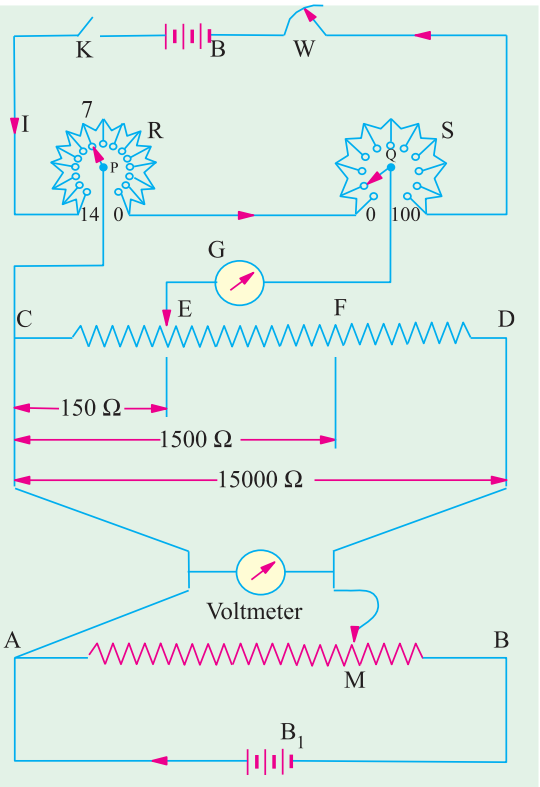


Fig. 10.81

If supply voltage is 150 V, then p.d. across *AB* is also 150 V and if *M* coincides with *B*, then p.d. across *CD* is also 150 V, so across *CF* is 15 volts and across *CE* is 1.5 V. Then p.d. across *CE* can be balanced over the potentiometer as shown in Fig. 10.81. Various voltages can be applied across the voltmeter by moving the contact point *M* on the resistance *AB*.

Suppose that *M* is so placed by voltmeter *V* reads 70 *V* and p.d. across *CE* is balanced by adjusting *P* and *Q*. If the readings on *P* and *Q* to give balance are 7 and 8.4 respectively, then p.d. across *CE* is 0.7084 V.

Hence, the true p.d. across *AM* or *CD* or voltmeter is 0.7048 × 100 = 70.84 V (because resistance of *CD* is 100 times greater than that of *CE*). In other words, the reading of the voltmeter is low by 0.84 V.

By shifting the position of *M* and then balancing the p.d. across *CE* on the potentiometer, the voltmeter can be calibrated throughout its range. By plotting the errors on a graph, a calibration curve of the instrument can also be drawn.

10.65. A.C. Potentiometer

An A.C. potentiometer basically works on the same principle as a d.c. potentiometer. However,

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there is one very important difference between the two. In d.c. potentiometer, only the ***magnitudes*** of the unknown e.m.f. and slide-wire voltage drop are made equal for obtaining balance. But in an a.c. potentiometer, not only the magnitudes but *phases* as well have to be equal for obtaining balance. Moreover, to avoid frequency and waveform errors, the a.c. supply for slide-wire must be taken from same source as the voltage or current to be measured.

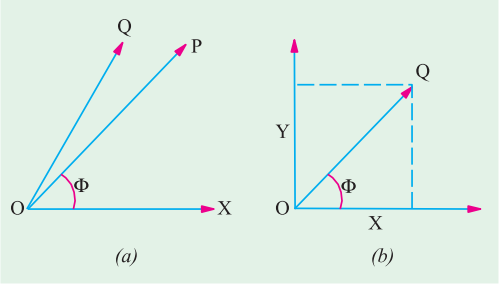


Fig. 10.82

A.C. potentiometers are of two general types differing in the manner in which the value of the unknown voltage is presented by the instrument dials or scales. The two types are :

(*i*) Polar potentiometers in which the unknown voltage is mesured in polar form *i.e.* in terms of magnitude and relative phase.

(*ii*) Co-ordinate potentiometers which measure the rectangular co-ordinates of the voltage under test.

The two products are illustrated in Fig. 10.82. In Fig. 10.82 (*a*), vector *OQ* denotes the test voltage whose magnitude and phase are to be imitated. In polar potentiometer, the length *r* of the vector *OP* can be varied with the help of a sliding contact on the slidewire while its phase ϕ is varied independently with the help of a phase-shifter. Drysdale potentiometer is of this type.

In co-ordinate type potentiometers, the unknown voltage vector *OQ* is copied by the adjustment of 'in phase' and 'quadrature' components *X* and *Y*. Their values are read from two scales of the potentiometer. The magnitude of the required vector is $= \sqrt{X^2 + Y^2}$ and its phase is given by $\phi = \tan^{-1}(X/Y)$. Examples of this type are **(*i*)** Gall potentiometer and **(*ii*)** Campbell-Larsen potentiometer.

10.66. Drysdale Polar Potentiometer

As shown in Fig. 10.83 for a.c. measurements, the slide-wire *MN* is supplied from a phase shifting circuit so arranged that magnitude of the voltage supplied by it remains constant while its

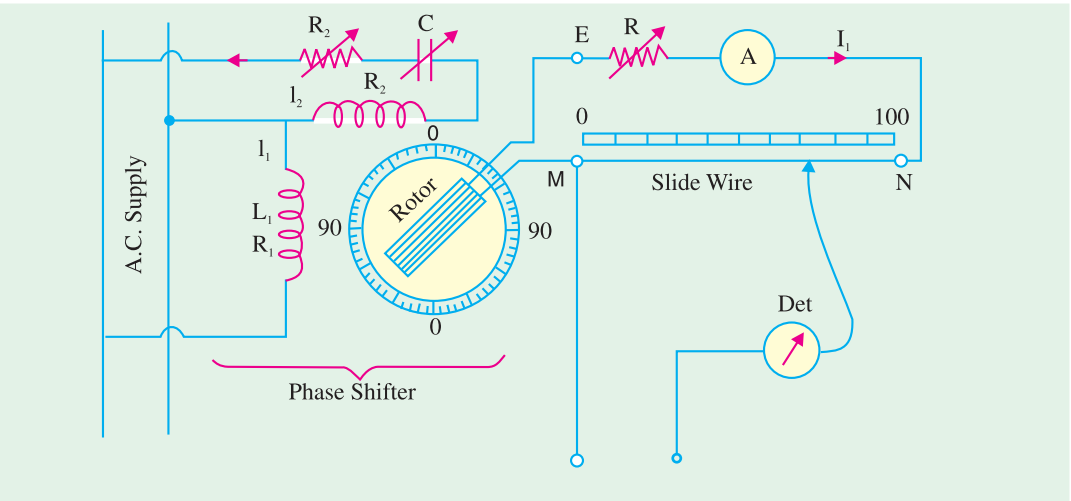


Fig. 10.83

phase can be varied through 360°. Consequently, slide-wire current *I* can be maintained constant in magnitude but varied in phase. The phase-shifting circuit consists of :

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(*i*) Two stator coils supplied from the same source in parallel. Their currents I_1 and I_2 are made to differ by 90° by using well-known phase-splitting technique.

(*ii*) The two windings produce a rotating flux which induces a secondary e.m.f. in the rotor winding which is of constant magnitude but the phase of which can be varied by rotating the rotor in any position either manually or otherwise. The phase of the rotor e.m.f. is read from the circular graduated dial provided for the purpose.

The ammeter *A* in the slide-wire circuit is of electrodynamic of thermal type. Before using it for a.c. measurements, the potentiometer is first calibrated by using d.c. supply for slidwire and a standard cell for test terminals T_1 and T_2.

The unknown alternating voltage to be measured is applied across test terminals I_2 and T_2 balance is effected by the alternate adjustment of the slide-wire contact and the position of phase-shifting rotor. The slide-wire reading represents the magnitude of the test voltage phase-shifter reading gives its phase with reference to an arbitrary reference vector.

10.67. Gall Co-ordinate Potentiometer

This potentiometer uses two slide-wires *CD* and *MN* with their currents I_1 and I_2 (Fig. 10.84) having a mutual phase difference of 90°. The two currents are obtained from the single phase supply through isolating transformers, the circuit for 'quadrature' slidewire *MN* incorporating a phase shifting arrangement.

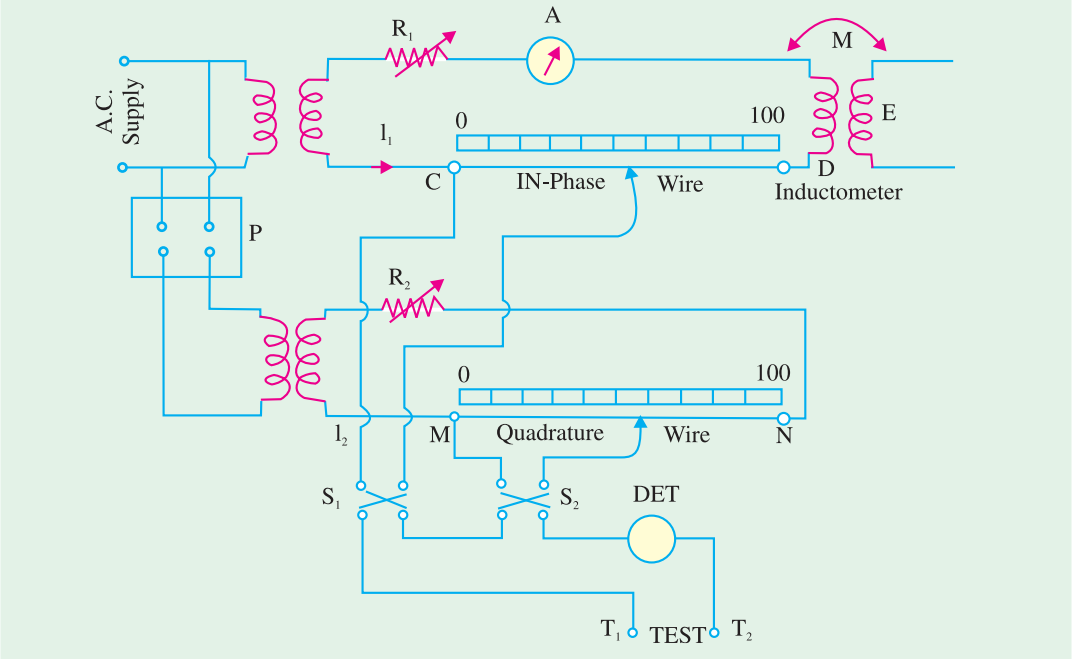


Fig. 10.84

Before use, then current I_1 is first standardised as described for Drysdale potentiometer (Art. 10.66). Next, current I_2 is standardised with the help of the mutually induced e.m.f. *E* in inductometer secondary. This e.m.f. $E = \omega M I_1$ and is in quadrature phase with I_1. Now, *E* is balanced against the voltage drop on slide-wire *MN*. This balance will be obtained only when I_2 is of correct magnitude and is in exact quadrature with I_1. Balance is achieved with the help of the phase-shifter and rheostat R_2.

The unknown voltage is applied across the test terminals T_1 and T_2. Slide-wire *MN* measures that component of the unknown voltage which is in phase with I_2. Similarly, slide-wire *CD* measures that component of the unknown voltage which is in phase with I_1. Since I_1 and I_2 are in quadrature, the

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two measured values are quadrature components of the unknown voltage. If V_1 and V_2 are these values, then

$$V = \sqrt{V_1^2 + V_2^2} \quad \text{and} \quad \phi = \tan^{-1}(V_2/V_1) \quad \text{—with respect to } I_1$$

Reversing switches S_1 and S_2 are used for measuring both positive and negative in-phase and quadrature components of the unknown-voltage.

10.68. Instrument Transformers

The d.c. circuits when large currents are to be measured, it is usual to use low-range ammeters with suitable shunts. For measuring high voltages, low-range voltmeters are used with high resistances connected in series with them. But it is neither convenient nor practical to use this method with alternating current and voltage instruments. For this purpose, specially constructed accurate-ratio instrument transformers are employed in conjunction with standard low-range a.c. instruments. Their purpose is to reduce the line current or supply voltage to a value small enough to be easily measured with meters of moderates size and capacity. In other words, they are used for extending the range of a.c. ammeters and voltmeters. Instruments transformers are of two types :

(*i*) current transformers (*CT*) —for measuring large alternating currents.
(*ii*) potential transformers (*VT*) —for measuring high alternating voltages.

Advantages of using instrument transformers for range extension of a.c. meters are as follows :

(1) the instrument is insulated from the line voltage, hence it can be grounded. **(2)** the cost of the instrument (or meter) together with the instrument transformer is less than that of the instrument alone if it were to be insulated for high voltages. **(3)** it is possible to achieve standardisation of instruments and meters at secondary ratings of 100–120 volts and 5 or 1 amperes **(4)** if necessary, several instruments can be operated from a single transformer and 5 power consumed in the measuring circuits is low.

In using instrument transformers for current (or voltage) measurements, we must know the ratio of primary current (or voltage) to the secondary current (or voltage). These ratios give us the multiplying factor for finding the primary values from the instrument readings on the secondary side.

However, for energy or power measurements, it is essential to know not only the transformation ratio but also the phase angle between the primary and secondary currents (or voltages) because it necessitates further correction to the meter reading.

For range extension on a.c. circuits, instrument transformers are more desirable than shunts (for current) and multipliers (for voltage measurements) for the following reasons :

1. time constant of the shunt must closely match the time constant of the instrument. Hence, a different shunt is needed for each instrument.
2. range extension is limited by the current-carrying capacity of the shunt *i.e.* upto a few hundred amperes at the most.
3. if current is at high voltage, instrument insulation becomes a very difficult problem.
4. use of multipliers above 1000 becomes almost impracticable.
5. insulation of multipliers against leakage current and reduction of their distributed capacitance becomes not only more difficult but expensive above a few thousand volts.

10.69. Ratio and Phase-angle Errors

For satisfactory and accurate performance, it is necessary that the ratio of transformation of the instrument transformer should be constant within close limits. However, in practice, it is found that neither current transformation ratio I_1/I_2 (in the case of current transformers) nor voltage transformation ratio V_1/V_2 (in the case of potential transformers) remains constant. The transformation ratio is found to depend on the exciting current as well as the current and the power factor of the secondary circuit. This fact leads to an error called ratio error of the transformer which depends on the working component of primary.

It is seen from Fig. 1.85 (*a*) that the phase angle between the primary and secondary currents is not exactly 180° but slightly less than this value. This difference angle β may be found by reversing

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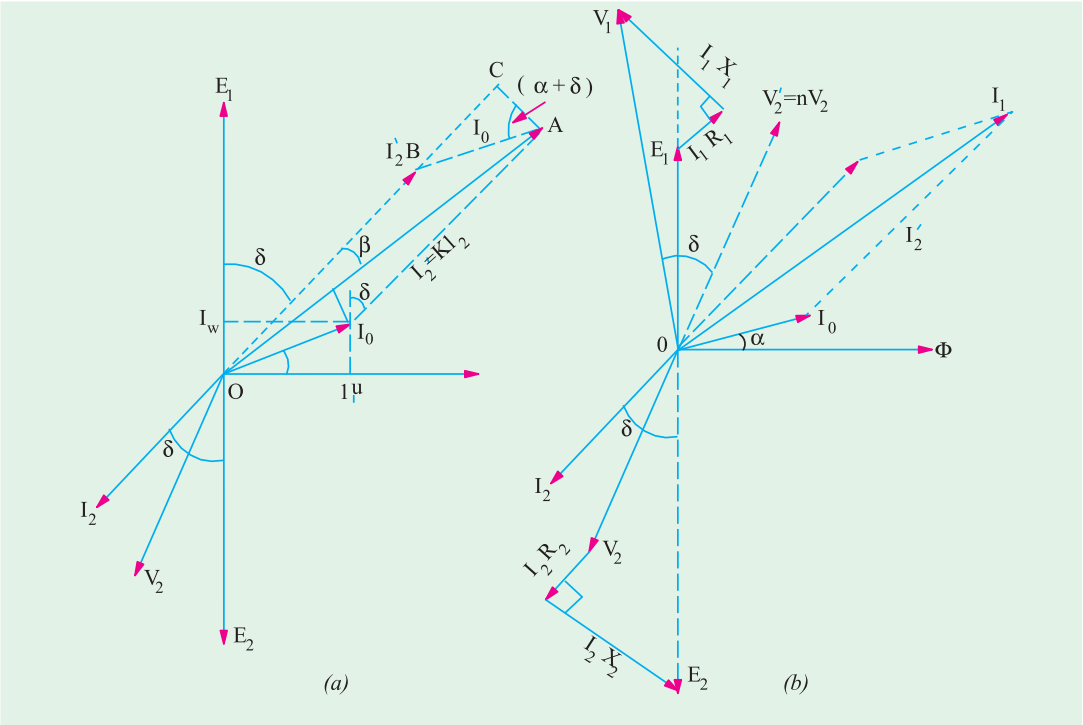


Fig. 10.85

vector I_2. The angular displacement between I_1 and I_2 reversed is called the phase angle error of the current transformer. This angle is reckoned positive if the reversed secondary current ***leads*** the primary current. However, on very low power factors, the phase angle may be negative. Similarly, there is an angle of γ between the primary voltage V_1 and secondary voltage reversed-this angle represents the phase angle error of a voltage transformer. In either case, the phase error depends on the magnetising component $I\mu$ of the primary current. It may be noted that ratio error is primarily due to the reason that the ***terminal*** voltage transformation ratio of a transformer is not exactly equal to its turn ratio. The divergence between the two depends on the resistance and reactance of the transformer windings as well as upon the value of the exciting current of the transformer. Accuracy of voltage ratio is of utmost importance in a voltage transformer although phase angle error does not matter if it is to be merely connected *to* ***a voltmeter.*** Phase-angle error becomes important only when voltage transformer supplies the voltage coil of a wattmeter *i.e.* in power measurement. In that case, phase angle error causes the wattmeter to indicate on a wrong power factor.

In the case of current transformers, constancy of current ratio is of paramount importance. Again, phase angle error is of no significance if the current transformer is merely feeding an ammeter but it assumes importance when feeding the current coil of a wattmeter. While discussing errors, it is worthwhile to define the following terms :

(i) Nominal transformation ratio (k_n). It is the ratio of the ***rated*** primary to the ***rated*** secondary current (or voltage).

$$k_n = \frac{\text{rated primary current } (I_1)}{\text{rated secondary current } (I_2)} \quad \text{—for } CT$$

$$= \frac{\text{rated primary voltage } (V_1)}{\text{rated secondary voltage } (V_2)} \quad \text{—for } VT$$

In the case of current transformers, it may be stated either as a fraction such as 500/5 or 100/1 or

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simply as the number representing the numerator of the reduced fraction *i.e.* 100. It is also known as ***marked*** ratio.

(ii) Actual transformation ratio (k). The actual transformation ratio or just ratio under any given *condition of loading is*

$$k = \frac{\text{primary current } (I_1)}{\text{corresponding secondary current } (I_2)}$$

In general, k differs from k_n except in the case of an ideal or perfect transformer when $k = k_n$ for all conditions of loading.

(iii) Ratio Error (σ). In most measurements it may be assumed that $I_1 = k_n I_2$ but for very accurate work, it is necessary to correct for the difference between k and k_n. It can be done with the help of ratio error which is defined as

$$\sigma = \frac{k_n - k}{k} = \frac{\text{nominal ratio-actual ratio}}{\text{actual ratio}}$$

Also,
$$\sigma = \frac{k_n \,.\, I_2 - kI_2}{k \,.\, I_2} = \frac{k_n \,.\, I_2 - I_1}{I_1}$$

Accordingly, ratio error may be defined as the difference ***between the primary current reading (assuming the nominal ratio) and the true primary current divided by the true primary current.***

(iv) Ratio Correction Factor (R.C.F.). It is given by

$$R.C.F. = \frac{\text{actual ratio}}{\text{nominal ratio}} \quad \frac{k}{k_n}$$

10.70. Current Transformer

A current transformer takes the place of shunt in d.c. measurements and enables heavy alternating current to be measured with the help of a standard 5-A range a.c. ammeter.

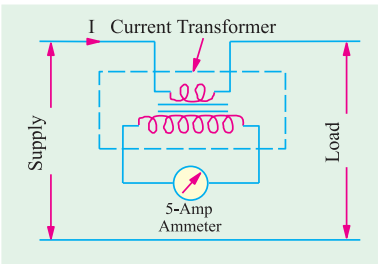


Fig. 10.86

As shown in Fig. 10.86, the current - or series-transformer has a primary winding of one or more turns of thick wire connected in series with the line carrying the current to be measured. The secondary consist of a large number of turns of fine wire and feeds a standard 5-A ammeter (Fig. 10.86) or the current coil of a watt-meter or watthour-meter (Fig. 10.87).

For example, a 1,000/5A current transformer with in single-turn primary will have 200 secondary turns. Obviously, it steps down the current in the 200 : 1 ratio whereas it steps up the voltage drop across the single-turn primary (an extremely small quantity) in the ratio 1 : 200. Hence if we know the current ratio of the transformer and the reading of the a.c. ammeter, the line current can be calculated.

It is worth noting that ammeter resistance being extremely low, a current transformer operates with its secondary under nearly short-circuit conditions. Should it be necessasry to remove the ammeter of the current coils of the wattmeter or a relay, the seondary winding must, first of all, be short-circuited *before* the instrument is disconnected.

If it is not done then due to the absence of counter ampere-turns of the secondary, the unopposed primary m.m.f. will set up an abnormally high flux in the core which will produce excessive core loss with subsequent heating of and damage of the transformer insulation and a high voltage across the secondary

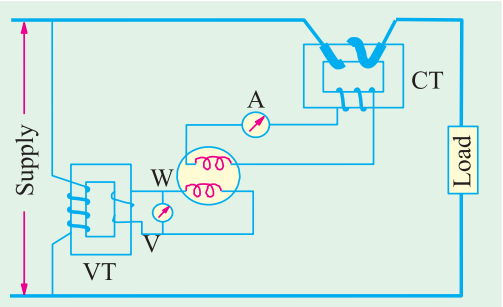


Fig. 10.87

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terminals. This is not the case with the ordinary constant-potential transformers because their primary current is determined by the load on their secondary whereas in a current transformer, primary current is determined entirely by the load on the system and not by the load on its own secondary. Hence, the secondary of a current transformer should ***never be left open under any circumstances.***

10.71. Theory of Current Transformer

Fig. 10.85 (*b*) represents the general phase diagram for a current transformer. Current I_0 has been exaggerated for clarity.

(a) Ratio Error. For obtaining an expression for the ratio error, it will be assumed that the turn ratio n (= secondary turns, N_2/primary turns N_1) is made equal to the nominal current ratio *i.e.* $n = k_n$. In other words, it will be assumed that $I_1/I_2 = n$ although actually $n = I_1/I_2'$. As seen from Art. 10.63.

$$\sigma = \frac{n I_2 - I_1}{I_1} = \frac{I_2' - I_1}{I_1} = \frac{OB - OA}{OA} \qquad \text{— } [\because n = k_n]$$

$$\equiv \frac{OB - OC}{OA} \qquad (\because \beta \text{ is very small angle})$$

$$= -\frac{BC}{OA} = -\frac{AB \sin(\alpha + \delta)}{OA} = -\frac{I_0 \sin(\alpha + \delta)}{I_1} = -\frac{I_0 \sin(\alpha + \delta)}{nI_2}$$

For most instrument transformers, the power factor of the secondary burden is nearly unity so that δ is very small. Hence, very approximately.

$$\sigma = \frac{I_0 \sin \alpha}{I_1} - \frac{I_\omega}{I_1}$$

where I_ω is the iron-loss or working or wattful component of the exciting current I_0

Note. The transformation ratio R may be found from Fig. 10.85 (*a*) as under :

$I_1 = OA = OB + BC = nI_2 \cos\beta + I_0 \cos[90 - (\delta + \beta + \alpha)] = nI_2 \cos\beta + I_0 \sin(\delta + \beta + \alpha)$

Now $\beta = (\alpha + \delta)$ hence $I_1 = nI_2 + I_0 \sin(\alpha + \delta)$ where n is the turn ratio of the transformer.

$$\therefore \qquad \text{ratio } R = \frac{I_1}{I_2} = \frac{nI_2 + I_0 \sin(\alpha + \delta)}{I_2} \text{ or } R = n + \frac{I_0 \sin(\alpha + \delta)}{I_2} \qquad \text{...(i)}$$

If δ is negligible small, then $R = n + \dfrac{I_0 \sin \alpha}{I_2} = n + \dfrac{I_\omega}{I_2}$

It is obvious from **(*i*)** above that ratio error can be eliminated if secondary turn are reduced by a number

$$= I_0 \sin(\alpha + \delta)/I_2$$

(*b*) Phase angle (β)

Again from Fig. 10.85 (*a*), we fine that

$$\beta \cong \sin\beta = \frac{AC}{OA} = \frac{AB \cos(\alpha + \delta)}{OA} = \frac{I_0 \cos(\alpha + \delta)}{I_1} = \frac{I_0 \cos(\alpha + \delta)}{nI_2}$$

Again, if the secondary power factor is nearly unity, then δ is very small, hence

$$\beta \cong \frac{I_0 \cos \alpha}{I_1} = \frac{I_\mu}{I_1} \text{ or } \frac{I_\mu}{nI_2}$$

where I_μ is the magnetising component of the exciting current I_0.

$$\therefore \qquad \beta = \frac{I_\mu}{I_1} \text{ —in radian; } = \frac{180}{\pi} \times \frac{I_\mu}{I_1} \text{ —in degrees}$$

Note. As found above, $\beta = \dfrac{I_0 \cos(\quad)}{I_1}$ $\dfrac{I_0 (\cos \quad \cos \quad \sin \quad \sin \quad)}{I_1}$

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$$= \frac{I_\mu \cos\delta - I_\omega \sin\delta}{I_1} = \frac{I_\mu \cos\delta - I_\omega \sin\delta}{nI_2} \text{ radian}$$

$$\therefore \quad \beta = \frac{180}{\pi} \times \frac{I_\mu \cos\delta - I_\omega \sin\delta}{nI_2} \text{ degrees.}$$

Dependence of ratio error on working component of I_0 and that of phase angle on the magnetising component is obvious. If R is to come closer to k and β is to become negligible small, then I_μ and I_ω and hence I_0 should be very small.

10.72. Clip-on Type Current Transformer

It has a laminated core which is so arranged that it can be opened out at a hinged section by merely pressing a trigger-like projection (Fig. 10.88). When the core is thus opened, it permits the admission of very heavy current-carrying bus-bars or feeders whereupon the trigger is released and the core is tightly closed by a spring. The current-carrying conductor of feeder acts as a single-turn primary whereas the secondary is connected across the standard ammeter conveniently mounted in the handle itself.

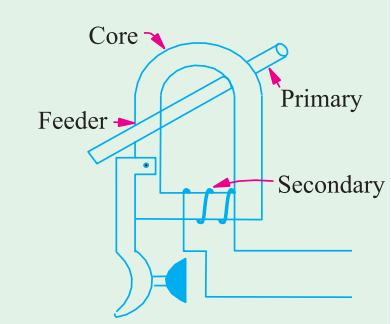


Fig. 10.88

10.73. Potential Transformers

These transformers are extremely accurate-ratio stepdown transformers and are used in conjunction with standard low-range voltmeters (100-120 V) whose deflection when divided by transformation ratio, gives the true voltage on the primary or high-voltage side. In general, they are of the shell type and do not differ much from the ordinary two-winding transformers except that their power rating is extremely small. Sine their secondary windings are required to operate instruments or relays or pilot lights, their ratings are usually of 40 to 100W. For safety, the secondary is completely insulated from the high voltage primary and is, in addition, grounded for affording protection to the operator. Fig. 10.89 shows the connection of such a transformer.



Potential transformer

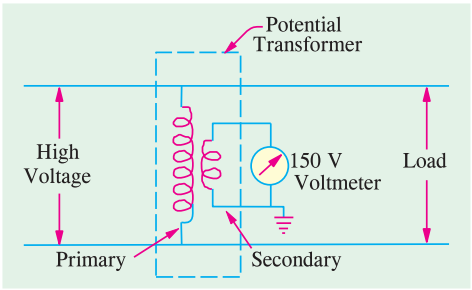


Fig. 10.89

10.74. Ratio and Phase-angle Errors

In the case of a potential transformer, we are interested in the ratio of the primary to the secondary terminal voltage and in the phase angle γ between the primary and reversed secondary terminal voltage V_2'.

The general theory of voltage transformer is the same as for the power transformers except that, as the current in the secondary burden is very small, the total primary current I_1 is not much greater than I_0.

In the phasor diagram of Fig. 10.90, vectors *AB, BC, CD* and *DE* represent small voltage drops due to resistances and reactances of the transformer winding (they have been exaggerated for the sake of clarity). Since the drops as well as the phase angle γ are small, the top portion of diagram 10.90 (*a*) can be drawn with negligible loss of accuracy as in Fig. 10.90 (*b*) where V_2' vector has been drawn parallel to the vector for V_1.

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In these diagrams, V_2' is the secondary terminal voltage as referred to primary assuming transformation without voltage drops. All actual voltage drops have been referred to the primary. Vector AB represents total resistive drop as referred to primary *i.e.* $I_2'\ R_{01}$. Similarly, BC represents total reactive drop as referred to primary *i.e.* $I_2'\ X_{01}$.

In a voltage transformer, the relatively large no-load current produces appreciable resistive drops which have been represented by vectors CD and DE respectively. Their values are I_0R_1 and I_0X_1 respectively.

(*a*) Ratio Error

In the following theory, n would be taken to represent the ratio of ***primary turns to secondary turns*** (Art. 10.69). Further, it would be assumed, as before, that n equals the nominal transformation ratio *i.e.* $n = k_n$.

In other words, it would be assumed that $V_1/V_2 = n$, although, actually, $V_1/V_2' = n$.

Then
$$\sigma = \frac{k_n - k}{k} = \frac{k_n . V_2 - k V_2}{kV_2} = \frac{V_2' - V_1}{V_1} = -\frac{EN}{OE} \quad \text{...Fig. 10.90 } (a)$$

$$= -\frac{AG + FC + LD + EM}{OE} \quad \text{...Fig. 10.90 } (b)$$

$$= -\frac{I_2' R_{02} \cos\delta + I_2' X_{02} \sin\delta + I_0R_1 \sin\alpha + I_0X_1 \cos\alpha}{V_1}$$

$$= -\frac{I_2' R_{02} \cos\delta + I_2' X_{02} \sin\delta + I_\mu R_1 + I_\mu X_1}{V_1}$$

where I_ω and I_μ are the iron-loss and magnetising components of the no-load primary current I_0.

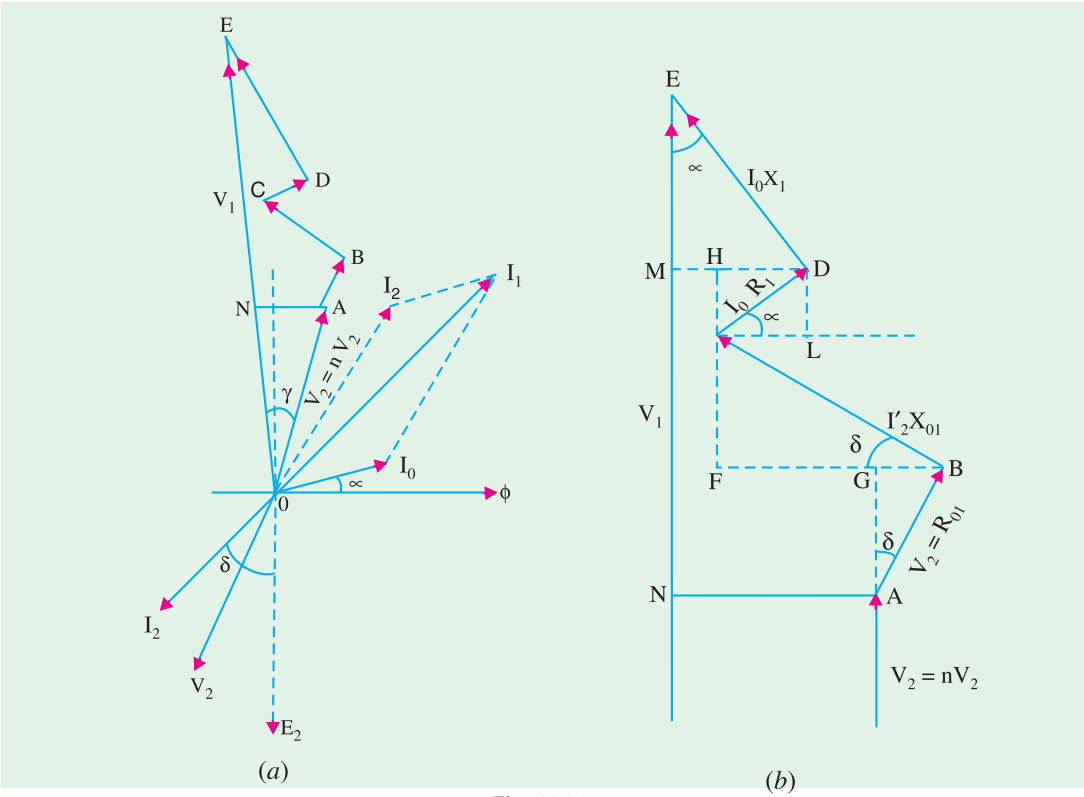


Fig. 10.90

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(b) Phase Angle (γ)

To a very close approximation, value of γ is given by $\gamma = AN/OA$ —in radian

Now, $OA \cong OE$ provided ratio error is neglected. In that case,

$$\gamma = \frac{AN}{OE} = -\frac{GF + HM}{OE} \quad \text{...Fig. 10.90 } (b)$$

$$= -\frac{(BE - BG) + (DM - DH)}{OE}$$

$$= -\frac{I_2' X_{01} \cos\delta - I_2' R_{01} \sin\delta + I_0 X_1 \sin\alpha - I_0 R_1 \cos\alpha}{V_1}$$

$$= -\frac{I_2' X_{01} \cos\delta - I_2' R_{01} \sin\delta + I_\omega X_1 - I_\mu R_1}{V_1}$$

The negative sign has been given because reversed secondary voltage *i.e.* V_2' lags behind V_1.

Example 10.44. *A current transformer with 5 primary turns and a nominal ratio of 1000/5 is operating with a total secondary impedance of 0.4 + j 0.3 Ω. At rated load, the iron loss and magnetising components of no-load primary current are 1.5 A and 6 A respectively. Calculate the ratio error and phase angle at rated primary current if the secondary has* ***(a)*** *1000 turns and* ***(b)*** *990 turns.*

Solution. Phasor diagram of Fig. 10.87 may please be referred to.

(a) $\tan\delta = 0.3/0.4$ or $\delta = \tan^{-1}(0.3/0.4) = 36°52'$

$\alpha = \tan^{-1}(1.5/6) = 14°2'$ $\quad \therefore (\alpha + \beta) = 50°54'$

$I_0 = \sqrt{I_\omega^2 + I_\mu^2} = \sqrt{1.5^2 + 6^2} = 6.186$ A

$BC = I_0 \sin(\alpha + \beta) = 6.186 \times \sin 50°54' = 4.8$ A

Since β is amall, $OC \cong OA = 1000$ A $\quad \therefore I_2' = OC - BC = 1000 - 4.8 = 995.2$ A

$$\therefore \quad \sigma = \frac{k_n I_2 - I_1}{I_1} = \frac{nI_2 - I_1}{I_1}$$ because in this case $n\ (= N_2/N_1 = 1000/5)$ is equal to nominal ratio k_n (= 1000/5 A).

or $$\sigma = \frac{I_2 \quad I_1}{I_1} \quad \frac{995.2 \quad 1000}{1000} \quad 0.0048 \text{ or } \mathbf{48\%}$$

$\beta = \tan^{-1}(AC/OC)$

Now $AC = I_0 \cos(\alpha + \beta) = 6.186 \cos 50°54' = 3.9$ A

centre $\beta = \tan^{-1}(AC/OC) = \tan^{-1}(3.9/1000) = \mathbf{0°13'}$

(b) In this case, $I_2 = 995.2 \times \frac{5}{990}$; $k_n . I_2 = \frac{1000}{5} \times 995.2 \times \frac{5}{990} = 1005.3$ A

$$\therefore \quad \sigma = \frac{k_n I_2 \quad I_1}{I_1} \quad \frac{1005.3 \quad 1000}{1000} \quad 0.0053 \text{ or } \mathbf{0.53\%}$$

The value of phase angle β would not be significantly different from the value obtained in (*a*) above.

Example 10.45. *A relay current-transformer has a bar primary and 200 secondary turns. The secondary burden is an ammeter of resistance 1.2 Ω and resistance of 0.5 Ω and the secondary winding has a resistance of 0.2 Ω and reactance of 0.3 Ω. The core requires the equivalent of 100 AT for magnetisation and 50 AT for core losses.*

(i) *Find the primary current and the ratio error when the secondary ammeter indicates 5.0 A.*

(ii) *By how many turns should the secondary winding be reduced to eliminate the ratio error for this condition ?* **(Electrical Measurements, Bombay Univ.)**

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Solution. Total secondary impedance is

$$\mathbf{Z_2} = 1.4 + j\,0.8 = 1.612 \angle 29°45' \quad \therefore \ \delta = 29°45'$$

$$\mathbf{I0} = 100 + j50 = 111.8 \angle 26°34' \quad \therefore \ \alpha = 26°34'$$

Turn ratio, $n = 200/I = 200$; Transformation ratio $R = n + \dfrac{I_0 \sin(\alpha + \delta)}{I_2}$

$$\therefore \quad R = 200 + \frac{111.8 \sin 56°19'}{5} = 218.6$$

(*i*) Primary current $= 5 \times 218.6 = \mathbf{1093\ A}$

Ratio error $\sigma = \dfrac{I_0 \sin(\quad)}{nI_2}$ $\dfrac{111.8 \quad 0.8321}{200 \quad 5}$ **0.093 or 9.3%**

(*ii*) No. of secondary turns to be reduced $= I_0 \sin(\alpha + \delta)/I_2 = 93/5 =$ **19 (approx).**

Example 10.46. *A current transformer has 3 primary turns and 300 secondary turns. The total impedance of the secondary is (0.583 + j 0.25) ohm. The secondary current is 5 A. The ampere-turns required to supply excitation and iron losses are respectively 10 and 5 per volt induced in the secondary.*

Determine the primary current and phase angle of the transformer.

(Elect Meas; M.S. Univ. Baroda)

Solution. Z2 $= 0.583 = j\,0.25 = 0.6343 \angle 23°\ 10' \ \therefore E_2 = I_2 Z_2 = 5 \times 0.6343 =$ **3.17 V**

Now, there are 10 magnetising *AT* per *secondary* volt induced in secondary.

$\therefore$ total magnetising $AT = 3.17 \times 10 = 31.7$; Similarly, iron-loss $AT = 3.17 \times 5 = 15.85$

Remembering that there are 3 primary turns, the magnetising and iron-loss components of primary current are as under :

Magnetising current, $I_\mu = 31.7/3 = 10.6$ A; iron-loss current $I_\omega = 15.85/3 = 5.28$ A

$$I_0 = \sqrt{10.6^2 + 5.28^2} = 11.84 \text{ A}$$

Now, $R = n \dfrac{I_0 \sin(\quad)}{I_2}$

Here, $n = 300/3 = 100$; $\alpha = \tan^{-1}(I_\omega/I_\mu) = \tan^{-1}(5.28/10.6)$

$= \tan^{-1}(0.498) = 26°30'$

δ = secondary load angle $= 23°10'$ —found earlier

$$\therefore \quad R = 100 + \frac{11.86}{5}(\sin 49°40') = 100 + 1.81 = 101.81$$

$$I_1 = R \times I_2 = 101.81 \times 5 = \mathbf{509.05\ A}$$

$$\beta = \frac{180}{\pi} \times \frac{I_\mu \cos\delta - I_\omega \sin\delta}{nI_2}$$

$$= \frac{180}{\quad} \quad \frac{10.6 \cos 23\ 10 \quad 5.28 \sin 23\ 10}{100 \quad 5} \quad \mathbf{0.88°}$$

Example 10.47. *A current transformer with a bar primary has 300 turns in its secondary winding. The resistance and reactance of the secondary circuit are 1.5 Ω and 1.0 Ω respectively including the transformer winding. With 5 A flowing in the secondary circuit the magnetising ampere-turns required are 100 and iron loss is 1.2 W. Determine the ratio error at this condition.*

(Elect. Measure, A.M.I.E. Sec. B, 1992)

Solution. Turn ratio $n = 300/1 = 300$

Secondary impedance is $Z_2 = 1.5 + j\,1.0 = 1.8 \angle 33°42'$

Secondary induced e.m.f. $E_2 = I_2 Z_2 = 5 \times 1.8 = 9$ V

$E_1 = E_2/n = 9/300 = 0.03$ V

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Let us now find the magnetising and working components of primary no-load current I_0

Magnetising $AT = 100$. Since threre is one primary turn, $\therefore I_\mu = 100/1 = 100$ A

Now, $E_1 I_\omega = 1.2 \quad \therefore I_\omega = 1.2/0.03 = 40$ A

$$I_0 = 100 + j\,40 = 107.7 \angle 21°48' \; ; \sigma = -\frac{I_0 \sin(\alpha + \delta)}{nI_2}$$

Now $\alpha = 21°48'$ and $\delta = 33°42'$

$$\therefore \quad \sigma = \frac{107.7 \sin 55\ 30}{300\quad 5} \qquad \textbf{0.0592 or 5.92\%}$$

Phase angle $$\beta = \frac{I_0 \cos(\alpha + \delta)}{nI_2} = \frac{107.7 \times \cos 55°30'}{1500}$$

$$= \frac{180\quad 107.7\quad 0.5664}{1500} \qquad \textbf{2°20}$$

Tutorial Problems No. 10.4

1. A current transformer with 5 primary turns has a secondary burden consisting of a resistance of 0.16 Ω and an inductive reactance of 0.12Ω. When primary current is 200 A, the magnetising current is 1.5 A and the iron-loss component is 0.4 A. Determine the number of secondary turns needed to make the current ratio 100/1 and also the phase angle under these conditions. **[407 : 0.275°]**
2. A current transformer having a 1-turn primary is rated at 500/5 A, 50 Hz, with an output of 1.5 VA. At rated load with the non-inductive burden, the in-phase and quadrature components (referred to the flux) of the exciting ampere-turns are 8 and 10 respectively. The number of turns in the secondary is 98 and and the resistance and leakage reactance of the secondary winding are 0.35 Ω and 0.3 Ω respectively. Calculate the current ratio and the phase angle error. **[501.95/5; 0.533°]** ***(Elect. Inst. and Meas, M.S. Univ. Baroda)***
3. A ring-core current transformer with a nominal ratio of 500/5 and a bar primary has a secondary resistance of 0.5 Ω and negligible secondary reactance. The resultant of the magnetising and iron-losss components of the primary current associated with a full-load secondary current of 5 A in a burden of 1.0 Ω (non-inductive) is 3 A at a power factor of 0.4. Calculate the true ratio and the phase-angle error of the transformer on full-load. Calculate also the total flux in the core, assuming that frequency is 50 Hz. **[501.2/5; 0.314°; 337 μWb]**
4. A current transformer has a single-turn primary and a 200-turn secondary winding. The secondary supplies a current of 5 A to a non-inductive burden of 1 Ω resistance, the requisite flux is set up in the core by 80 AT. The frequency is 50 Hz and the net cross-section of the core is 10 cm^2. Calculate the ratio and phase angle and the flux density in the core. **[200.64; 4°35′ 0.079 Wb/m2]** ***(Electrical Measurements, Osmania Univ.)***
5. A potential transformer, ratio 1000/100-V, has the following constants :

   primary resistance = 94.5 Ω ; secondary resistance = 0.86 Ω
   primary reactance = 66.2 Ω ; equivalent reactance = 66.2 Ω
   magnetising current = 0.02 A at 0.4 p.f.

   Calculate (*i*) the phase angle at no-load between primary and secondary voltages (*ii*) the load in VA at u.p.f. at which the phase angle would be zero. **[(i) 0°4′ (ii) 18.1 VA]**
6. PMMC instrument has FSD current of 50 milliampere and 2 ohm resistance. How the instrument can be converted to
   (*i*) 0.5 A range Ammeter (*ii*) 0.100 V range Voltmeter? ***(Nagpur University, Summer 2002)***
7. What are the essential torques of an indicating instruments? Justify their necessity. ***(Nagpur University, Winter 2002)***
8. Discuss the necessity of damping in Indicating instrument and explain eddy current damping. ***(U.P. Technical University 2002) (Nagpur University, Summer 2003)***
9. PMMC instrument has FSD current of 50 milliampere and 2 ohm resistance. How the instrument can be converted to (*i*) 0-5 A range Ammeter (*ii*) 0-100 V range Voltmeter? ***(Nagpur University, Summer 2003)***

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10. What are the essential requirements of indicating type instruments? Explain each of them. ***(U.P. TechnicalUniversity 2002) (Nagpur University, Winter 2003)***

11. What are the different operating systems required in an instrument? Explain damping system in detail. ***(U.P. TechnicalUniversity 2002) (Nagpur University, Summer 2004)***

12. If a shunt for a moving coil instrument is a have a multiplicationfactor m, shown that its shunt resistance is given by

$$R_{sh} = \frac{R_m}{m - 1}$$

where R_m →resistance of the meter. ***(Nagpur University, Summer 2004)***

13. Find the value of a series resistance to be connected to a basic d' Arsonval movement with internal resistance R_m = 100 ohm and full scale deflection current is 1 mA, for conversion into 0–500 volt. ***(Nagpur University, Summer 2004)***

14. What is the basic principle of Induction type instruments? ***(Anna University, April 2002)***

15. What is damping torque in instruments? ***(Anna University, April 2002)***

16. Why does MI instruments have non-linear scale? ***(Anna University, April 2002)***

17. Explain how can you measure power by the Ammeters. ***(Anna University, April 2002)***

18. What is creeping in energy meter? ***(Anna University, April 2002)***

19. What is the advantage of Induction type meters? ***(Anna University, April 2002)***

20. What is a trivector meter? ***(Anna University, April 2002)***

21. What is the difference between PMMC instrument and ballistic galvanometer? ***(Anna University, April 2002)***

22. What is the basic principle of operation of reed type frequency meters? ***(Anna University, April 2002)***

23. Why can't you measure low resistance with meggar? ***(Anna University, April 2002)***

24. What are the advantages of electronic meters? ***(Anna University, April 2002)***

25. Explain the principle of operation, construction and the expression for deflection of a moving iron instrument. ***(Anna University, April 2002)***

26. Draw and explain the working of 2 element wattmeter. Explain how does the 2 element wattmeter reads total power in a 3 ϕ circuit. ***(Anna University, April 2002)***

27. Write the equations for wattmeter readings W_1 and W_2 in 3 phase power measurement and therefrom for power factor. ***(Anna University, Wimter 2002)***

28. Explain how power can be measured in a three phase circuit with the help of two wattmeters, for a balanced star connected load. Draw the phasor diagram. ***(Anna University Winter 2002)***

29. State the working principle of a dynamometer type wattmeter and show its connections. ***(V.T.U., Belgaum Karnataka University, February 2002)***

30. A.D.C. milliammeter having a resistance of 2Ω gives full scale deflection when the current is 50mA. How can it be used to measure (*i*) a current of 5A (*ii*) a voltage of 500 V. ***(V.T.U., Belgaum Karnataka University, February 2002)***

31. With the help of a neat diagram, explain the construction and principle of operation of single phase energy meter. ***(V.T.U., Belgaum Karnataka University, Wimter 2003)***

32. With a neat sketch explain the construction and working of a single phase induction type energymeter. ***(V.T.U., Belgaum Karnataka University, Summer 2003)***

OBJECTIVE TESTS—10

1. The kWh meter can be classified as a/an-instrument :
(*a*) deflecting (*b*) digital
(*c*) recording (*d*) indicating

2. The moving system of an indicating type of electrical instrument is subjected to :
(*a*) a deflecting torque
(*b*) a controlling torque
(*c*) a damping torque
(*d*) all of the above

3. The damping force acts on the moving system of an indicating instrument only when it is :
(*a*) moving (*b*) stationary
(*c*) near its full deflection
(*d*) just starting to move.

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4. The most efficient form of damping employed in electrical instruments is :
   (*a*) air friction
   (*b*) fluid friction
   (*c*) eddy currents
   (*d*) none of the above.
5. Moving iron instruments can be used for measuring :
   (*a*) direct currents and voltages
   (*b*) alternating current and voltages
   (*c*) radio frequency currents
   (*d*) both (*a*) and (*b*).
6. Permanent-magnet moving-coil ammetere have uniform scales because :
   (*a*) of eddy current damping
   (*b*) they are spring-controlled
   (*c*) their deflecting torque varies directly as current
   (*d*) both (*b*) and (*c*).
7. The meter that is suitable for *only* direct current measurements is :
   (*a*) moving-iron type
   (*b*) permanent-magnet type
   (*c*) electrodynamic type
   (*d*) hot-wire type.
8. A moving coil voltmeters measures—
   (*a*) only a.c. voltages
   (*b*) only d.c. voltages
   (*c*) both a.c. and d.c. voltages
   (Principles of Elect. Engg. Delhi Univ.)
9. The reading of the voltmeter in Fig. 10.91 would be nearest to–volt :
   (*a*) 80
   (*b*) 120
   (*c*) 200
   (*d*) 0

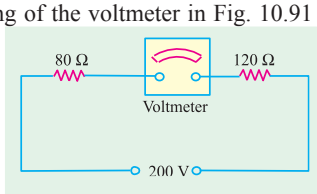


Fig. 10.91

10. The hot-wire ammeter :
   (*a*) is used only for d.c. circuits
   (*b*) is a high precision instrument
   (*c*) is used only for a.c. circuits
   (*d*) reads equally well on d.c. and/or a.c. circuits.

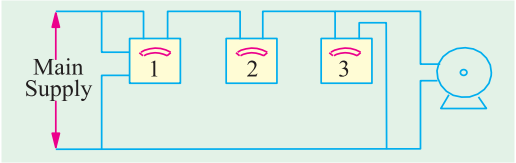


Fig. 10.92

11. If an energy meter disc makes 10 revolutions in 100 seconds when a load of 450 w is connected to it, the meter constant (in rev/k Wh) is
   (*a*) 1000 (*b*) 500
   (*c*) 1600 (*d*) 800
   (GATE 2001)
12. The inductance of a certain moving-iron ammeter is expressed as $L = 10 + 3\theta - \frac{\theta^2}{4}$ μH, where θ is the deflection in radians from the zero position. The control spring torque in 25×10^{-6} Nm/radian. The deflection of the pointer in radian when the meter carries a current of 5A, is
   (*a*) 2.4 (*b*) 2.0
   (*c*) 1.2 (*d*) 1.0
   (GATE 2003)
13. A dc potentiometer is designed to measure up to about 2 V with a slide wire of 800 mm. A standard cell of emf 1.18 V obtains balance at 600 mm. A test cell is seen to obtain balance at 680 mm. The emf of the test cell is
   (*a*) 1.00 V (*b*) 1.34 V
   (*c*) 1.50 V (*d*) 1.70 V
   (GATE 2004)
14. A galvanometer with a full scale current of 10 mA has a resistance of 1000 Ω. The multiplying power (the ratio of measured current to galvanometer current) of a 100 Ω shunt with this galvanometer is
   (*a*) 110 (*b*) 100
   (*c*) 11 (*d*) 10
   (GATE 2004)
15. A moving coil of a meter has 100 turns, and a length and depth of 10 mm and 20 mm respectively. It is positioned in a uniform radial flux density of 200mT. The coil carries a current of 50 mA. The torque on the coil is
   (*a*) 200 μNm (*b*) 100 μNm
   (*c*) 2 μNm (*d*) 1 μNm
   (GATE 2004)
16. A dc A-h meter is rated for 15 A, 250 V. The meter constant is 14.4 A-sec/rev. The meter constant at rated voltage may be expressed as
   (*a*) 3750 rev/kWh (*b*) 3600 rev/kWh
   (*c*) 1000 rev/kWh (*d*) 960 rev/kWh
   (GATE 2004)

ANSWERS

1. *c* **2.** *d* **3.** *a* **4.** *c* **5.** *d* **6.** *d* **7.** *b* **8.** *b* **9.** *c* **10.** *d*

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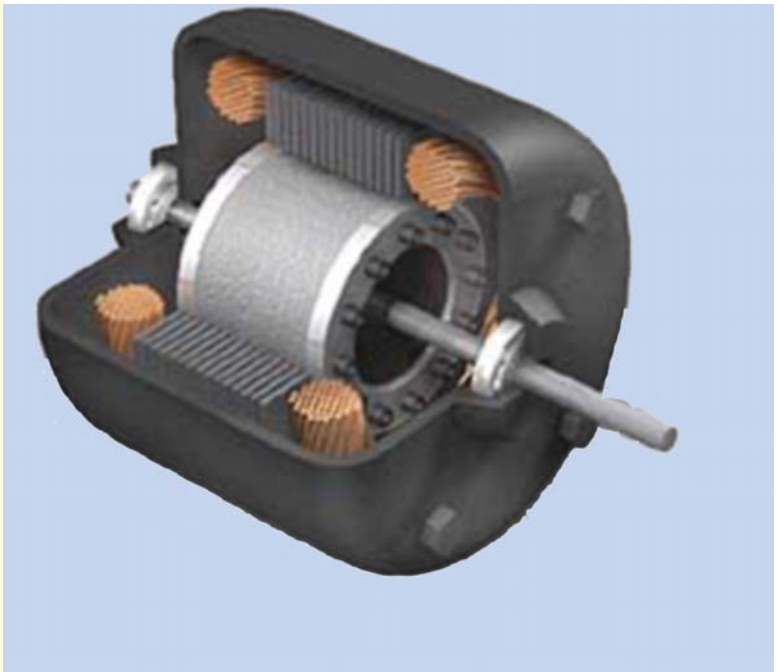
Contents

CHAPTER 11

A.C. FUNDA-MENTALS

Learning Objectives

- Generation of Alternating Voltages and Currents
- Alternate Method for the Equations of Alternating Voltages and currents
- Simple Waveforms
- Cycle
- Different Forms of E.M.F. Equation
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- Complex Voltage Applied to Pure Inductance
- A.C. through Capacitance alone



Alternating current circuits improves the versatility and usefulness of electrical power system. Alternating current plays a vital role in today's energy generation

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11.1. Generation of Alternating Voltages and Currents

Alternating voltage may be generated by rotating a coil in a magnetic field, as shown in Fig. 11.1 (*a*) or by rotating a magnetic field within a stationary coil, as shown in Fig. 11.1 (*b*).

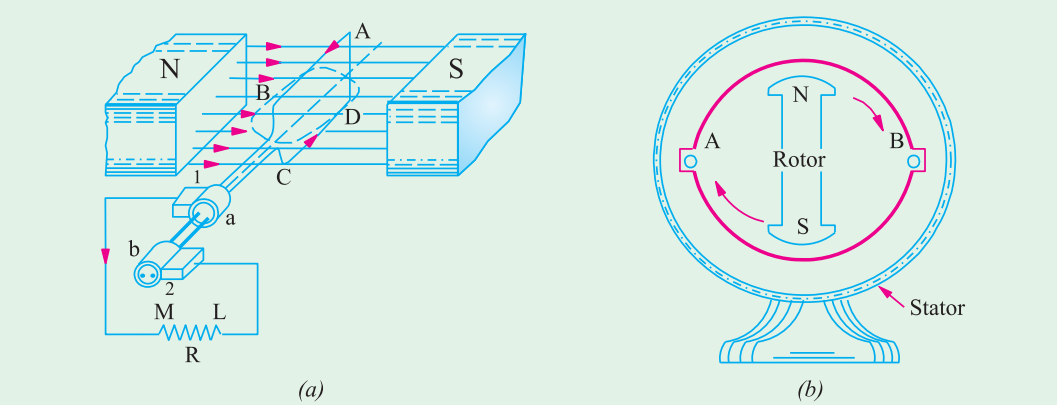


Fig. 11.1

The value of the voltage generated depends, in each case, upon the number of turns in the coil, strength of the field and the speed at which the coil or magnetic field rotates. Alternating voltage may be generated in either of the two ways shown above, but rotating-field method is the one which is mostly used in practice.

11.2. Equations of the Alternating Voltages and Currents

Consider a rectangular coil, having N turns and rotating in a uniform magnetic field, with an angular velocity of ω radian/second, as shown in Fig. 11.2. Let time be measured from the X-axis. Maximum flux Φ_m is linked with the coil, when its plane coincides with the X-axis. In time t seconds, this coil rotates through an angle $\theta = \omega t$. In this deflected position, the component of the flux which is perpendicular to the plane of the coil, is $\Phi = \Phi_m \cos \omega t$. Hence, ***flux linkages*** of the coil at any time are $N\Phi = N\Phi_m \cos \omega t$.

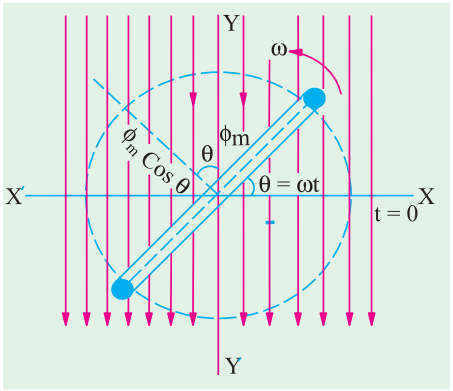


Fig. 11.2

According to Faraday's Laws of Electromagnetic Induction, the e.m.f. induced in the coil is given by the rate of change of flux-linkages of the coil. Hence, the value of the induced e.m.f. at this instant (*i.e.* when $\theta = \omega t$) or the instantaneous value of the induced e.m.f. is

$$e = -\frac{d}{dt}(N\Phi) \text{ volt} = -N.\frac{d}{dt}(\Phi_m \cos \omega t) \text{ volt} = -N\Phi_m \omega(-\sin \omega t) \text{ volt}$$

$$= \omega N \Phi_m \sin \omega t \text{ volt} = \omega N \Phi_m \sin \theta \text{ volt} \quad \text{...(i)}$$

When the coil has turned through 90° *i.e.* when $\theta = 90°$, then $\sin \theta = 1$, hence e has maximum value, say E_m. Therefore, from Eq. (*i*) we get

$$E_m = \omega N \Phi_m = \omega N B_m A = 2\pi f N B_m A \text{ volt} \quad \text{...(ii)}$$

where B_m = maximum flux density in Wb/m^2 ; A = area of the coil in m^2

f = frequency of rotation of the coil in rev/second

Substituting this value of E_m in Eq. (*i*), we get $e = E_m \sin \theta = E_m \sin \omega t$ ...(*iii*)

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Similarly, the equation of induced alternating current is $i = I_m \sin \omega t$ ...(iv)

provided the coil circuit has been closed through a resistive load.

Since $\omega = 2\pi f$, where f is the frequency of rotation of the coil, the above equations of the voltage and current can be written as

$$e = E_m \sin 2\pi f t = E_m \sin \left(\frac{2\pi}{T}\right) t \quad \text{and} \quad i = I_m \sin 2\pi f t = I_m \sin \left(\frac{2\pi}{T}\right) t$$

where T = time-period of the alternating voltage or current = $1/f$

It is seen that the induced e.m.f. varies as sine function of the time angle ωt and when e.m.f. is plotted against time, a curve similar to the one shown in Fig. 11.3 is obtained. This curve is known as sine curve and the e.m.f. which varies in this manner is known as ***sinusoidal*** e.m.f. Such a sine curve can be conveniently drawn, as shown in Fig. 11.4. A vector, equal in length to E_m is drawn. It rotates in the counter-clockwise direction with a velocity of ωradian/second, making one revolution while the generated e.m.f. makes two loops or one cycle. The projection of this vector on Y-axis gives the instantaneous value e of the induced e.m.f. *i.e.* $E_m \sin \omega t$.

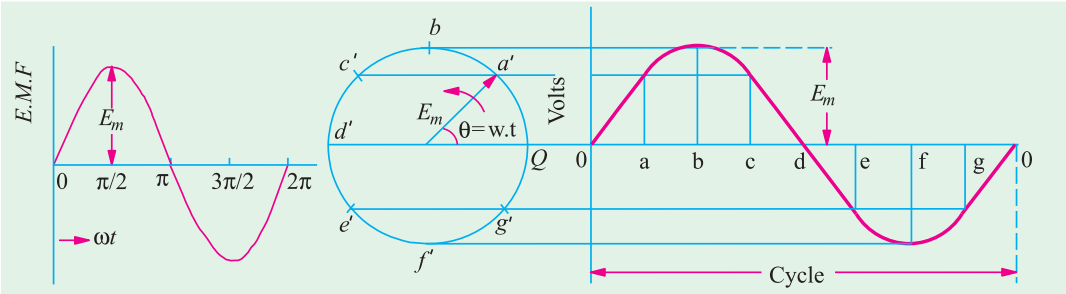


Fig. 11.3

Fig. 11.4

To construct the curve, lay off along X-axis equal angular distance oa, ab, bc, cd etc. corresponding to suitable angular displacement of the rotating vector. Now, erect coordinates at the points a, b, c and d etc. (Fig. 11.4) and then project the free ends of the vector E_m at the corresponding positions a', b', c', etc to meet these ordinates. Next draw a curve passing through these intersecting points. The curve so obtained is the graphic representation of equation ***(iii)*** above.

11.3. Alternate Method for the Equations of Alternating Voltages and Currents

In Fig. 11.5 is shown a rectangular coil AC having N turns and rotating in a magnetic field of flux density B Wb./m^2. Let the length of each of its sides A and C be l meters and their peripheral velocity v metre/second. Let angle be measured from the horizontal position *i.e.* from the X-axis. When in horizontal position, the two sides A and C move parallel to the lines of the magnetic flux. Hence, no flux is cut and so no e.m.f. is generated in the coil.

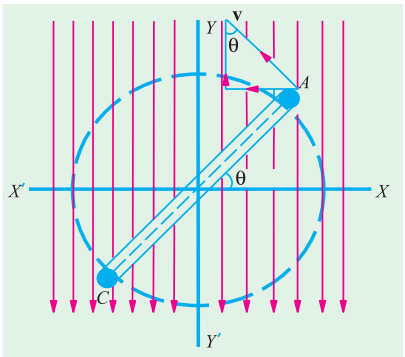


Fig. 11.5

When the coil has turned through angle θ, its velocity can be resolved into two mutually perpendicular components ***(i)*** $v \cos \theta$ component-parallel to the direction of the magnetic flux and ***(ii)*** $v \sin \theta$ component-perpendicular to the direction of the magnetic flux. The e.m.f. is generated due entirely to the perpendicular component *i.e.* $v \sin \theta$.

Hence, the e.m.f. generated in one side of the coil which contains N conductors, as seen from Art. 7.7, is given by, $e = N \times Bl\, v \sin \theta$.

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Total e.m.f. generated in both sides of the coil is

$$e = 2BNl\,v \sin\theta \text{ volt} \qquad \text{...(i)}$$

Now, e has maximum value of E_m (say) when $\theta = 90°$. Hence, from Eq. (i) above, we get,

$E_m = 2\,B\,N\,l\,v$ volt. Therefore Eq. **(i)** can be rewritten as $e = E_m \sin\theta$ ...as before

If b = width of the coil in meters ; f = frequency of rotation of coil in Hz, then $v = \pi\,bf$

$$\therefore \quad E_m = 2\,B\,N\,l \times \pi\,b\,f = 2\,\pi\,f\,N\,B\,A \text{ volts} \qquad \text{...as before}$$

Example 11.1. *A square coil of 10 cm side and 100 turns is rotated at a uniform speed of 1000 revolutions per minute, about an axis at right angles to a uniform magnetic field of 0.5 Wb/m^2. Calculate the instantaneous value of the induced electromotive force, when the plane of the coil is* ***(i)*** *at right angles to the field* ***(ii)*** *in the plane of the field.*

(Electromagnetic Theory, A.M.I.E. Sec B, 1992)



square coils

Solution. Let the magnetic field lie in the vertical plane and the coil in the horizontal plane. Also, let the angle θ be measured from X-axis.

Maximum value of the induced e.m.f., $E_m = 2\,\pi\,f\,N\,B_m\,A$ volt.

Instantaneous value of the induced e.m.f. $e = E_m \sin\theta$

Now $f = 100/60 = (50/3)$ rps, $N = 100$, $B_m = 0.5$ Wb/m^2, $A = 10^{-2}$ m^2

(i) In this case, $\theta = 0°$

$\therefore \quad e = 0$

(ii) Here $\theta = 90°$, $\quad \therefore \; e = E_m \sin 90° = E_m$

Substituting the given values, we get

$$e = 2\pi \times (50/3) \times 100 \times 0.5 \times 10^{-2} = \mathbf{52.3\ V}$$

11.4. Simple Waveforms

The shape of the curve obtained by plotting the instantaneous values of voltage or current as the ordinate against time as a abscissa is called its waveform or wave-shape.

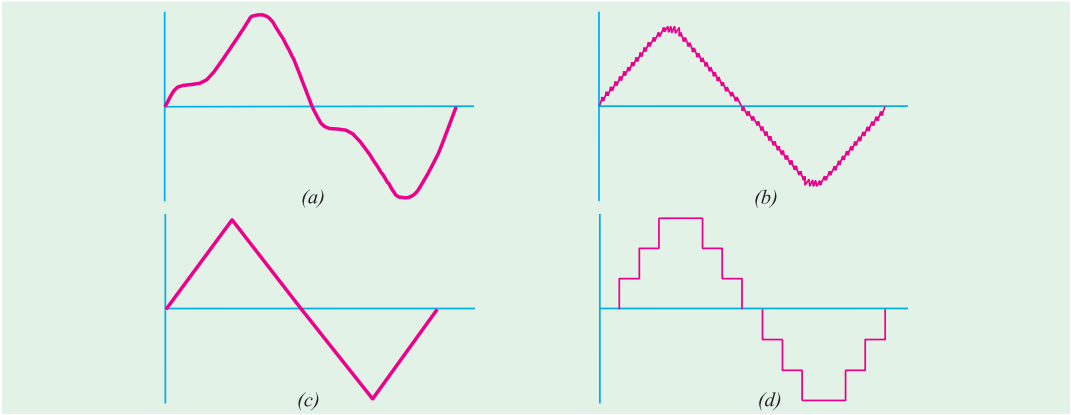


Fig. 11.6

An alternating voltage or current may not always take the form of a systematical or smooth wave such as that shown in Fig. 11.3. Thus, Fig. 11.6 also represents alternating waves. But while it is scarcely possible for the manufacturers to produce sine-wave generators or alternators, yet sine wave is the ideal form sought by the designers and is the accepted standard. The waves deviating from the standard sine wave are termed as distorted waves.

In general, however, ***an alternating current or voltage is one the circuit direction of which reverses at regularly recurring intervals.***

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11.5. Complex Waveforms

Complex waves are those which depart from the ideal sinusoidal form of Fig. 11.4. All alternating complex waves, which are periodic and have equal positive and negative half cycles can be shown to be made up of a number of pure sine waves, having different frequencies but all these frequencies are integral multiples of that of the lowest alternating wave, called the ***fundamental*** (or first harmonic). These waves of higher frequencies are called ***harmonics.*** If the fundamental frequency is 50 Hz, then the frequency of the second ***harmonic*** is 100 Hz and of the third is 150 Hz and so on. The complex wave may be composed of the fundamental wave (or first harmonic) and any number of other harmonics.

In Fig. 11.7 is shown a complex wave which is made up of a fundamental sine wave of frequency of 50 Hz and third harmonic of frequency of 150 Hz. It is seen that

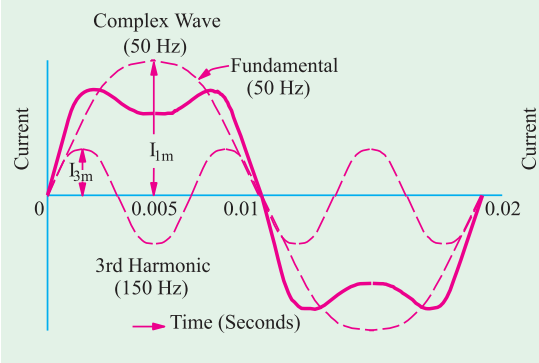


Fig. 11.7

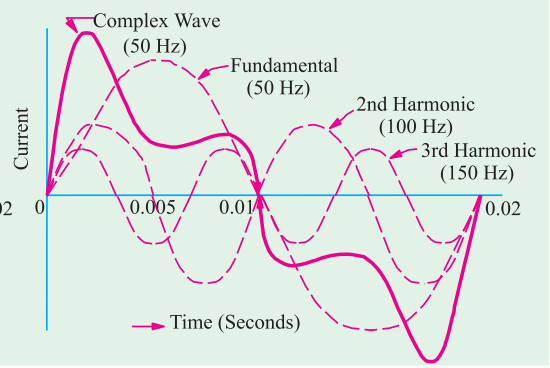


Fig. 11.8

(i) the two halves of the complex wave are identical in shape. In other words, there is no distortion. This is always the case when only ***odd*** harmonic (3rd, 5th, 7th, 9th etc.) are present.

(ii) frequency of the complex wave is 50 Hz *i.e.* the same as that of the fundamental sine wave.

In Fig. 11.8 is shown a complex wave which is a combination of fundamental sine wave of frequency 50 Hz and 2nd harmonic of frequency 100 Hz and 3rd harmonic of frequency 150 Hz. It is seen that although the frequency of the complex wave even now remains 50 Hz, yet :

(i) the two halves of the complex wave are not identical. It is always so when *even* harmonics (2nd, 4th, 6th etc.) are present.

(ii) there is distortion and greater departure of the wave shape from the purely sinusoidal shape.

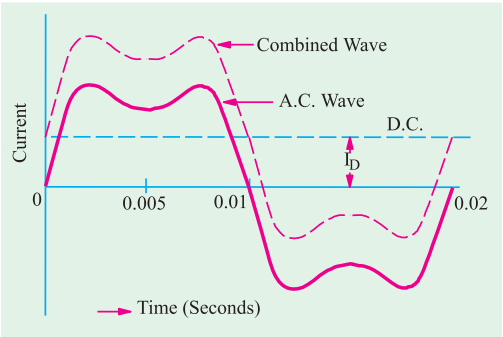


Fig. 11.9

Sometimes, a combination of an alternating and direct current flows simultaneously through a circuit In Fig. 11.9 is shown a complex wave (containing fundamental and third harmonic) combined with a direct current of value I_D. It is seen that the resultant wave remains undistorted in shape but is raised above the axis by an amount I_D. It is worth noting that with reference to the original axis, the two halves of the combined wave are not equal in area.

11.6. Cycle

One complete set of positive and negative values of alternating quantity is known as cycle. Hence, each diagram of Fig. 11.6 represents one complete cycle.

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A cycle may also be sometimes specified in terms of angular measure. In that case, one complete cycle is said to spread over 360° or 2π radians.

11.7. Time Period

The time taken by an alternating quantity to complete one cycle is called its time period T. For example, a 50-Hz alternating current has a time period of 1/50 second.

11.8. Frequency

The number of cycles/second is called the frequency of the alternating quantity. Its unit is hertz (Hz).

In the simple 2-pole alternator of Fig. 24.1 (*b*), one cycle of alternating current is generated in one revolution of the rotating field. However, if there were 4 poles, then two cycles would have been produced in each revolution. In fact, the frequency of the alternating voltage produced is a function of the speed and the number of poles of the generator. The relation connecting the above three quantities is given as

$f = PN/120$ where N = revolutions in r.p.m. and P = number of poles

For example, an alternator having 20 poles and running at 300 r.p.m. will generate alternating voltage and current whose frequency is $20 \times 300/120 = 50$ hertz (Hz).

It may be noted that the frequency is given by the reciprocal of the time period of the alternating quantity.

$$\therefore \qquad f = 1/T \quad \text{or} \quad T = 1/f$$

11.9. Amplitude

The maximum value, positive or negative, of an alternating quantity is known as its amplitude.

11.10. Different Forms of E.M.F. Equation

The standard form of an alternating voltage, as already given in Art. 11.2, is

$$e = E_m \sin \theta = E_m \sin \omega t = E_m \sin 2\pi f t = E_m \sin \frac{2\pi}{T} t$$

By closely looking at the above equations, we find that

(i) the maximum value or peak value or amplitude of an ***alternating voltage is given by the coefficient of the sine of the time angle.***

(ii) the frequency f is given by the coefficient of time divided by 2π..

For example, if the equation of an alternating voltage is given by $e = 50 \sin 314t$ then its maximum value of 50 V and its frequency is $f = 314/2\pi = 50$ Hz.

Similarly, if the equation is of the form $e = I_m \sqrt{(R^2 + 4\omega^2 L^2)} \sin 2\omega t$, then its maximum value is $E_m = I_m \sqrt{(R^2 + 4\omega^2 L^2)}$ and the frequency is $2\omega/2\pi$ or ω/π Hz.

Example 11.2. *The maximum values of the alternating voltage and current are 400 V and 20 A respectively in a circuit connected to 50 Hz supply and these quantities are sinusoidal. The instantaneous values of the voltage and current are 283 V and 10 A respectively at t = 0 both increasing positively.*

(i) *Write down the expression for voltage and current at time t.*

(ii) *Determine the power consumed in the circuit.* **(Elect. Engg. Pune Univ.)**

Solution. *(i)* In general, the expression for an a.c. voltage is $v = V_m \sin(\omega t + \phi)$ where ϕ is the phase difference with respect to the point where $t = 0$.

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Now, v = 283 V ; V_m = 400 V. Substituting $t = 0$ in the above equation, we get

283 = 400 (sin ω × 0 + ϕ) ∴sin ϕ = 283/400 = 0.707 ; ∴ ϕ = 45° or π/4 radian.

Hence, general expression for voltage is

v = 400 (sin 2π × 50 × t + π/4)

= 400 sin (100 π t + π/4)

Similarly, at t = 0, 10 = 20 sin (ω × 0 + ϕ)

∴ sin ϕ = 0.5 ∴ ϕ = 30° or π/6 radian

Hence, the general expression for the current is

i = 20 (sin 100 π t + 30°) = 20 sin (100 π t + π/6)

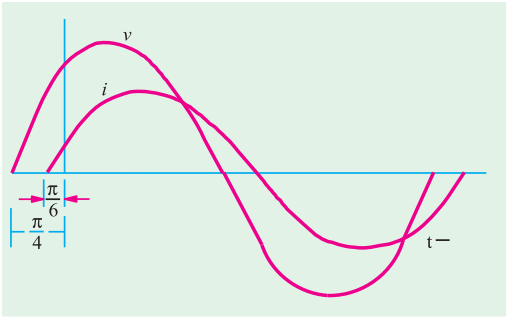


Fig. 11.10

(ii) $P = VI \cos\theta$ where V and I are rms values and θ is the phase difference between the voltage and current.

Now, $V = V_m/\sqrt{2} = 400/\sqrt{2}$; $I = 20/\sqrt{2}$; θ = 45° –30° = 15° (See Fig. 11.10)

∴ $P = (400/\sqrt{2}) \times (20/\sqrt{2}) \times \cos 15° =$ **3864 W**

Example 11.3. *An alternating current of frequency 60 Hz has a maximum value of 120 A. Write down the equation for its instantaneous value. Reckoning time from the instant the current is zero and is becoming positive, find* ***(a)*** *the instantaneous value after 1/360 second and* ***(b)*** *the time taken to reach 96 A for the first time.*

Solution. The instantaneous current equation is

$i = 120 \sin 2\pi ft = 120 \sin 120\pi t$

Now when t = 1/360 second, then

(a) $i = 120 \sin (120 \times \pi \times 1/360)$ ...angle in radians

$= 120 \sin (120 \times 180 \times 1/360)$ ...angle in degree

$= 120 \sin 60° = 103.9$ A

(b) $96 = 120 \times \sin 2 \times 180 \times 60 \times t$ ...angle in degree

or $\sin (360 \times 60 \times t) = 96/120 = 0.8$ ∴ $360 \times 60 \times t = \sin^{-1} 0.8 = 53°$ (approx)

∴ $t = \theta/2\pi f = 53/360 \times 60 =$ **0.00245 second.**

11.11. Phase

By phase of an alternating current is meant the fraction of the time period of that alternating current which has elapsed since the current last passed through the zero position of reference. For example, the phase of current at point A is $T/4$ second, where T is time period or expressed in terms of angle, it is π/2 radians (Fig. 11.11). Similarly, the phase of the rotating coil at the instant shown in Fig. 11.1 is ωt which is, therefore, called its phase angle.

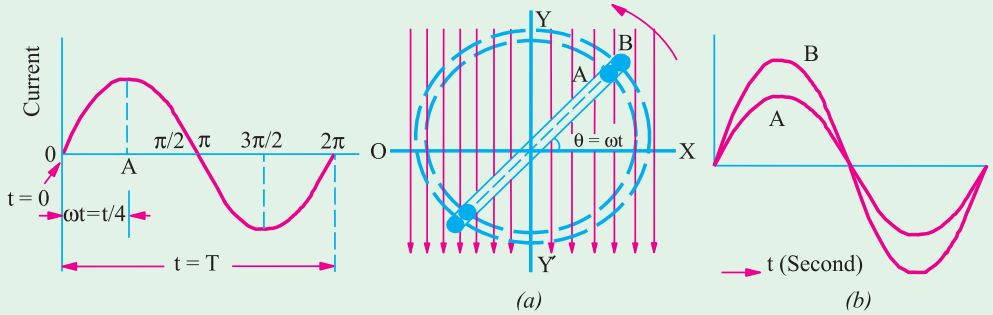


Fig. 11.11 Fig. 11.12

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In electrical engineering, we are, however, more concerned with relative phases or phase differences between different alternating quantities, rather than with their absolute phases. Consider two single-turn coils of different sizes [Fig. 11.12 (*a*)] arranged radially in the same plane and rotating with the same angular velocity in a common magnetic field of uniform intensity. The e.m.fs. induced in both coils will be of the same frequency and of sinusoidal shape, although the values of instantaneous e.m.fs. induced would be different. However, the two alternating e.m.fs. would reach their maximum and zero values at the same time as shown in Fig. 11.12 (*b*). Such alternating voltages (or currents) are said to be in phase with each other. The two voltages will have the equations

$$e_1 = E_{m1} \sin \omega t \quad \text{and} \quad e_2 = E_{m2} \sin \omega t$$

11.12. Phase Difference

Now, consider three similar single-turn coils displaced from each other by angles α and β and rotating in a uniform magnetic field with the same angular velocity [Fig. 11.13 (*a*)].

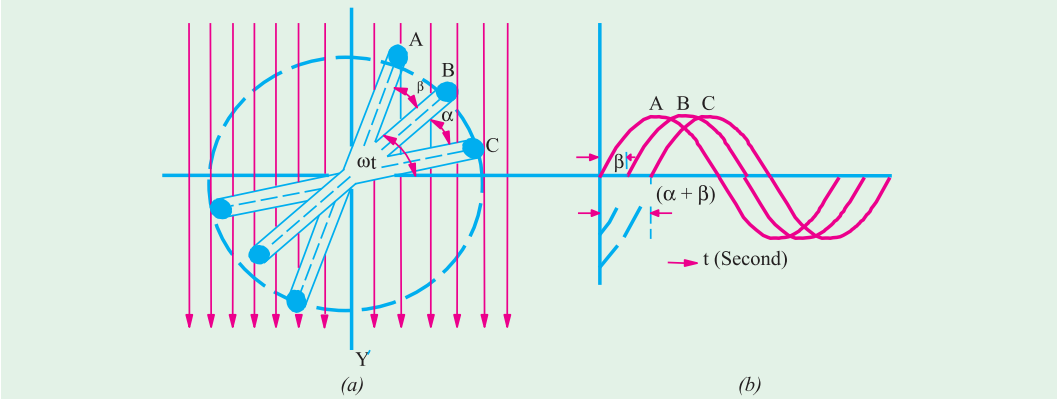


Fig. 11.13

In this case, the value of induced e.m.fs. in the three coils are the same, but there is one important difference. The e.m.fs. in these coils do not reach their maximum or zero values simultaneously but one after another. The three sinusoidal waves are shown in Fig. 11.13 (*b*). It is seen that curves *B* and *C* are displaced from curve *A* and angles β and (α + β) respectively. Hence, it means that phase difference between *A* and *B* is β and between *B* and *C* is α but between *A* and *C* is (α + β). The statement, however, does not give indication as to which e.m.f. reaches its maximum value first. This deficiency is supplied by using the terms ***'lag'*** or ***'lead'***.

A leading alternating quantity is one which reaches its maximum (or zero) value earlier as compared to the other quantity.

Similarly, a lagging alternating quantity is one which reaches its maximum or zero value later than the other quantity. For example, in Fig. 11.13 (*b*), *B* lags behind *A* by β and *C* lags behind *A* by (α + β) because they reach their maximum values later.

The three equations for the instantaneous induced e.m.fs. are (Fig. 11.14)

$e_A = E_m \sin \omega t$ ...reference quantity

$e_B = E_m \sin (\omega t - \beta)$

$e_C = E_m \sin [\omega t - (\alpha + \beta)]$

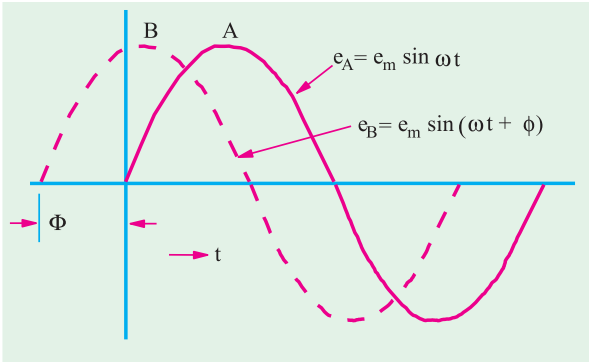


Fig. 11.14

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In Fig. 11.14, quantity B leads A by an angle ϕ. Hence, their equations are

$e_A = E_m \sin \omega t$ ...reference quantity

$e_B = E_m \sin (\omega t - \phi)$

A plus (+) sign when used in connection with phase difference denotes 'lead' whereas a minus (–) sign denotes 'lag'.

11.13. Root-Mean-Square (R.M.S.) Value

The r.m.s. value of an alternating current ***is given by that steady (d.c.) current which when flowing through a given circuit for a given time produces the same*** **heat** ***as produced by the alternating current when flowing through the same circuit for the same time.***

It is also known as the ***effective*** or ***virtual*** value of the alternating current, the former term being used more extensively. For computing the r.m.s. value of symmetrical sinusoidal alternating currents, either mid-ordinate method or analytical method may be used, although for symmetrical but non-sinusoidal waves, the mid-ordinate method would be found more convenient.

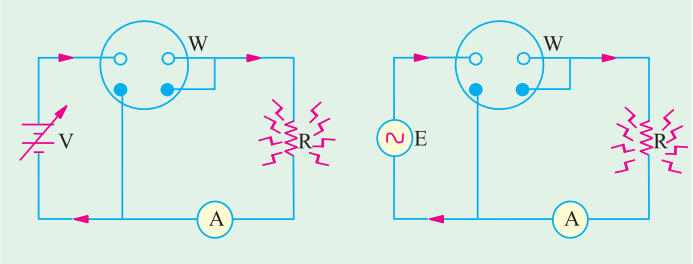


Fig. 11.15

A simple experimental arrangement for measuring the equivalent d.c. value of a sinusoidal current is shown in Fig. 11.15. The two circuits have identical resistances but one is connected to battery and the other to a sinusoidal generator. Wattmeters are used to measure heat power in each circuit. The voltage applied to each circuit is so adjusted that heat power production in each circuit is the same. In that case, the direct current will equal $I_m/\sqrt{2}$ which is called r.m.s. value of the sinusoidal current.

11.14. Mid-ordinate Method

In Fig. 11.16 are shown the positive half cycles for both symmetrical sinusoidal and non-sinusoidal alternating currents. Divide time base 't' into n equal intervals of time each of duration t/n seconds. Let the average values of instantaneous currents during these intervals be respectively i_1, i_2, i_3 .... i_n (*i.e.* mid-ordinates in Fig. 11.16). Suppose that this alternating current is passed through a circuit of resistance R ohms. Then,

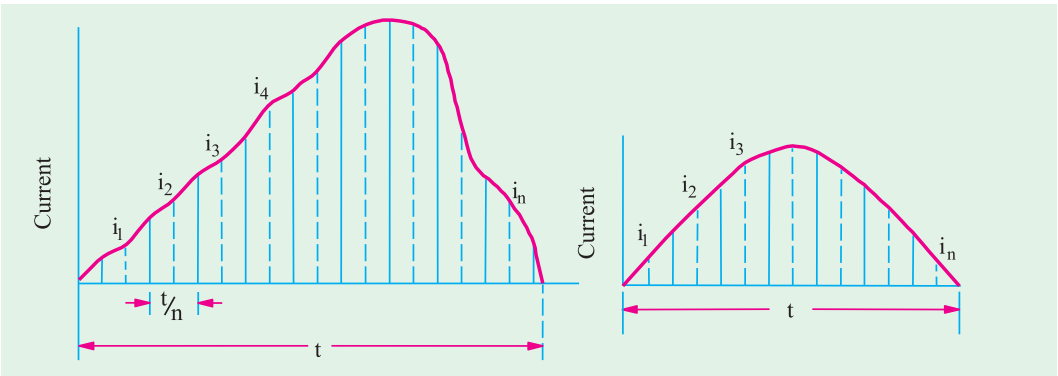


Fig. 11.16

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Heat produced in 1st interval $= 0.24 \times 10^{-3}\, i_1^2\, Rt/n$ kcal $\quad (\because 1/J = 1/4200 = 0.24 \times 10^{-3})$

Heat produced in 2nd interval $= 0.24 \times 10^{-3}\, i_2^2\, Rt/n$ kcal

: : : : :

: : : : :

: : : : :

Heat produced in nth interval $= 0.24 \times 10^{-3}\, i_n^2\, Rt/n$ kcal

Total heat produced in t seconds is $= 0.24 \times 10^{-3}\, Rt \left(\dfrac{i_1^2 + i_2^2 + \ldots + i_n^2}{n} \right)$ kcal

Now, suppose that a direct current of value I produces the same heat through the same resistance during the same time t. Heat produced by it is $= 0.24 \times 10^{-3}\, I^2 Rt$ kcal. By definition, the two amounts of heat produced should be equal.

$$\therefore \quad 0.24 \times 10^{-3} I^2 Rt = 0.24 \times 10^{-3}\, Rt \left(\frac{i_1^2 + i_2^2 + \ldots + i_n^2}{n} \right)$$

$$\therefore \quad I^2 = \frac{i_1^2 + i_2^2 + \ldots + i_n^2}{n} \quad \therefore \; I = \sqrt{\left(\frac{i_1^2 + i_2^2 + \ldots + i_n^2}{n} \right)}$$

= square root of the mean of the squares of the instantaneous currents

Similarly, the r.m.s. value of alternating voltage is given by the expression

$$V = \sqrt{\left(\frac{v_1^2 + v_2^2 + \ldots + v_n^2}{n} \right)}$$

11.15. Analytical Method

The standard form of a sinusoidal alternating current is $i = I_m \sin \omega t = I_m \sin \theta$.

The mean of the squares of the instantaneous values of current over one complete cycle is (even the value over half a cycle will do).

$$= \int_0^{2\pi} \frac{i^2\, d\theta}{(2\pi - 0)}$$

The square root of this value is $= \sqrt{\left(\int_0^{2\pi} \frac{i^2\, d\theta}{2\pi} \right)}$

Hence, the r.m.s. value of the alternating current is

$$I = \sqrt{\left(\int_0^{2\pi} \frac{i^2 d\theta}{2\pi} \right)} = \sqrt{\left(\frac{I_m^2}{2\pi} \int_0^{2\pi} \sin^2 \theta\, d\theta \right)} \qquad (\text{put } i = I_m \sin \theta)$$

Now, $\cos 2\theta = 1 - 2 \sin^2 \theta \quad \therefore \; \sin^2 \theta = \dfrac{1 - \cos 2\theta}{2}$

$$\therefore \quad I = \sqrt{\left(\frac{I_m^2}{4\pi} \int_0^{2\pi} (1 - \cos 2\theta)\, d\theta \right)} = \sqrt{\left(\frac{I_m^2}{4\pi} \left| \theta - \frac{\sin 2\theta}{2} \right|_0^{2\pi} \right)}$$

$$= \sqrt{\frac{I_m^2}{4}\; 2} \qquad \sqrt{\frac{I_m^2}{2}} \quad \therefore \; I = \frac{I_m}{\sqrt{2}} = 0.707\, I_m$$

Hence, we find that for a symmetrical sinusoidal current

r.m.s. value of current = 0.707 × max. value of current

The r.m.s. value of an alternating current is of considerable importance in practice, because the ammeters and voltmeters record the r.m.s. value of alternating current and voltage respectively. In

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electrical engineering work, ***unless indicated otherwise, the values of the given current and voltage are always the r.m.s. values.***

It should be noted that the average heating effect produced during one cycle is

$$= I^2R = (I_m/\sqrt{2})^2 R = \frac{1}{2} I_m^2 R$$

11.16. R.M.S. Value of a Complex Wave

In their case also, either the mid-ordinate method (when equation of the wave is not known) or analytical method (when equation of the wave is known) may be used. Suppose a current having the equation $i = 12 \sin \omega t + 6 \sin (3\omega t - \pi/6) + 4 \sin (5\omega t + \pi/3)$ flows through a resistor of R ohm. Then, in the time period T second of the wave, the effect due to each component is as follows :

Fundamental ..... $(12/\sqrt{2})^2 \; RT$ watt

3rd harmonic ..... $(6/\sqrt{2})^2 \; RT$ watt

5th harmonic ..... $(4/\sqrt{2})^2 \; RT$ watt

$\therefore$ Total heating effect $= RT\,[(12/\sqrt{2})^2 + (6/\sqrt{2})^2 + (4/\sqrt{2})^2]$

If I is the r.m.s. value of the complex wave, then equivalent heating effect is I^2RT

$$\therefore \quad I^2RT = RT\,[(12/\sqrt{2})^2 + (6/\sqrt{2})^2 + (4/\sqrt{2})^2]$$

$$\therefore \quad I = \sqrt{[(12/\sqrt{2})^2 + (6/\sqrt{2})^2 + (4/\sqrt{2})^2]} = 9.74 \text{ A}$$

Had there been a direct current of (say) 5 amperes flowing in the circuit also*, then the r.m.s. value would have been

$$= \sqrt{(12/\sqrt{2})^2 \quad (6/\sqrt{2})^2 \quad (4/\sqrt{2})^2 \quad 5^2} = 10.93 \text{ A}$$

Hence, for complex waves the rule is as follows : ***The r.m.s. value of a complex current wave is equal to the square root of the sum of the squares of the r.m.s. values of its individual components.***

11.17. Average Value

The average value I_a of an alternating current is expressed ***by that steady current which transfers across any circuit the same charge as is transferred by that alternating current during the same time.***

In the case of a symmetrical alternating current (*i.e.* one whose two half-cycles are exactly similar, whether sinusoidal or non-sinusoidal), the average value over a complete cycle is zero. Hence, in their case, the average value is obtained by adding or integrating the instantaneous values of current over one half-cycle only. ***But in the case of an unsymmetrical alternating current (like half-wave rectified current) the average value must always be taken over the whole cycle.***

(*i*) Mid-ordinate Method

With reference to Fig. 11.16, $I_{av} = \dfrac{i_1 + i_2 + \ldots + i_n}{n}$

This method may be used both for sinusoidal and non-sinusoidal waves, although it is specially convenient for the latter.

(*ii*) Analytical Method

The standard equation of an alternating current is, $i = I_m \sin \theta$

$$I_{av} = \int_0^{\pi} \frac{i d\theta}{(\pi - 0)} = \frac{I_m}{\pi} \int_0^{\pi} \sin \theta \, d\theta \qquad \text{(putting value of } i\text{)}$$

---

* The equation of the complex wave, in that case, would be,
$i = 5 + 12 \sin \omega t + 6 \sin (3\omega t - \pi/6) + 4 \sin (5\omega t + \pi/3)$

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$$= \frac{I_m}{\pi}\left|-\cos\theta\right|_0^{\pi} = \frac{I_m}{\pi}\left|+1-(-1)\right| = \frac{2I_m}{\pi} = \frac{I_m}{\pi/2} = \frac{\text{twice the maximum current}}{\pi}$$

$\therefore$ $I_{av} = I_m/\frac{1}{2}\pi = 0.637\, I_m$ $\therefore$ **average value of current = 0.637 × maximum value**

Note. R.M.S. value is always greater than average value except in the case of a rectangular wave when both are equal.

11.18. Form Factor

It is defined as the ratio, $K_f = \dfrac{\text{r.m.s. value}}{\text{average value}} = \dfrac{0.707\, I_m}{0.637\, I_m} = 1.1$. (for sinusoidal alternating currents only)

In the case of sinusoidal alternating voltage also, $K_f = \dfrac{0.707\, E_m}{0.637\, E_m} = 1.11$

As is clear, the knowledge of form factor will enable the r.m.s. value to be found from the arithmetic mean value and ***vice-versa.***

11.19. Crest or Peak or Amplitude Factor

It is defined as the ratio $K_a = \dfrac{\text{maximum value}}{\text{r.m.s. value}} = \dfrac{I_m}{I_m/\sqrt{2}} = \sqrt{2} = 1.414$ (for sinusoidal a.c. only)

For sinusoidal alternating voltage also, $K_a = \dfrac{E_m}{E_m/\sqrt{2}} = 1.414$

Knowledge of this factor is of importance in dielectric insulation testing, because the dielectric stress to which the insulation is subjected, is proportional to the maximum or peak value of the applied voltage. The knowledge is also necessary when measuring iron losses, because the iron loss depends on the value of maximum flux.

Example 11.4. *An alternating current varying sinusoidally with a frequency of 50 Hz has an RMS value of 20 A. Write down the equation for the instantaneous value and find this value* ***(a)*** *0.0025 second* ***(b)*** *0.0125 second after passing through a positive maximum value. At what time, measured from a positive maximum value, will the instantaneous current be 14.14 A ?*

(Elect. Science-I Allahabad Univ. 1992)

Solution. $I_m = 20\sqrt{2} = 28.2$ A, $\omega = 2\pi \times 50 = 100\,\pi$ rad/s.

The equation of the sinusoidal current wave with reference to point O (Fig. 11.17) as zero time point is $i = 28.2 \sin 100\,\pi t$ **ampere**

Since time values are given from point *A* where voltage has positive and maximum value, the equation may itself be referred to point *A*. In the case, the equation becomes :

$$i = 28.2 \cos 100\,\pi t$$

(i) When $t = 0.0025$ second

$i = 28.2 \cos 100\pi \times 0.0025$ ...angle in radian

$= 28.2 \cos 100 \times 180 \times 0.0025$ ...angle in degrees

$= 28.2 \cos 45° = 20$ A ...point *B*

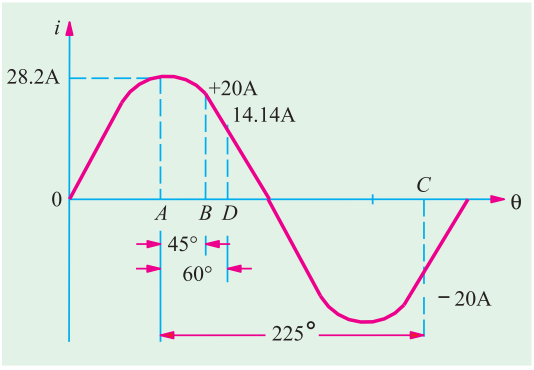


Fig. 11.17

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(ii) When t = 0.0125 second

$i = 28.2 \cos 100 \times 180 \times 0.0125$

$= 28.2 \cos 225^\circ = 28.2 \times (-1/\sqrt{2})$

$= -20$ A ...point C

(iii) Here i = 14.14 A

$\therefore \quad 14.14 = 28.2 \cos 100 \times 180\, t \quad \therefore \cos 100 \times 180\, t = \frac{1}{2}$

or $100 \times 180\, t = \cos^{-1}(0.5) = 60^\circ$, t = **1/300 second** ...point D

Example 11.5. *An alternating current of frequency 50 Hz has a maximum value of 100 A. Calculate* ***(a)*** *its value 1/600 second after the instant the current is zero and its value decreasing thereafter* ***(b)*** *how many seconds after the instant the current is zero (increasing thereafter wards) will the current attain the value of 86.6 A ?* **(Elect. Technology. Allahabad Univ. 1991)**

Solution. The equation of the alternating current (assumed sinusoidal) with respect to the origin O (Fig. 11.18) is

$$i = 100 \sin 2\pi \times 50t = 100 \sin 100\,\pi t$$

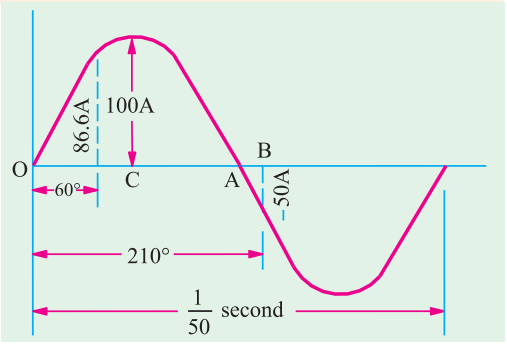


Fig. 11.18

(a) It should be noted that, in this case, time is being measured from point A (where current is zero and decreasing thereafter) and not from point O.

If the above equation is to be utilized, then, this time must be referred to point O. For this purpose, half time-period *i.e.* 1/100 second has to be added to 1/600 second. The given time as referred to point O becomes

$$= \frac{1}{100} + \frac{1}{600} = \frac{7}{600} \text{ second}$$

$\therefore \quad i = 100 \sin 100 \times 180 \times 7/600 = 100 \sin 210^\circ$

$= 100 \times -1/2 =$ **–50 A** ...point B

(b) In this case, the reference point is O.

$\therefore \quad 86.6 = 100 \sin 100 \times 180\, t$ or $\sin 18{,}000\, t = 0.866$

or $18{,}000\, t = \sin^{-1}(0.866) = 60^\circ \quad \therefore \quad t = 60/18{,}000 =$ **1/300 second**

Example 11.6. *Calculate the r.m.s. value, the form factor and peak factor of a periodic voltage having the following values for equal time intervals changing suddenly from one value to the next : 0, 5, 10, 20, 50, 60, 50, 20, 10, 5, 0, –5, –10 V etc. What would be the r.m.s value of sine wave having the same peak value ?*

Solution. The waveform of the alternating voltage is shown in Fig. 11.19. Obviously, it is not sinusoidal but it is symmetrical. Hence, though r.m.s value may be full one cycle, the average value has necessarily to be considered for half-cycle only, otherwise the symmetrical negative and positive half-cycles will cancel each other out.

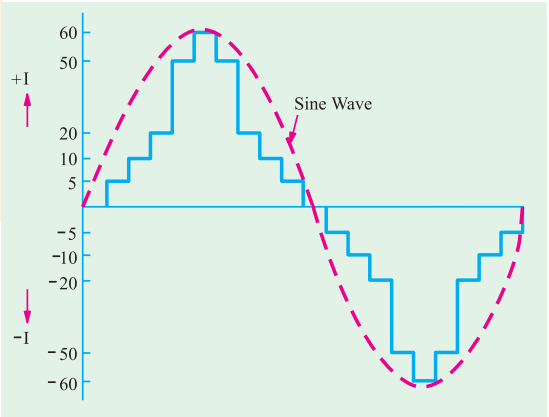


Fig. 11.19

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Mean value of $v^2 = \dfrac{0^2 + 5^2 + 10^2 + 20^2 + 50^2 + 60^2 + 50^2 + 20^2 + 10^2 + 5^2}{10} = \mathbf{965\ V}$

$\therefore$ r.m.s. value $= \sqrt{965} = \mathbf{31\ V\ (approx.)}$

Average value (half-cycle) $= \dfrac{0 + 5 + 10 + 20 + 50 + 60 + 50 + 20 + 10 + 5}{10} = \mathbf{23\ V}$

Form factor $= \dfrac{\text{r.m.s. value}}{\text{average value}} = \dfrac{31}{23} = \mathbf{1.35.}$ Peak factor $= 60/31 = \mathbf{2\ (approx.)}$

R.M.S. value of a sine wave of the same peak value $= 0.707 \times 60 = \mathbf{42.2\ V.}$

Alternative Solution

If 't' be the regular time interval, then area of the half-cycle is

$= (5t + 10t + 20t + 50t)\ 2 + 60t = 230t$, Base $= 10t$ $\therefore$ Mean value $= 230t/10t = 23$ V.

Area when ordinates are squared $= (25t + 100t + 400t + 2500t)\ 2 + 3600t = 9650t$, Base $= 10t$

$\therefore$ Mean height of the squared curve $= 9650t/10\ t = 965$

$\therefore$ r.m.s. value $= \sqrt{965} = 31$ V

Further solution is as before.

Example 11.7. *Calculate the reading which will be given by a hot-wire voltmeter if it is connected across the terminals of a generator whose voltage waveform is represented by*

$$v = 200 \sin \omega t + 100 \sin 3\omega t + 50 \sin 5\omega t$$

Solution. Since hot-wire voltmeter reads only r.m.s value, we will have to find the r.m.s. value of the given voltage. Considering one complete cycle,

R.M.S. value $V = \sqrt{\dfrac{1}{2\pi}\int_0^{2\pi} v^2 d\theta}$ where $\theta = \omega t$

or $V^2 = \dfrac{2}{2\pi}\int_0^{2\pi} (200 \sin \theta + 100 \sin 3\theta + 50 \sin 5\theta)^2\, d\theta$

$= \dfrac{1}{2\pi}\int_0^{2\pi} (200^2 \sin^2\theta + 100^2 \sin^2 3\theta + 50^2 \sin^2 5\theta$

$+ 2 \times 200.100 \sin\theta . \sin 3\theta + 2 \times 100.50 . \sin 3\theta . \sin 5\theta$

$+ 2 \times 50.200 . \sin 5\theta . \sin\theta)\, d\theta$

$= \dfrac{1}{2\pi}\left(\dfrac{200^2}{2} + \dfrac{100^2}{2} + \dfrac{50^2}{2}\right) 2\pi = 26{,}250$

$\therefore$ $V = \sqrt{26{,}250} = \mathbf{162\ V}$

Alternative Solution

The r.m.s. value of individual components are $(200/\sqrt{2})$, $(100/\sqrt{2})$ and $(50/\sqrt{2})$. Hence, as stated in Art. 11.16,

$$V = \sqrt{V_1^2 + V_2^2 + V_3^2} = \sqrt{(200/\sqrt{2})^2 + (100/\sqrt{2})^2 + (50/\sqrt{2})^2} = \mathbf{162\ V}$$

11.20. R.M.S. Value of H.W. Rectified Alternating Current

Half-wave (H.W.) rectified alternating current is one whose one half-cycle has been suppressed *i.e.* one which flows for half the time during one cycle. It is shown in Fig. 11.20 where suppressed half-cycle is shown dotted.

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As said earlier, for finding r.m.s. value of such an alternating current, summation would be carried over the period for which current ***actually*** flows *i.e.* from 0 to π, though it would be averaged for the whole cycle *i.e.* from 0 to 2π.

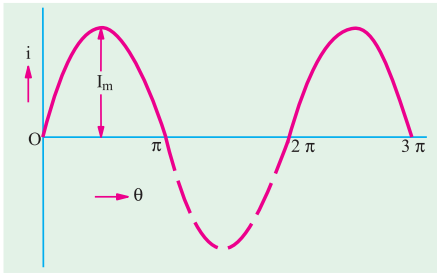


Fig. 11.20

∴ R.M.S. current

$$I = \sqrt{\left(\int_0^{\pi} \frac{i^2 d\theta}{2\pi}\right)} = \sqrt{\left(\frac{I_m^2}{2\pi}\int_0^{\pi} \sin^2\theta\, d\theta\right)}$$

$$= \sqrt{\frac{I_m^2}{4\pi}\int_0^{\pi}(1-\cos 2\theta)d\theta}$$

$$= \sqrt{\left(\frac{I_m^2}{4\pi}\left|\theta - \frac{\sin 2\theta}{2}\right|_0^{\pi}\right)} = \sqrt{\left(\frac{I_m^2}{4\pi}\times\pi\right)} = \sqrt{\left(\frac{I_m^2}{4}\right)} \quad \therefore\ I = \frac{I_m}{2} = \mathbf{0.5I_m}$$

11.21. Average Value of H.W. Rectified Alternating Current

For the same reasons as given in Art. 11.20, integration would be carried over from 0 –π

$$\therefore \quad I_{av} = \int_0^{\pi} \frac{i d\theta}{2\pi} = \frac{I_m}{2\pi}\int_0^{\pi} \sin\theta\, d\theta \qquad (\because\ i = I_m \sin\theta)$$

$$= \frac{I_m}{2\pi}\left|-\cos\theta\right|_0^{\pi} = \frac{I_m}{2\pi}\times 2 = \frac{I_m}{\pi}$$

11.22. Form Factor of H.W. Rectified Alternating Current

$$\text{Form factor} = \frac{\text{r.m.s. value}}{\text{average value}} = \frac{I_m/2}{I_m/\pi} = \frac{\pi}{2} = 1.57$$

Example 11.8. *An alternating voltage e = 200 sin 314t is applied to a device which offers an ohmic resistance of 20 Ω to the flow of current in one direction, while preventing the flow of current in opposite direction. Calculate RMS value, average value and form factor for the current over one cycle.* **(Elect. Engg. Nagpur Univ. 1992)**

Solution. Comparing the given voltage equation with the standard form of alternating voltage equation, we find that V_m = 200 V, R = 20 Ω, I_m = 200/20 = 10 A. For such a half-wave rectified current, RMS value = $I_m/2$ = 10/2 = **5A.**

Average current = I_m/π = 10/π = **3.18 A ;** Form factor = 5/3.18 = **1.57**

Example 11.9. *Compute the average and effective values of the square voltage wave shown in Fig. 11.21.*

Solution. As seen, for $0 < t < 0.1$ *i.e.* for the time interval 0 to 0.1 second, v = 20 V. Similarly, for $0.1 < t < 0.3$, v = 0. Also time-period of the voltage wave is 0.3 second.

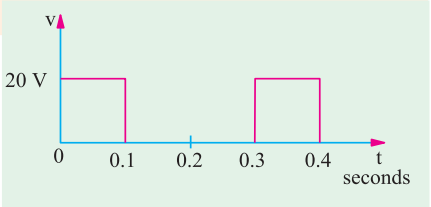


Fig. 11.21

$$\therefore \quad V_{av} = \frac{1}{T}\int_0^T v\ dt = \frac{1}{0.3}\int_0^{0.1} 20\ dt$$

$$= \frac{1}{0.3}(20\times 0.1) = \mathbf{6.67\ V}$$

$$V^2 = \frac{1}{T}\int_0^T v^2 dt = \frac{1}{0.3}\int_0^{0.1} 20^2\ dt = \frac{1}{0.3}(400\times 0.1) = 133.3;\ V = \mathbf{11.5\ V}$$

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Example 11.10. *Calculate the RMS value of the function shown in Fig. 11.22 if it is given that for $0 < t < 0.1$, $y = 10(1 - e^{-100t})$ and $0.1 < t < 0.2$, $y = 10\, e^{-50(t-0.1)}$*

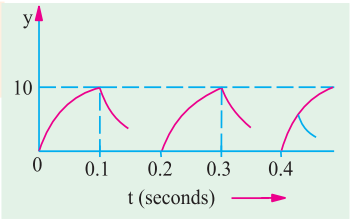


Fig. 11.22

Solution.

$$Y^2 = \frac{1}{0.2}\left\{\int_0^{0.1} y^2 dt + \int_{0.1}^{0.2} y^2 dt\right\}$$

$$= \frac{1}{0.2}\left\{\int_0^{0.1} 10^2\,(1 - e^{-100t})^2 dt + \int_{0.1}^{0.2} (10e^{-50(t-0.1)})^2 dt\right\}$$

$$= \frac{1}{0.2}\left\{\int_0^{0.1} 100\,(1 + e^{-200t} - 2e^{-100t}) dt + \int_{0.1}^{0.2} 100\, e^{-100(t-0.1)} dt\right\}$$

$$= 500\left\{\left[t - 0.005e^{-200t} + 0.02\, e^{-100t}\right]_0^{0.1} + \left[-0.0\, e^{-100(t-0.1)}\right]_{0.1}^{0.2}\right\}$$

$$= 500\left\{\left[\left(0.1 - 0.005e^{-20} + 0.02e^{-10}\right) - \left(0 - 0.005 + 0.02\right)\right] + \left[\left(-0.01e^{-10}\right) - \left(-0.01\right)\right]\right\}$$

$$= 500 \times 0.095 = 47.5 \quad \therefore \quad Y = \sqrt{47.5} = \mathbf{6.9}$$

Example 11.11. *The half cycle of an alternating signal is as follows : It increases uniformly from zero at 0° to F_m at $\alpha°$, remains constant from $\alpha°$ to $(180 - \alpha)°$, decreases uniformaly from F_m at $(180 - \alpha)°$ to zero at 180°. Calculate the average and effective values of the signal.*

(Elect. Science-I, Allahabad Univ. 1992)

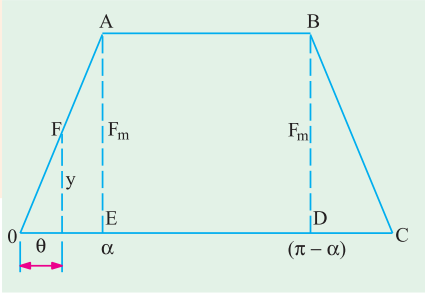


Fig. 11.23

Solution. For finding the average value, we would find the total area of the trapezium and divide it by π (Fig. 11.23).

Area $= 2 \times \Delta OAE +$ rectangle $ABDE = 2 \times (1/2) \times F_m\alpha + (\pi - 2\alpha)\, F_m = (\pi - \alpha)\, F_m$

$\therefore$ average value $= (\pi - \alpha)\, F_m/\pi$

RMS Value From similar triangles, we get $\frac{y}{\theta} = \frac{F_m}{\alpha}$ or $y^2 \quad \frac{Fm^2}{2}$ 2

This gives the equation of the signal over the two triangles *OAE* and *DBC*. The signal remains constant over the angle α to $(\pi - \alpha)$ *i.e.* over an angular distance of $(\pi - \alpha) - \alpha = (\pi - 2\alpha)$

Sum of the squares $= \frac{2F_m^2}{\alpha^2}\int_\theta^\alpha \theta^2\, d\theta + F_m^2(\pi - 2\alpha) = F_m^2(\pi - 4\alpha/3)$.

The mean value of the squares is $= \frac{1}{\pi}F_m^2\left(\pi - \frac{4\alpha}{3}\right) = F_m^2\left(1 - \frac{4\alpha}{3\pi}\right)$

$$\text{r.m.s. value} = F_m\sqrt{\left(1 - \frac{4\,\alpha}{3\pi}\right)}$$

Example 11.12. *Find the average and r.m.s values of the a.c. voltage whose waveform is given in Fig. 11.24 (a)*

Solution. It is seen [(Fig. 11.24 (a)] that the time period of the waveform is 5s. For finding the average value of the waveform, we will calculate the net area of the waveform over one period and then find its average value for one cycle.

$$A_1 = 20 \times 1 = 20\ V\text{–}s,\ A_2 = -5 \times 2 = -10\ \text{V –s}$$

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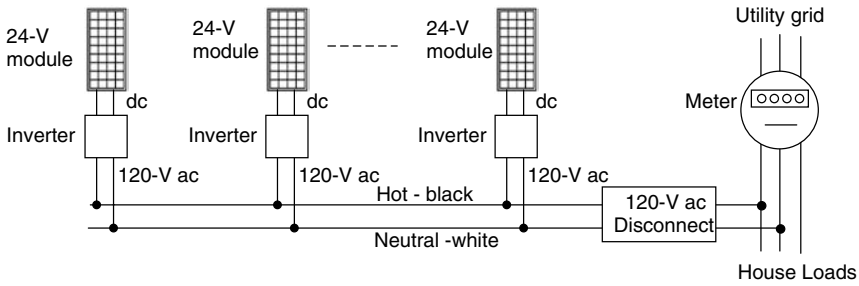


Figure 9.21 AC modules each have their own inverters mounted on the backside of the collector, allowing simple system expansion at any time.

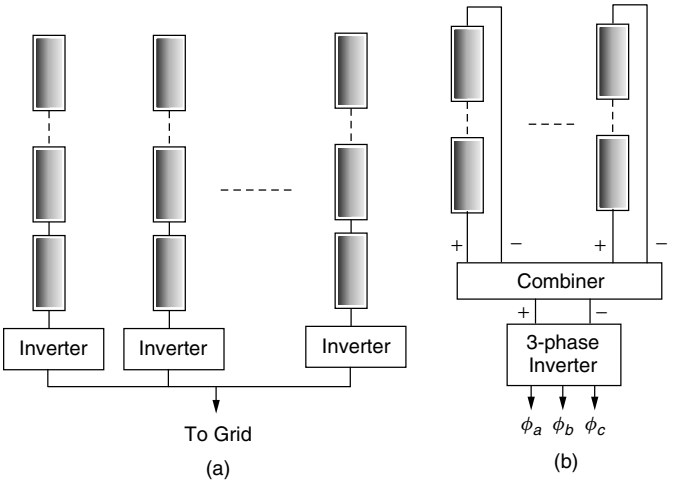


Figure 9.22 Large grid-connected systems may use an individual inverter for each string (a) or may incorporate a large, central inverter system to provide three-phase power (b).

is also minimized making the installation potentially cheaper than a large, central inverter. Large, central inverter systems providing three-phase power to the grid are also an option (Fig. 9.22b).

9.3.1 Interfacing with the Utility

The ac output of a grid-connected PV system is fed into the main electrical distribution panel of the house, from which it can provide power to the house or put power back onto the grid as shown in Fig. 9.23. In most cases, whenever the PV system delivers more power than the home needs at that moment, the electric meter runs backwards, building up a credit with the utility. At other times, when demand exceeds that supplied by the PVs, the grid provides supplementary power. This arrangement, in which a single electric meter runs in both directions,

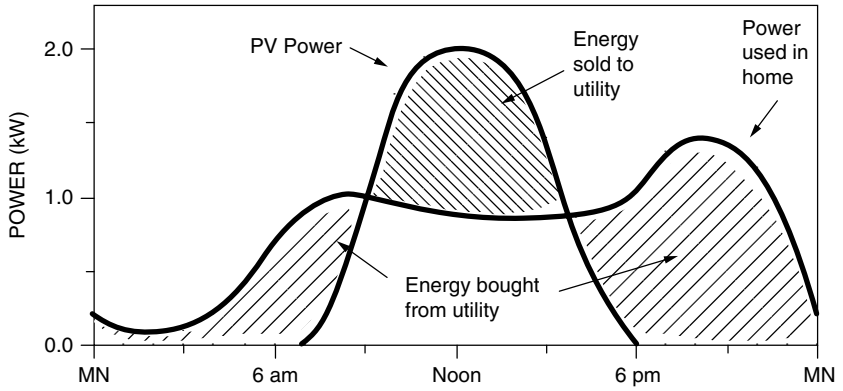


Figure 9.23 During the day, excess power from the array is sold to the utility; at night, the deficit is purchased from the utility.

is called *net metering* since the customer's monthly electric bill is only for that net amount of energy that the PV system is unable to supply. For an example of one version of net metering, refer back to Example 5.2 in Chapter 5. It is also possible to use two ratcheted meters, one to measure power you sell to the grid and the other to measure power you buy back from the grid. The two-meter arrangement is not only more cumbersome, it also can mean that the customer may have to sell electricity at a wholesale price and buy it back at the utility's more expensive retail price. From a PV owner's perspective, the net metering approach is usually preferred.

The power conditioning unit absolutely must be designed to quickly and automatically drop the PV system from the grid in the event of a utility power outage. When there is an outage, breakers automatically isolate a section of the utility lines in which the fault has occurred, creating what is referred to as an "island." A number of very serious problems may occur if, during such an outage, a self-generator, such as a grid-connected PV system, supplies power to that island.

Most faults are transient in nature, such as a tree branch brushing against the lines, and so utilities have automatic procedures that are designed to limit the amount of time the outage lasts. When there is a fault, breakers trip to isolate the affected lines, and then they are automatically reclosed a few tenths of a second later. It is hoped that in the interim the fault clears and customers are without power for just a brief moment. If that doesn't work, the procedure is repeated with somewhat longer intervals until finally, if the fault doesn't clear, workers are dispatched to the site to take care of the problem. If a self-generator is still on the line during such an incident, even for less than one second, it may interfere with the automatic reclosing procedure, leading to a longer-than-necessary outage. And if a worker attempts to fix a line that has supposedly been disconnected from all energy sources, but it is not, then a serious hazard has been created.

When a grid-connected system must provide power to its owners during a power outage, a small battery back-up system may be included. If the users really need uninterruptible power for longer periods of time, the battery system can be augmented with a generator such as has been suggested in Fig. 9.2.

9.3.2 DC and AC Rated Power

Grid-connected systems consist of an array of modules and a power conditioning unit that includes an inverter to convert dc from the PVs into ac required by the grid. A good starting point to estimate system performance is the rated dc power output of an individual module under standard test conditions (STC)—that is, 1-sun, AM 1.5 and 25°C cell temperature. Then we can try to estimate the actual ac power output under varying conditions.

When a PV system is put into the field, the actual ac power delivered at 1-sun, call it P_{ac}, can be represented as the following product:

$$P_{ac} = P_{dc,STC} \times \text{(Conversion Efficiency)} \tag{9.10}$$

where $P_{dc(STC)}$ is the dc power of the array obtained by simply adding the individual module ratings under standard test conditions. The conversion efficiency accounts for inverter efficiency, dirty collectors, mismatched modules, and differences in ambient conditions. Even in full sun, the impact of these losses can easily derate the power output by 20–40%.

Consider first the impact of slight variations in $I-V$ curves for modules in an array. Figure 9.24 shows a simple example consisting of two mismatched

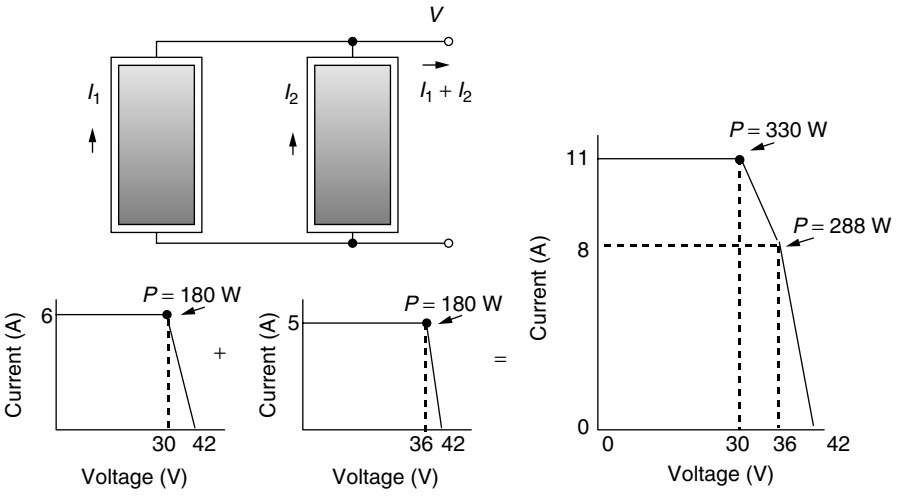


Figure 9.24 Illustrating the loss due to mismatched modules. Each module is rated at 180 W, but the parallel combination yields only 330 W at the maximum power point.

180-W modules wired in parallel. Their somewhat idealized $I-V$ curves have been drawn so that one produces 180 W at 30 V and the other does so at 36 V. As shown, the sum of their $I-V$ curves shows that the maximum power of the combined modules is only 330 W instead of the 360 W that would be expected if their $I-V$ curves were identical. In addition, not all modules coming off the very same production line will have exactly the same rated output. Some 100-W modules may really be 103 W and others 97 W, for example. In other words, production tolerances can reduce array output as well. These two module mismatch factors can easily drop the array output by several percent.

An even more important factor that reduces module power below the rated value is cell temperature. In the field, the cells are likely to be much hotter than the 25°C at which they are rated and we know that as temperature increases, power decreases. To help account for the change in module power caused by elevated cell temperatures, another rating system has been evolving that is based on field tests performed as part of an extensive monitoring program called PVUSA. The PVUSA test conditions (PTC) are defined as 1-sun irradiance in the plane of the array, 20°C ambient temperature, and a wind-speed of 1 m/s. The ac output of an array under PTC conditions $P_{ac(PTC)}$ is a much better indicator of the actual power delivered to the building in full sun than is the more commonly used $P_{dc(STC)}$. For its PV rebate programs, California, for example, has chosen to use the PTC rating of collectors.

Finally there is the efficiency of the inverter itself, which varies depending on the load, as is suggested in Fig. 9.25. Good grid-connect inverters have efficiencies above 90% when operating at all but very low loads.

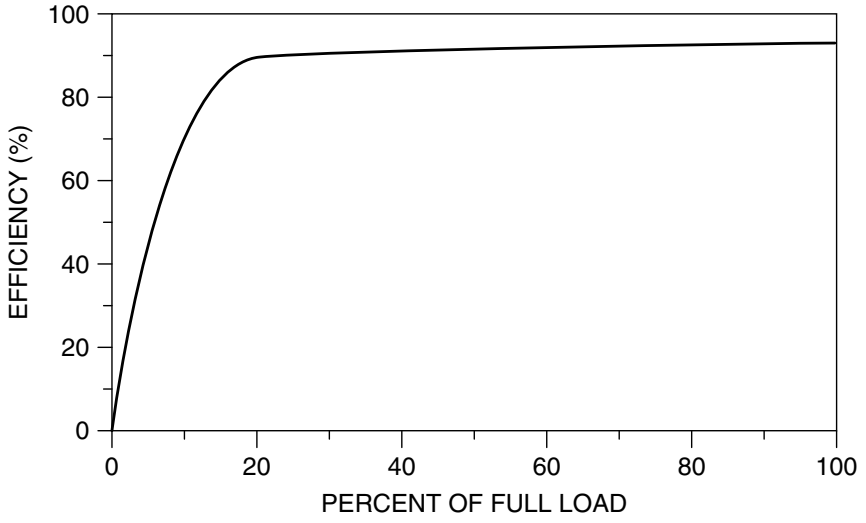


Figure 9.25 The efficiency of an inverter depends on the fraction of its rated power at which it operates.

Example 9.3 Derating a PV Array to a PTC, AC Rating. Consider a PV array rated at 1 kW under standard test conditions. Module nominal operating cell temperature (NOCT) is 47°C (see Section 8.6). DC power output at the MPP drops by 0.5%/°C above the STC temperature of 25°C.

Estimate its ac output under PTC conditions if there is a 3% array loss due to mismatched modules, dirt loss is 4%, and the inverter has an efficiency of 90%.

Solution. Inserting PTC conditions (ambient 20°C, insolation $S = 1$ kW(m^2) into (8.24) gives us an estimated cell temperature of

$$T_{\text{cell}} = T_{\text{amb}} + \left(\frac{\text{NOCT} - 20}{0.8}\right) \cdot S$$

$$= 20 + \left(\frac{47 - 20}{0.8}\right) \cdot 1 = 53.8^{\circ}\text{C}$$

With power loss at 0.5% per degree above 25°C, the dc rated power of the array would be

$$P_{dc(PTC)} = 1 \text{ kW}[1 - 0.005(53.8 - 25)] = 0.856 \text{ kW}$$

Including mismatch, dirt, and inverter efficiencies will result in an estimated ac rated power at PTC of

$$P_{ac(PTC)} = 0.856 \text{ kW} \times 0.97 \times 0.96 \times 0.90 = 0.72 \text{ kW}$$

That's a pretty sizably reduced indicator of rated power, but it is much more realistic in terms of actual field experience than the 1-kW dc STC rating that would often be quoted.

PV systems have been traditionally described in terms of their dc output under standard test conditions. In the above example, the system would likely be sold as a "1-kW system." But in that example the array will deliver only 72% of that as ac power to the load under the much more realistic PTC conditions. It should probably be called a 0.72-kW(ac) system. And in fact studies of real systems in the field under PTC conditions suggest that the ratio of $P_{ac(PTC)}$ to $P_{dc(STC)}$ may be lower still. Scheuermann et al. (2002) measured 19 PV systems in California and found the ratio to be between 53% and 70%.

Whether the ac PTC rating system will be adopted as the standard remains to be seen. It is much more realistic than the usual STC rating, but it has the disadvantage of being less easy to define since it depends on the specific inverter that has been chosen. In the meantime, it should be clear that describing a system based on its dc performance under STC without also including corrections for temperature and inverter is misleading. Even the PTC system is crude since it too is based on arbitrary ambient conditions. In cold climates, for example, where

PVs perform better, the PTC would underestimate energy delivered, and in hot climates the opposite would occur.

9.3.3 The "Peak-Hours" Approach to Estimating PV Performance

Predicting performance is a matter of combining the characteristics of the major components—the PV array and the inverter—with local insolation and temperature data. After having adjusted dc power under STC to expected ac from the inverter, the second key factor is the amount of sun available at the site. Chapter 7 was devoted to developing equations for clear sky insolation, and tables of values are given in Appendices C and D. In Appendix E there are tables of estimated average insolations for a number of locations in the United States and Appendix F presents seasonal averages around the globe.

When the units for daily, monthly, or annual average insolation are specifically kWh/m^2-day, then there is a very convenient way to interpret that number. Since 1-sun of insolation is defined as 1 kW/m^2, we can think of an insolation of say 5.6 kWh/m^2-day as being the same as 5.6 h/day of 1-sun, or 5.6 h of "peak sun." So, if we know the ac power delivered by an array under 1-sun insolation (P_{ac}), we can just multiply that rated power by the number of hours of peak sun to get daily kWh delivered.

To see whether this simple approach is reasonable, consider the following analysis. We can write the energy delivered in a day's time as

$$\text{Energy (kWh/day)} = \text{Insolation}\left(\frac{\text{kWh/m}^2}{\text{day}}\right) \cdot A\ (\text{m}^2) \cdot \overline{\eta} \tag{9.11}$$

where A is the area of the PV array and $\overline{\eta}$ is the average system efficiency over the day.

When exposed to 1-sun of insolation, we can write for ac power from the system

$$P_{ac}(\text{kW}) = \left(\frac{1\ \text{kW}}{\text{m}^2}\right) \cdot A\ (\text{m}^2) \cdot \eta_{1-\text{sun}} \tag{9.12}$$

where η_{1-sun} is the system efficiency at 1-sun. Combining (9.11) and (9.12) gives

$$\text{Energy (kWh/day)} = P_{ac}(\text{kW}) \cdot \left[\frac{\text{Insolation (kWh/m}^2\text{/day)}}{1\ \text{kW/m}^2}\right] \cdot \left(\frac{\overline{\eta}}{\eta_{1-\text{sun}}}\right) \tag{9.13}$$

If we assume that the average efficiency of the system over a day's time is the same as the efficiency when it is exposed to 1-sun, then the energy collected is what we hoped it would be

$$\text{Energy (kWh/day)} = P_{ac}(\text{kW}) \cdot (\text{h/day of "peak sun"}) \tag{9.14}$$

The key assumption in (9.14) is that system efficiency remains pretty much constant throughout the day. The main justification is that these grid-connected systems have maximum power point trackers that keep the operating point near the knee of the $I-V$ curve all day long. Since power at the maximum point is nearly directly proportional to insolation, system efficiency should be reasonably constant. Cell temperature also plays a role, but it is less important. Efficiency might be a bit higher than average in the morning, when it is cooler and there is less insolation, but all that will do is make (9.14) slightly conservative.

The value of something like the ac PTC rating system is now quite apparent. Using it to indicate system ac power at 1-sun, coupled with the interpretation of kWh/m^2-day as hours of peak sun, gives us an easy way to estimate energy production.

Example 9.4 Annual Energy Using the Peak-Sun Approach. Estimate the annual energy delivered by the 1-kW (dc, STC) array described in Example 9.3 if it located in Madison, WI, is south-facing, and has a tilt angle equal to its latitude minus 15°. Use the PTC ac rating.

Solution. Appendix E shows the annual insolation in Madison at L-15 is 4.5 kWh/m^2-day. Using the de-rated ac output of 0.72 kW (ac, PTC) that was found in Example 9.3, along with 4.5 h/day of peak sun, gives

$$\text{Energy} = 0.72 \text{ kW} \times 4.5 \text{ h/day} \times 365 \text{ day/yr} = 1183 \text{ kWh/yr}$$

The PTC rating assumes a nominal ambient of 20°C, which is a pretty good average estimate for many locations in the United States. We could expect it to overpredict performance when it is hotter than that, and underpredict when it is cooler. To test this, let us rework Example 9.4 on a monthly basis using Madison ambient temperature instead of the assumed 20°C.

Example 9.5 Correcting Predicted Performance for Temperature Effects. Estimate the energy that the 1-kW (dc, STC) array described in Example 9.3 would deliver in Madison in January. Assume south-facing with tilt = L-15 and use the average daily maximum temperature instead of the 20°C assumed by PTC. The nominal operating cell temperature (NOCT) was given as 47°C for this array.

Solution. In Appendix E, the average daily maximum temperature for Madison in January is given as −4.0°C. When it is that cold, (8.24) estimates cell temperature at 1-sun to be

$$T_{\text{cell}} = T_{\text{amb}} + \left(\frac{\text{NOCT} - 20}{0.8}\right) \cdot S$$

$$= -4.0 + \left(\frac{47 - 20}{0.8}\right) \cdot 1 = 29.8^{\circ}\text{C}$$

With power loss at 0.5% per degree above 25°C, the dc rated power of the array without dirt and mismatched modules would be

$$P_{dc} = 1 \text{ kW}[1 - 0.005(29.8 - 25)] = 0.98 \text{ kW}$$

For comparison, in Example 9.3 the cell temperature at PTC was a much warmer 53.8°C and the dc power was

$$P_{dc(PTC)} = 1 \text{ kW}[1 - 0.005(53.8 - 25)] = 0.856 \text{ kW}$$

In other words, the array will perform considerably better than PTC would have predicted because it is so cold in Madison.

Including mismatch, dirt and inverter efficiencies given in Example 9.3, yields an estimated ac rated power at of

$$P_{ac} = 0.98 \text{ kW} \times 0.97 \times 0.96 \times 0.90 = 0.82 \text{ kW}$$

Appendix E gives January insolation at L-15 in Madison as 3.0 kWh/m^2 or 3.0 h/day of 1-sun. So we estimate this 1 kW array will deliver

$$\text{Energy} = 0.82 \text{ kW} \times 3.0 \text{ h/day} \times 31 \text{ day/mo} = 76 \text{ kWh/mo}$$

It is easy to systematize calculations of the sort demonstrated in Example 9.5 so that actual site temperature data can be used instead of the 20°C assumed in PTC. A spreadsheet for the 1-kW (dc) Madison system using the same dirt, mismatch, and inverter assumptions is demonstrated in Table 9.2. Notice that the PTC estimate for annual energy found in Example 9.4 was 1183 kWh/yr, and this more carefully done spreadsheet gives 1202 kWh. The difference is negligible. A similar comparison for Phoenix, which is a much hotter place than Madison, results in a 7% overestimate using PTC. These differences are so small that it

TABLE 9.2 Estimated Energy Delivered by a 1-kW (dc, STC) PV Array in Madison, WI, Using Average Maximum Monthly Temperatures to Compute Performance Degradation[a]

Madison, WI, South L-15	
dc Power	1 kW at STC
Temp. coef.	0.5%/°C
Mismatch	0.03
Dirt	0.04
Inverter	0.90
NOCT	47°C

Month	Insolation (kWh/m^2-day)	Avg Max Temp. (°C)	Cell Temp. (°C)	Array dc Power (kW)	Array ac Power (kW)	Energy (kWh/mo)
Jan	3.0	−4.0	29.8	0.98	0.82	76
Feb	3.9	−1.1	32.7	0.96	0.81	88
Mar	4.5	5.3	39.1	0.93	0.78	109
Apr	5.1	13.7	47.5	0.89	0.74	114
May	5.8	20.5	54.3	0.85	0.72	129
Jun	6.2	25.7	59.5	0.83	0.69	129
July	6.2	28.0	61.8	0.82	0.68	131
Aug	5.7	26.4	60.2	0.82	0.69	122
Sept	4.8	21.9	55.7	0.85	0.71	102
Oct	3.8	15.5	49.3	0.88	0.74	87
Nov	2.5	6.7	40.5	0.92	0.77	58
Dec	2.3	−1.2	32.6	0.96	0.81	57
Avg:	4.5	13.2			kWh/yr = 1202	

[a]Inverter, mismatch, and dirt losses from Example 9.3 are included.

would appear that using the simpler PTC approach is quite reasonable under a wide range of temperature conditions.

One of the most useful ways to describe energy production from a PV array is in terms of kWh/yr delivered per kW of STC rated power. Table 9.3 presents such data for a number of U.S. cities. The table has been derived following the methods presented in Example 9.5. Temperature and insolation data are from Appendix E. A 90% efficient inverter, along with 3% module mismatch and 4% dirt depreciation were assumed. Annual energy delivered for south-facing modules tilted at the latitude -15° ranges from about 1000 kWh per kW (dc, STC) in Seattle, to over 1600 kWh/kW in Albuquerque. When a single-axis E–W tracker with polar tilt angle is used, annual production increases by 24–36%.

Table 9.3 suggests a very crude rule-of-thumb for annual performance of a grid-connected system in a pretty good location:

> Each 1 kW of dc, STC-rated PV will deliver an average of roughly 1400 kWh per year.

TABLE 9.3 Annual Energy Production in Various Cities per kW (dc, STC) of Installed PV Capacity[a]

	South Facing, L-15 Fixed			1-Axis, Polar Mount		
Location	Average High Temp. (°C)	Insolation (kWh/ m^2-d)	Annual kWh/kW	Insolation (kWh/ m^2-d)	Annual kWh/kW	Ratio 1-axis/ Fixed
Seattle, WA	15.3	3.8	1006	4.7	1247	1.24
New York, NY	16.8	4.5	1195	5.6	1479	1.24
Madison, WI	13.2	4.5	1202	5.7	1519	1.26
Boston, MA	15.0	4.5	1209	5.7	1529	1.26
Atlanta, GA	21.8	5.0	1294	6.4	1639	1.27
Honolulu, HI	29.1	5.5	1373	7.4	1834	1.34
Boulder, CO	17.9	5.3	1404	7.2	1885	1.34
Los Angeles, CA	21.3	5.5	1420	7.0	1808	1.27
El Paso, TX	25.3	6.3	1583	8.6	2159	1.36
Albuquerque, NM	21.2	6.3	1618	8.5	2199	1.36

[a] Assumed inverter efficiency 90%, mismatching loss 3%, dirt loss 4%

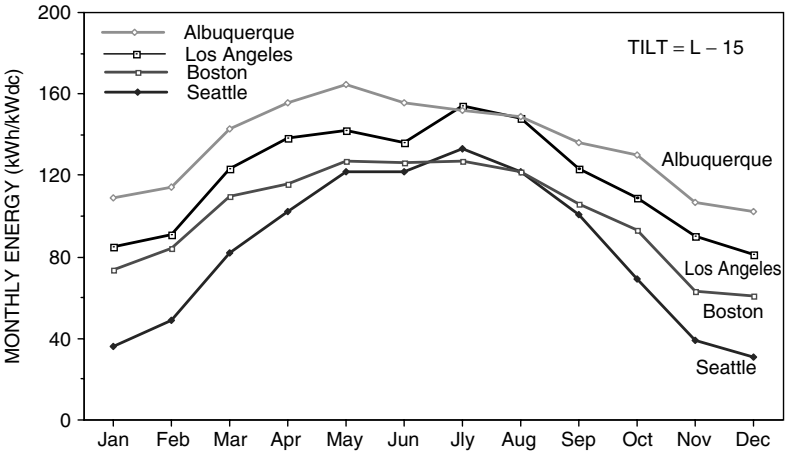


Figure 9.26 Monthly energy production for four cities in kWh per kW (dc, STC) for fixed south-facing, L-15 tilt. Assumed inverter efficiency 90%, mismatch loss 3%, dirt loss 4%. Includes local temperature impacts.

Figure 9.26 shows monthly values of kWh/kW for four U.S. cities that pretty much represent the range of likely performance across the coterminous states. A couple of features are worth pointing out. In Albuquerque, the decline in output after it peaks in May is due in large part to the high summer temperatures. At the other extreme, Seattle, with latitude about as high as it gets in the coterminous

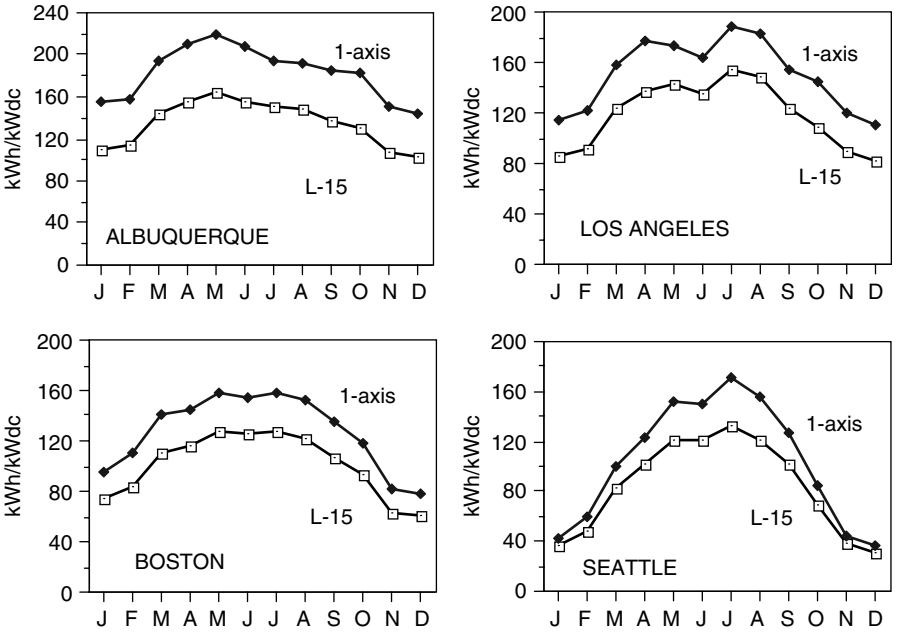


Figure 9.27 Comparing energy delivered from fixed L-15 tilt with single-axis polar tracking.

states, does quite well in the summer due to long, clear, relatively cool days. Figure 9.27 shows the improvement in monthly performance for those four cities when a single-axis polar mount tracker is incorporated into the system.

9.3.4 Capacity Factors for PV Grid-Connected Systems

A simple way to present the energy delivered by any electric power generation system is in terms of its rated ac power and its capacity factor (CF). If the system delivered full, rated power continuously, the CF would be unity. A CF of 0.4, for example, could mean that the system delivers full-rated power 40% of the time and no power at all the rest of the time, but that is not the only interpretation. It could also deliver 40% of rated power all of the time and still have CF = 0.4, or any of a number of other combinations.

The governing equation for annual performance in terms of CF is simply

$$\text{Energy (kWh/yr)} = P_{ac}(\text{kW}) \cdot \text{CF} \cdot 8760(\text{h/yr}) \tag{9.15}$$

where 8760 is the product of 24 hours per day times 365 days per year. Monthly or daily capacity factors are similarly defined.

Combining (9.14) and (9.15) leads to the simple interpretation of capacity factor for grid-connected PV systems:

$$\text{Capacity factor (CF)} = \frac{\text{(h/day of "peak sun")}}{24 \text{ h/day}} \tag{9.16}$$

Notice that the complication associated with temperature is included in the definition of P_{ac} so it doesn't affect CF. Capacity factors for a number of U.S. cities are shown in Fig. 9.28. They range from 0.16 to 0.26 for fixed, south-facing panels at tilt L-15 and range from 0.20 to 0.36 for single-axis polar-mount trackers.

9.3.5 Grid-Connected System Sizing

With the utility there to provide energy storage and back-up power, sizing grid-connected systems is not nearly as critical as it is for stand-alone systems. Moreover, there are few economies of scale with these systems—it costs roughly twice as much to install a system that will deliver twice as much energy. Sizing grid-connected systems therefore is more a matter of how much area is conveniently available on the building, and the budget of the buyer, than it is trying to match supply to demand. It is, nonetheless, very important to be able to predict as accurately as possible the annual energy delivered by the system in order to decide whether it makes economic sense. Certain components will dictate some of the details, but what has already been developed on rated ac power and "peak hours" of insolation provides a good start to system design.

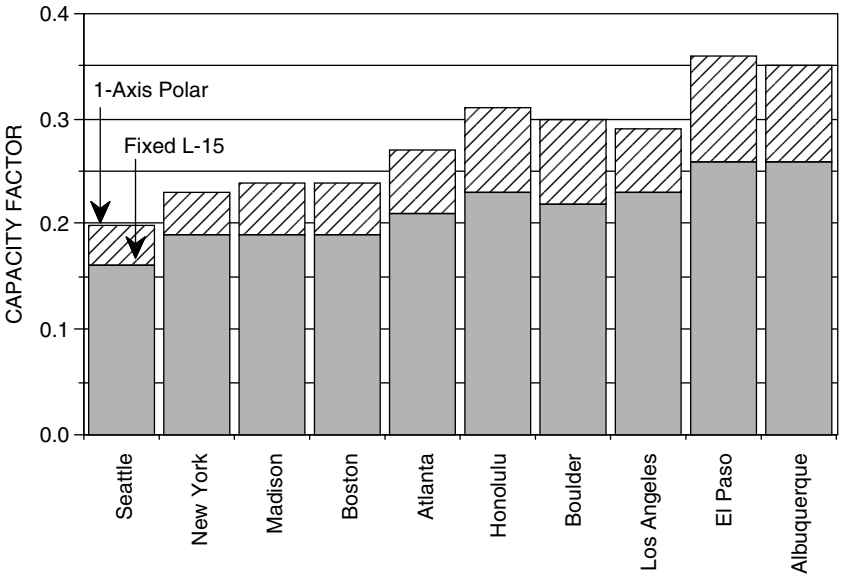


Figure 9.28 AC photovoltaic capacity factors for a number of U.S. cities.

As an academic exercise, system sizing is straightforward. How many kWh/yr are required? How many peak watts of dc PV power are needed to provide that amount? How much area will that system require? The realities of design, however, revolve around real components, which are available only in certain sizes and which have their own design constraints. It is hard to find a collector that is rated at 103.45 W. Available rooftop areas and orientations, whether a pole-mount is physically and aesthetically acceptable, is a collector rack in the yard viable, all affect system sizing. Some decisions require not only technical data but cost data as well, such as whether a tracking system is more cost effective than a fixed array. And certainly budget constraints dominate every decision.

Example 9.6 System Sizing in Fresno, CA: A First Cut. An energy efficient house in Fresno is to be fitted with a rooftop PV array that will annually displace all of the 3600 kWh/yr of electricity that the home uses. How many kW (dc, STC) of panels will be required and what area will be needed? Make assumptions as needed.

Solution. We'll assume the roof is south-facing with a moderate tilt angle. Aesthetically, the array will look best if it is mounted at the same angle as the roof, which means a shallow pitch. Data in Appendix E indicate 5.7 kWh/m^2-day of annual insolation for L-15, which at Fresno's latitude of 37° means a tilt of 22°. That's probably a good estimate for the roof insolation.

Using the peak hour approach, we can write

$$\text{Energy (kWh/yr)} = P_{ac}\text{(kW)} \cdot \text{(h/day @1-sun)} \cdot 365 \text{ days/yr}$$

Solving

$$P_{ac} = \frac{3600 \text{ kWh/yr}}{5.7 \text{ h/day} \times 365 \text{ days/yr}} = 1.73 \text{ kW}$$

Previous examples have shown how to individually estimate the impacts of temperature, inverter efficiency, module mismatch, and dirt to come up with conversion efficiency from dc to ac. Those results suggests that a de-rating of about 25%, or an efficiency of 75%, is in the ballpark, so we'll use that to estimate the STC rated dc power of the array:

$$P_{dc,STC} = \frac{P_{ac}}{\text{Conversion efficiency}} = \frac{1.73 \text{ kW}}{0.75} = 2.3 \text{ kW}$$

If we can estimate collector efficiency, we can find collector area from the following:

$$P_{dc,STC} = 1 \text{ kW/m}^2 \text{ insolation} \cdot A \text{ (m}^2\text{)} \cdot \eta$$

Assuming crystalline silicon modules, Table 8.3 suggests that an efficiency of about 12.5% is reasonable, resulting in an area estimate of

$$A = \frac{2.3 \text{ kW}}{1 \text{ kW/m}^2 \cdot 0.125} = 18.4 \text{ m}^2 \ (198 \text{ ft}^2)$$

The procedure outlined in Example 9.6 illustrates the first step in grid-connected system design, which is to estimate the rated power and area required for the PV array. Local insolation in kWh/m^2-day was a key site parameter, and the resulting calculations included rough estimates for a de-rating from dc to ac power along with the module efficiency. Figure 9.29 shows how the estimate of annual energy production depends on the de-rating and local insolation, while Fig. 9.30 shows how area required depends on module efficiency.

Example 9.7 Fresno House with a 1-Axis Polar Tracker. Use Figs. 9.29 and 9.30 to estimate the module rated power and area needed to deliver 3600 kWh/yr in Fresno if a single-axis, polar mount tracker is used.

Solution. From Appendix E, annual insolation on a 1-axis tracker in Fresno is 7.6 kWh/m^2-day. From Fig. 9.29 with the 75% dc-to-ac efficiency, the energy delivered per kW (dc, STC) is about 2100 kWh/yr. This suggests that we need

$$P_{dc,STC} = \frac{3600 \text{ kWh/yr}}{2100 \text{ kWh/yr/kW}_{dc,STC}} = 1.7 \text{ kW}$$

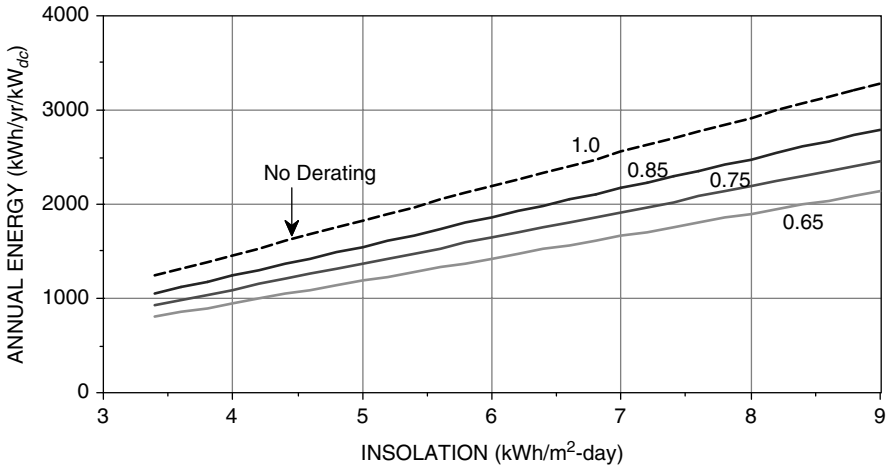


Figure 9.29 Annual energy delivered by a 1 kW(dc, STC) PV array, with dc to ac conversion efficiency as a parameter.

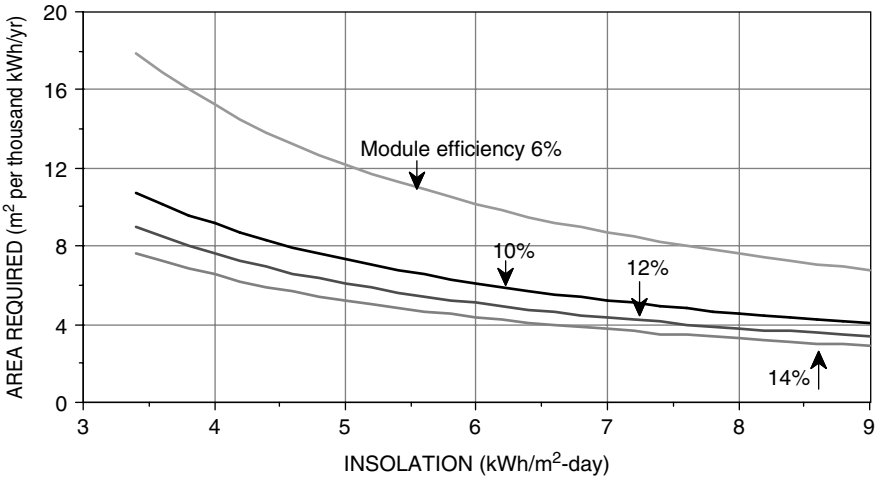


Figure 9.30 Area required to deliver 1000 kWh/yr with module efficiency as a parameter. Assumes a conversion efficiency from dc to ac of 75%.

This is considerably less than the 2.3 kW needed for the nontracking array in Example 9.6. The extra cost of the tracker will need to be compared to the reduction in cost of both the modules and the inverter.

From Fig. 9.30, the area of collectors at the assumed efficiency of 12.5% looks like about 3.9 m^2 per 1000 kWh/yr. That is, the area would be about

$$\text{Area} = 3600 \text{ kWh/yr} \times 3.9 \text{ m}^2/(1000 \text{ kWh/yr}) = 14 \text{ m}^2$$

Examples 9.6 and 9.7 illustrate the first step in grid-connected system design, which is to estimate the rated power and area required for the PV array. The next step is to explore the interactions between the choice of PV modules and inverters and how those impact the layout of the PV array. Finally, we need to consider details about voltage and current ratings for fuses, switches, and conductors.

Most traditional collectors on the market have 36 or 72 series cells in order to satisfy 12- or 24-V battery charging applications. Higher-voltage, higher-power modules are now becoming popular in grid-connected systems, for which battery-voltage constraints no longer apply. The key characteristics for a number of high-power modules intended for grid connections are given in Table 9.4.

Similarly, inverters for grid-connected systems are also different from those designed for battery-charging applications. Grid-connected inverters, for example, accept much higher input voltages and, as we shall see, those voltage constraints very much affect how the PV array is configured. The most important parameters for a number of inverters intended for grid-connected applications are given in Table 9.5.

TABLE 9.4 Important Characteristics of Several High-Power PV Modules

Module:	Sharp NE-K125U2	Kyocera KC158G	Shell SP150	Uni-Solar SSR256
Material:	Poly Crystal	Multicrystal	Monocrystal	Triple junction a-Si
Rated power $P_{dc,STC}$:	125 W	158 W	150 W	256 W
Voltage at max power:	26.0 V	23.2 V	34 V	66.0 V
Current at max power:	4.80 A	6.82 A	4.40 A	3.9
Open-circuit voltage V_{OC}:	32.3 V	28.9 V	43.4 V	95.2
Short-circuit current I_{SC}:	5.46 A	7.58 A	4.8 A	4.8
Length:	1.190 m	1.290 m	1.619 m	11.124 m
Width:	0.792 m	0.990 m	0.814 m	0.420 m
Efficiency:	13.3%	12.4%	11.4%	5.5%

TABLE 9.5 Example Inverter Characteristics for Grid-Connected Systems

Manufacturer:	Xantrex	Xantrex	Xantrex	Sunny Boy	Sunny Boy
Model:	STXR1500	STXR2500	PV 10	SB2000	SB2500
AC power:	1500 W	2500 W	10,000 W	2000 W	2500 W
AC voltage:	211–264 V	211–264 V	208 V, 3Φ	198–251 V	198–251 V
PV voltage range MPPT:	44–85 V	44–85 V	330–600 V	125–500 V	250–550 V
Max input voltage:	120 V	120 V	600 V	500 V	600 V
Max input current:	—	—	31.9 A	10 A	11 A
Maximum efficiency:	92%	94%	95%	96%	94%

To explore the interactions between modules, inverters, and the PV array, let us continue the design started in Example 9.6. For this example, let us try the Kyocera KC158G 158-W module with the Xantrex STXR2500 inverter. Begin by determining the number of modules required. Since they are rated at 158 W each and we need 2300 dc, STC watts, we have

$$\text{Number of modules} = \frac{2300 \text{ W}}{158 \text{ W/module}} = 14.6$$

To help us decide between 14 modules or 15 modules, consider how they might be arranged into an array. With two modules per string, the STC rated voltage would be $2 \times 23.2 = 46.4$ V, which just barely falls into the MPPT range of 44–85 V for the inverter picked. At higher temperatures, the module voltage could drop below 44 V, which isn't so good. At three modules per string, the rated voltage becomes $3 \times 23.2 = 69.6$ V, which fits nicely with the MPPT range. This suggests using an array with five strings of three modules each, for a total of 15 modules.

It is important to estimate the maximum open-circuit voltage of the array to be sure that it doesn't violate the highest dc voltage that the inverter can accept, which in this case is 120 V. With three modules in series, each having a V_{OC} at STC of 28.9 V, the string voltage could reach $3 \times 28.9 = 86.7$ V. This is well below the 120-V limit. But, remember that V_{OC} increases when cell temperature is below the STC assumption of 25°C. We could imagine that on a cold morning, with a strong, cold wind and low sunlight, cell temperature might be close to ambient, and that might be well below 25°C. With V_{OC} increasing by 0.38%/°C below 25°C (for crystalline silicon), the open-circuit voltage could then be well above its STC value.

Suppose that it is −5°C on the coldest morning in Fresno, and assume that cell temperature and ambient temperature are the same. The three-module string V_{OC} would now be

$$V_{OC,max} = 86.7\ V \cdot [1 + 0.0038(25 + 5)] = 86.7 \times 1.114 = 97\ \text{V}$$

which is still below the 120-V limit of this inverter. That 1.114 multiplier is, of course, temperature-dependent, and Fig. 9.31 shows the relationship.

Another reason for checking the open-circuit voltage of a residential grid-connected array under the coldest possible ambient conditions at the site is that the National Electrical Code restricts all voltages in one- and two-family dwellings to no more than 600 V (Wiles, 2001). Our array easily satisfies this constraint, so we will choose a design with 15 modules arranged in five strings of three modules each.

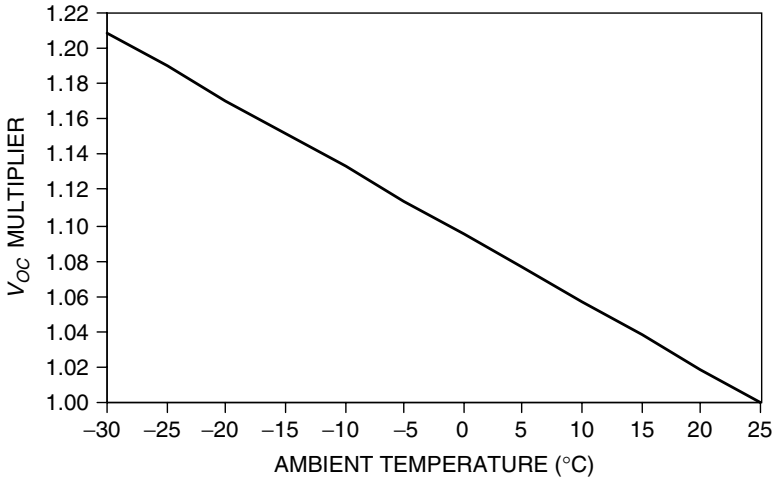


Figure 9.31 The V_{OC} multiplier for crystalline silicon assuming cells are at ambient temperature.

Example 9.8 Fresno House Example: Second Pass. Find the roof area required for the 15-module Fresno system with the fixed orientation collector, and estimate the annual energy delivered.

Solution. Collector dimensions are given in Table 9.4. With 15 modules, the collector area will be

$$\text{Area} = 15 \text{ modules} \times 1.29 \text{ m} \times 0.99 \text{ m} = 19.1 \text{ m}^2 \ (206 \text{ ft}^2)$$

which is just a bit larger than our original estimate of 18.4 m^2.

The dc STC rated power of the array will be

$$P_{dc,STC} = 158 \text{ W/module} \times 15 \text{ modules} = 2370 \text{ W}$$

Assuming a 25% de-rating for ac and using the 5.7-kWh/m^2-day average insolation for Fresno means that the system is predicted to deliver

$$\text{Energy} = 2.37 \text{ kW} \times 0.75 \times 5.7 \text{ h/day} \times 365 \text{ day/yr} = 3698 \text{ kWh/yr}$$

So, our goal of 3600 kWh/yr has been met.

In addition to the 600-V maximum voltage limit for one- and two-family dwellings, the proposed National Electrical Code for PV systems specifies other constraints on the choice of wires, fuses, and disconnect switches. They must be capable of withstanding 1.25 times the expected dc voltage. The NEC also recommends multiplying all PV currents by 1.25 to account for two possibilities: (1) the potential for insolation to exceed the 1-sun level of 1000 W/m^2 and (2) the increase in short-circuit current (approximately 0.1%/°C) caused by cell temperatures above the 25°C standard. In addition, the NEC requires that the continuous current of any circuit be multiplied by 1.25 to ensure that fuses, switches, and other disconnects as well as conductors are not operated above 80% of their rating. The two 125% current factors are independent and must be multiplied ($1.25 \times 1.25 = 1.56$) to properly size conductors and switchgear.

Finally, the allowable current that a conductor can handle (called its *ampacity*) is related to the type of insulation it has and the ambient temperature to which it is exposed. Standard ampacity ratings for cables assume an ambient temperature of 30°C. Given the possibility of higher ambients, exposure to sunlight, and nearness to hot collector surfaces, cable ampacity should be de-rated by factors of 0.33 to 0.58 depending on cable type, installation method (free air or conduit), and the temperature rating of the insulation (Wiles, 2001).

Example 9.9 Current and Voltage Ratings for the Fresno House Breakers. For our PV array consisting of five parallel strings of three Kyocera KC158G modules each, what minimum current rating should fuses have in the combiner, the array disconnect switch, and the utility breaker from the inverter? What maximum voltage should switchgear be rated for if the minimum daytime ambient temperature is expected to be -5°C?

Solution. From Table 9.4, the STC short-circuit current for each string of panels is 7.58 A. Applying the 125% factor for the effects of higher insolation and lower temperature, plus the 125 percent factor to afford a safe oversizing margin, results in combiner fuses that must allow at least

$$\text{Combiner fuses} > 7.58\ \text{A} \times 1.25 \times 1.25 = 11.8\ \text{A}$$

The array disconnect fuse must accommodate five such strings, so it must handle

$$\text{Array disconnect fuse} > 11.8\ \text{A} \times 5 = 59.2\ \text{A}$$

Applying the NEC 1.25 current multiplier to the 2500-W, 240-V inverter says its fuse must be rated to handle

$$\text{Inverter fuse} > 1.25 \times \frac{2500\ \text{W}}{240\ \text{V}} = 13\ \text{A}$$

Notice only a single 1.25 multiplier is used for the inverter since the PV temperature and insolation adjustments don't apply to the ac portions of the system.

The final system design is for the Fresno house is shown in Fig. 9.32.

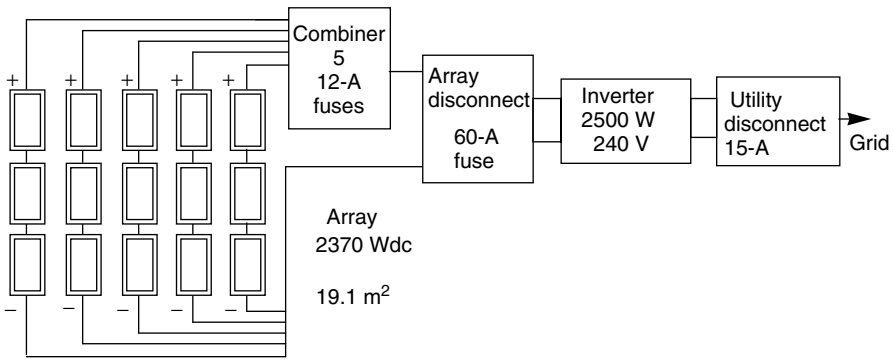


Figure 9.32 System design for the Fresno house example.

TABLE 9.6 Performance Adjustment Factors for Various Roof Pitches and Directions for 40° Latitude

Tilt Angle:	Flat	20	30	40	50	90
South:	0.84	0.97	1.00	1.00	0.97	0.61
SE, SW:	0.84	0.92	0.93	0.92	0.88	0.56
E, W:	0.84	0.80	0.77	0.72	0.67	0.40

Reference condition is 30° tilt, south facing.

Since most residential PV systems are mounted at the same pitch as the roof, and since those roofs seldom face due south, it is handy to have an adjustment factor to account for various collector orientations. The clear sky insolation tables in Appendix D can be used to estimate these adjustments. An example presentation in which a due south array at a 30° pitch is the reference condition is given in Table 9.6.

9.4 GRID-CONNECTED PV SYSTEM ECONOMICS

We now have the tools to allow us to estimate the energy delivered by a grid-connected PV system, so the next step is to explore its economic viability. Two types of economic analyses need to be made. One helps to make decisions between different system options—for example, whether to use a tracking system or a fixed mount, which collectors and which inverter are most cost effective, and so forth—and the other helps a buyer decide whether the investment is worthwhile.

9.4.1 System Trade-offs

To illustrate the decision between system options, consider the trade-off between the benefits of higher irradiance with a tracking mount compared to a simple fixed, roof mount. The following example demonstrates a process that can be used, but realistic decisions depend on current and accurate cost estimates for the equipment and installation.

Example 9.10 Should a House in Boulder Use a 1-Axis Tracker? A PV system for a house in Boulder, CO, is to be designed to deliver about 4000 kWh/yr. Given the following costs, decide whether to recommend a fixed array at tilt L-15 or a single-axis tracker. Assume 12%-efficient PVs and a 0.75 dc-to-ac efficiency factor.

Component	Cost
PVs	\$4.20/Wdc
Inverter	\$1.20/ W
Tracker	\$400 + \$100/m^2
Installation, BOS	\$3800

Solution

1-Axis Tracker: From Appendix E, annual insolation with a 1-axis tracker is 7.2 kWh/m^2-day. Using the peak-hours approach combined with the derating from dc, STC to ac:

$$\text{kWh/yr ac} = P_{dc,STC}(\text{kW}) \cdot (\text{Conversion efficiency}) \cdot \left(\frac{\text{h}}{\text{day @1-sun}}\right) \cdot 365 \text{ days/yr}$$

$$P_{dc,STC} = \frac{4000 \text{ kWh/yr}}{0.75 \times 7.2 \text{ h/day} \times 365 \text{ days/yr}} = 2.03 \text{ kW}$$

which would cost about \$4.20/W × 2030 W = \$8524 for the PVs.

At \$1.20/W, a 2.03-kW inverter would cost \$2435.

The area of collector needed at 12% efficiency would be

$$A(\text{m}^2) = \frac{P_{dc,STC}}{1 \text{ (kW/m}^2) \cdot \eta} = \frac{2.03 \text{ kW}}{1 \times 0.12} = 16.9 \text{ m}^2$$

The extra cost of the tracker would be \$400 + \$100/m^2 × 16.9 m^2 = \$2091.

Fixed Array: The insolation at L-15 is 5.4 kWh/m^2-day (Appendix E).

$$P_{dc,STC} = \frac{4000 \text{ kWh/yr}}{0.75 \times 5.4 \text{ h/day} \times 365 \text{ day/yr}} = 2.71 \text{ kW}$$

costing \$4.20/W × 2710 W = \$11,365 for the PVs. Notice that a bigger inverter is needed to accommodate the higher power rating of the collectors. A 2.71-kW inverter would cost \$1.20/W × 2710 W = \$3247.

The cost summary is thus:

Component	Tracker	Fixed Tilt
PVs	\$8,524	\$11,365
Inverter	\$2,435	\$3,247
Tracker	\$2,091	\$0
Installation, BOS	\$3,800	\$3,800
Total	\$16,850	\$18,412

It looks like there might some advantage to using the tracker, but only a more careful analysis based on real components could affirm the validity of that conclusion.

9.4.2 Dollar-per-Watt Ambiguities

The most important inputs to any economic analysis of a PV system are the initial cost of the system and the amount of energy it will deliver each year. Whether the system is economically viable depends on other factors—most especially, the price of the energy displaced by the system, whether there are any tax credits or other economic incentives, and how the system is to be paid for. A detailed economic analysis will include: estimates of operation and maintenance costs; future costs of utility electricity; loan terms and income tax implications if the money is to be borrowed, or personal discount rates if the owner purchases it outright; system lifetime; costs or residual value when the system is ultimately removed; and so forth.

Begin with the installed cost of the system. For individual buyers, it is total dollars for their system that matters, but when an overall snapshot of the industry is being presented, it is common practice to describe installed costs in dollars per watt of peak power. There are two ambiguities with the \$/W indicator, however, which must be made clear for the parameter to have any meaning. One is whether the watts are based on dc power from the PVs or ac power from the inverter. The other is whether or not a tracker has been used. To illustrate these differences, Table 9.7 summarizes the costs for the tracker versus fixed-array analysis just derived in Example 9.10. As can be seen, even though the tracker delivers the same kWh/yr and it is cheaper than the fixed-tilt array, it appears to have a higher cost when expressed as \$/Wdc or \$/Wac. Clearly, some factor is needed to account for the greater energy delivered by a module in a tracking system.

When a PV system uses tracking, an *Energy Production Factor* (EPF) must be included in order to make the simple \$/Wdc or \$/Wac descriptors directly comparable to those systems that have a fixed orientation. The EPF is essentially the ratio of insolation on the tracking surface to that on a fixed surface (for the example in Table 9.7, EPF = 7.2/5.4 = 1.333), which leads to the following:

$$\text{Tracker}(\$/\text{W}) = \frac{\$/\text{W}}{\text{EPF}} = \frac{\$/\text{W}}{(\text{Tracking insolation/Fixed insolation})} \tag{9.17}$$

When the EPF is included, the tracking array in Table 9.7 does correctly appear to be the least-cost system.

So, having resolved the \$/W ambiguities, how much do systems cost? Data gathered by the Utility Photovoltaic Group (UPVG) on actual costs of over 600 residential PV systems installed between 1994 and 2000 has been summarized in Fig. 9.33. Of the total \$6.80/Wdc, a little over 60% is modules with the remainder roughly equally divided between other components and installation.

TABLE 9.7 Illustrating the Ambiguities Associated with $/W Cost Indicators[a]

Parameter	Tracker	Fixed Tilt
Energy (kWh/yr)	4000	4000
Insolation (kWh/m^2-day)	7.2	5.4
System cost ($)	$16,850	$18,412
$P_{dc,STC}$ (W)	2029	2706
P_{ac} (W)	1522	2029
Cost $/Wdc	$8.30	$6.80
Cost $/Wac	$11.07	$9.07
Cost $/Wdc (EPF)	$6.23	
Cost $/Wac (EPF)	$8.30	

[a] Without an energy production factor (EPF) correction in $/W, the tracking system incorrectly appears to be more expensive. (Data are based on Example 9.10.)

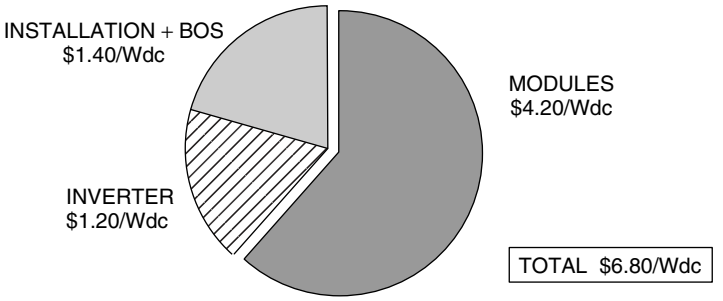


Figure 9.33 Average installed cost for 625 residential PV systems installed between 1994 and 2000. From Chang (2000).

9.4.3 Amortizing Costs

A simple way to estimate the cost of electricity generated by a PV system is to imagine taking out a loan to pay for the system and then using annual payments divided by annual kWh delivered to give ¢/kWh. If an amount of money, or principal, P ($), is borrowed over a period of n (years) at an interest rate of i (decimal fraction/yr), then the annual loan payments, A ($/yr), will be

$$A = P \cdot \mathrm{CRF}(i, n) \tag{9.18}$$

where $\mathrm{CRF}(i, n)$ is the capital recovery factor given by

$$\mathrm{CRF}(i, n) = \frac{i(1+i)^n}{(1+i)^n - 1} \tag{9.19}$$

Example 9.11 Cost of PV Electricity for the Boulder House. The tracking PV system of Example 9.10 costs $16,850 to deliver 4000 kWh/yr. If the system is paid for with a 6%, 30-year loan, what would be the cost of electricity?

Solution. The capital recovery factor for this loan would be

$$\text{CRF}(i, n) = \frac{i(1+i)^n}{(1+i)^n - 1} = \frac{0.06(1.06)^{30}}{(1.06)^{30} - 1} = 0.07265/\text{yr}$$

So the annual payments would be

$$\text{A} = \text{P CRF}(i, n) = \$16{,}850 \times 0.07265/\text{yr} = \$1224/\text{yr}$$

The cost per kWh is therefore

$$\text{Cost of electricity} = \frac{\$1224/\text{yr}}{4000 \text{ kWh/yr}} = \$0.306/\text{kWh}$$

A significant factor that was ignored in the cost calculation of Example 9.11 is the impact of the income tax benefit that goes with a home loan. As was described in Chapter 5, interest on such loans is tax deductible, which means that a person's gross income is reduced by the loan interest, and it is only the resulting net income that is subject to income taxes. The tax benefit that results depends on the marginal tax bracket (MTB) of the homeowner. For married couples filing jointly, the 2002 MTB for federal taxes is shown in Table 9.8. For example, a married couple earning $120,000 per year is in the 30.5% marginal tax bracket, which means that a one-dollar tax deduction reduces their income tax by 30.5 cents. The value of the tax deduction will be even greater when it reduces the homeowner's state income tax as well. For example, the same taxpayer in California would be in the 37% marginal tax bracket when both state and federal taxes are considered.

TABLE 9.8 Federal Income Tax for Married Couples Filing Jointly (2002)

Income Over	But Not Over	Tax Is	of the Amount Over
$0	$45,200	15%	$0
45,200	109,250	$6,780 + 27.5%	45,200
109,250	166,500	24,394 + 30.5%	109,250
166,500	297,350	41,855 + 35.5%	166,500
297,350	—	88,307 + 39.1%	297,350

During the first years of a long-term loan, almost all of the annual payments will be interest, with very little left to reduce the principal, while the opposite occurs toward the end of the loan. This means that the tax benefit of interest payments varies from year to year. For example, in the first year, interest is owed on the entire amount borrowed and the tax benefit is

$$\text{First-year tax benefit} = i \times P \times \text{MTB} \tag{9.20}$$

where MTB is the marginal tax bracket.

Example 9.12 Cost of PV Energy Including Income Tax Benefit. The PV system for the house in Boulder costs \$16,850 and annual payments on its 6%, 30-year loan are \$1224. If the homeowner is in the 30.5% marginal tax bracket, what is the cost of PV energy in the first year?

Solution. From (9.20) the reduction in federal income tax will be

$$\text{First-year tax benefit} = 0.06 \times \$16{,}850 \times 0.305 = \$308$$

The net cost of the PV system in that first year will therefore be

$$\text{First-year cost of PV} = \$1224 - \$308 = \$916$$

which means that its electricity cost will be

$$\text{Cost of electricity} = \frac{\$916/\text{yr}}{4000\ \text{kWh/yr}} = \$0.23/\text{kWh}$$

Figure 9.34 shows how the first-year cost of electricity varies as a function of annual insolation and system cost. A 30-year, 6% loan for a family in the 30.5% tax bracket has been used to derive the figure. For a reasonably good site, with annual insolation above 5.5 kWh/m^2-day, the installed cost of a PV system needs to drop to below about \$3/Wdc to make PV electricity competitive with 10¢/kWh electricity from the grid. At the \$6.80/Wdc average cost at the turn of the century, significant cost reductions are needed to make PVs competitive with the local utility. To help encourage the PV industry, some states have initiated sizable rebates and tax credits. California, for example, has a buy-down program funded by utility ratepayers that offers a \$4.50/Wac rebate (about \$3.50/Wdc) which pretty much closes the gap making PVs cost effective right now.

A very effective way to present the economics of a residential grid-connected PV system is with a spreadsheet that incorporates the price of avoided electricity

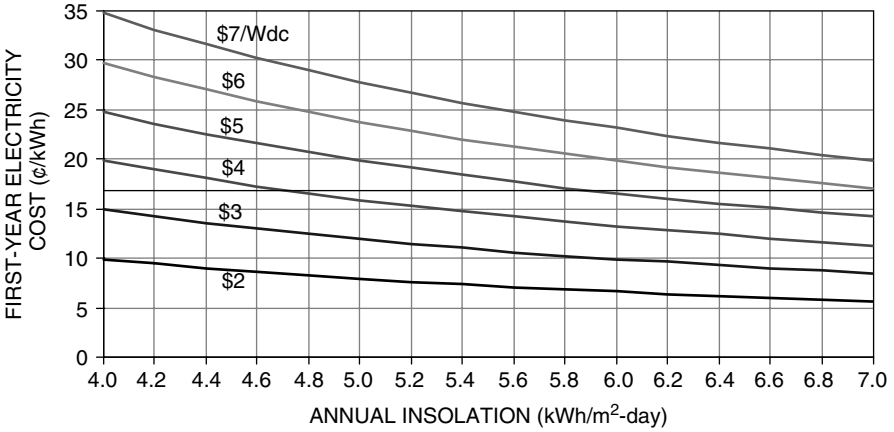


Figure 9.34 First-year cost of electricity with $/Wdc (STC) as the parameter. Assumptions: 6%, 30-yr loan, MTB 0.305, 75% conversion from dc (STC) to ac (PTC).

purchases, loan terms, including the homeowner MTB and income tax benefits, and any rebates or other monetary features that affect the decision.

Example 9.13 Financing a PV System. A 3-kWac PV system is predicted to deliver 6000 kWh/yr to a house that currently pays $0.12/kWh for electricity. The system, which costs $27,000, is eligible for a rebate of $4.50/Wac. If the balance is paid for with a 6%, 30-year loan and the owner is in the 37% tax bracket (combined state and federal), what is the cost of PV electricity in the first year and what would be the net economic benefit in the first year?

Solution. The net cost of the system after the rebate is

$$P = \$27{,}000 - \$4.50/\text{W} \times 3000\ \text{W} = \$13{,}500$$

From Example 9.11, CRF(0.06, 30 yr) is 0.07265/yr, so the annual loan payment is

$$\text{Loan} = 0.07265/\text{yr} \times \$13{,}500 = \$980.78$$

The reduction in income taxes on the first-year's interest portion of the loan payment is

$$\text{Tax savings} = 0.06 \times \$13{,}500 \times 0.37 = \$299.70$$

The first-year cost of electricity from the PV system is therefore

$$\text{Cost of electricity} = \frac{\$980.78 - \$299.70/\text{yr}}{6000\ \text{kWh/yr}} = \$0.1135/\text{kWh}$$

The net economic benefit in the first year is

$$\text{Benefit} = 6000\ \text{kWh/yr} \times (0.12 - 0.1135)\$/\text{kWh} = \$39/\text{yr}$$

It is easy to set up a spreadsheet to show the year-by-year economic value of a PV system as has been done in Table 9.9 for Example 9.13. In that spreadsheet, the price of utility electricity is assumed to escalate at 2%/yr, with the result that

TABLE 9.9 Annual Cash Flow for the PV System of Example 9.13[a]

Year	Loan Balance	Loan Payment	Loan Interest	Delta P	Delta tax	Annual Cost	PV cost ¢/kWh	Utility ¢/kWh	Savings $/yr
0	13,500	981	810	171	300	681	11.4	12.0	39
1	13,329	981	800	181	296	685	11.4	12.2	50
2	13,148	981	789	192	292	689	11.5	12.5	60
3	12,956	981	777	203	288	693	11.6	12.7	71
4	12,753	981	765	216	283	698	11.6	13.0	82
5	12,537	981	752	229	278	702	11.7	13.2	93
6	12,309	981	739	242	273	708	11.8	13.5	103
7	12,067	981	724	257	268	713	11.9	13.8	114
8	11,810	981	709	272	262	719	12.0	14.1	125
9	11,538	981	692	288	256	725	12.1	14.3	136
10	11,249	981	675	306	250	731	12.2	14.6	147
11	10,943	981	657	324	243	738	12.3	14.9	157
12	10,619	981	637	344	236	745	12.4	15.2	168
13	10,276	981	617	364	228	753	12.5	15.5	179
14	9,911	981	595	386	220	761	12.7	15.8	189
15	9,525	981	572	409	211	769	12.8	16.2	200
16	9,116	981	547	434	202	778	13.0	16.5	210
17	8,682	981	521	460	193	788	13.1	16.8	220
18	8,223	981	493	487	183	798	13.3	17.1	230
19	7,735	981	464	517	172	809	13.5	17.5	240
20	7,218	981	433	548	160	821	13.7	17.8	249
21	6,671	981	400	581	148	833	13.9	18.2	259
22	6,090	981	365	615	135	846	14.1	18.6	268
23	5,475	981	328	652	122	859	14.3	18.9	276
24	4,823	981	289	691	107	874	14.6	19.3	284
25	4,131	981	248	733	92	889	14.8	19.7	292
26	3,398	981	204	777	75	905	15.1	20.1	300
27	2,622	981	157	823	58	923	15.4	20.5	306
28	1,798	981	108	873	40	941	15.7	20.9	313
29	925	981	56	925	21	960	16.0	21.3	318
30	0	0	0	0	0	0	0.0	21.7	1304

[a]Buyer is in the 37% tax bracket and utility electricity is projected to grow at 2%/yr.

annual savings with the PV system grows year-by-year, reaching $318/yr in the last year of the 30-year loan.

9.5 STAND-ALONE PV SYSTEMS

Since grid-connected systems use the utility for back-up, there is no need for battery storage unless power outages are a problem. This makes them relatively simple and relatively inexpensive. Having the grid right there, however, means that they have to compete with relatively inexpensive utility power, which makes it hard to justify the PV system unless significant subsidies are provided. When the grid isn't nearby, electricity suddenly becomes much more valuable and the extra cost and complexity of a totally self-sufficient, stand-alone power system can provide enormous benefit. Instead of competing with 10-cent utility power, a PV–battery system competes with 50-cent gasoline or diesel-powered generators. Or it competes with the cost of bringing the grid to the site, which may run many thousands of dollars per mile. In developing countries, a panel and battery to run a few lights and a radio can change people's lives.

Figure 9.35 suggests a quite general system that includes a generator backup as well as the possibility for some loads to be served directly with more-efficient dc and others with ac. A combination charger–inverter is shown, which has the capability to convert ac to dc or vice versa. As a charger, it converts ac from the generator into dc to charge the batteries; as an inverter, it converts dc from the batteries into ac needed by the load. The charger–inverter unit may include an automatic transfer switch that allows the generator to supply ac loads directly whenever it is running.

Off-grid systems must be designed with great care to assure satisfactory performance. Users must be willing to check and maintain batteries, they must be willing to adjust their energy demands as weather and battery charge vary, they may have to fuel and fix a noisy generator, and they must take responsibility for the safe operation of the system. The reward is electricity that is truly valued.

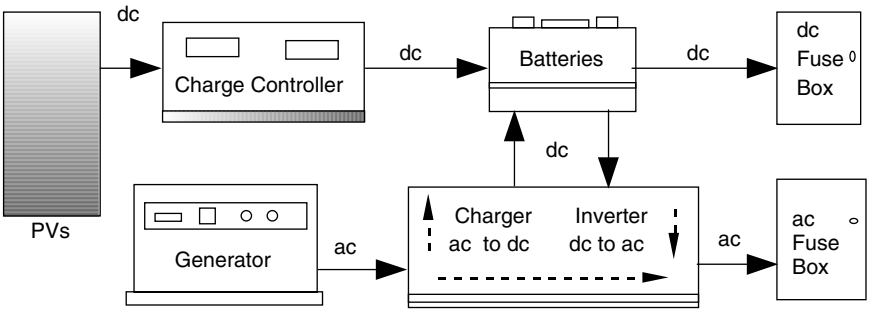


Figure 9.35 A stand-alone system with back-up generator and separate outputs for dc and ac loads.

9.5.1 Estimating the Load

The design process for stand-alone systems begins with an estimate of the loads that are to be provided for. As with all design processes, a number of iterations may be required. On the first pass, the user may try to provide the capability to power anything and everything that normal, grid-connected living allows. Various iterations will follow in which trade-offs are made between more expensive, but more efficient, appliances and devices in exchange for fewer PVs and batteries. Lifestyle adjustments need to be considered in which some loads are treated as essentials that must be provided for, and others are luxuries to be used only when conditions allow. A key decision involves whether to use all dc loads to avoid the inefficiencies associated with inverters, or whether the convenience of an all ac system is worth the extra cost, or perhaps a combination of the two is best. Another important decision is whether to include a generator back-up system and, if so, what fraction of the load it will have to supply.

The simplest of systems will incorporate only devices that run directly on dc. A rather large market already exists for such dc equipment to meet the needs of the boating and recreational vehicle communities. In addition, catalogs of such equipment are readily available. For example, Real Goods Trading Corporation in Ukiah, California, publishes its continuously updated *Solar Living Source Book*, which not only provides detailed descriptions of specific equipment, but also presents theory and perspectives on system design and equipment selection. At the other extreme, off-grid homes can be as conventional as any other, with ac appliances and devices purchased from mainstream suppliers.

Power needed by a load, as well as energy required over time by that load, is important for system sizing. In the simplest case, energy (watt-hours or kilowatt-hours) is just the product of some nominal power rating of the device multiplied by the hours that it is in use. The situation is often more complicated, however. For example, an amplifier needs more power when the volume is increased, and many appliances, such as refrigerators and washing machines, use different amounts of power during different portions of their operating cycle. An especially important consideration for household electronic devices—TVs, VCRs, computers, portable phones, and so on—is the power consumed when the device is in its standby or charging modes. Many devices, such as TVs, use power even while they are turned off since some circuits remain energized awaiting the turn-on signal from the remote. Consumer electronics now account for about 10% of all U.S. residential electricity, and researchers at Lawrence Berkeley National Labs conclude that almost two-thirds of this energy occurs when these devices are not actually being used (Rosen and Meier, 2000). Major appliances and shop tools have another complication caused by the surge of power required to start their electric motors. While that large initial spike doesn't add much to the energy used by a motor, it has important implications for sizing inverters, wires, fuses, and other ancillary electrical components in the system.

Table 9.10 lists examples of power used by a number of household electrical loads. Some of these are simply watts of power, which can be multiplied by hours of use to get watt-hours of energy. Many of the devices listed in the

TABLE 9.10 Power Requirements of Typical Loads

Kitchen Appliances	*Power*
Refrigerator: ac EnergyStar, 14 cu. ft	300 W, 1080 Wh/day
Refrigerator: ac EnergyStar, 19 cu. ft	300 W, 1140 Wh/day
Refrigerator: ac EnergyStar, 22 cu. ft	300 W, 1250 Wh/day
Refrigerator: dc Sun Frost, 12 cu. ft	58 W, 560 Wh/day
Freezer: ac 7.5 cu. ft	300 W, 540 Wh/day
Freezer: dc Sun Frost, 10 cu. ft	88 W, 880 Wh/day
Electric range (small burner)	1250 W
Electric range (large burner)	2100 W
Dishwasher: cool dry	700 W
Dishwasher: hot dry	1450 W
Microwave oven	750–1100 W
Coffeemaker (brewing)	1200 W
Coffeemaker (warming)	600 W
Toaster	800–1400 W
General Household	
Clothes washer: vertical axis	500 W
Clothes washer: horizontal axis	250 W
Dryer (gas)	500 W
Vacuum cleaner	1000–1400 W
Furnace fan: 1/4 hp	600 W
Furnace fan: 1/3 hp	700 W
Furnace fan: 1/2 hp	875 W
Ceiling fan	65–175 W
Whole house fan	240–750 W
Air conditioner: window, 10,000 Btu	1200 W
Heater (portable)	1200–1875 W
Compact fluorescent lamp (100-W equivalent)	30 W
Compact fluorescent lamp (60-W equivalent)	16 W
Electric blanket, single/double	60/100 W
Clothes iron	1000–1800 W
Electric clock	4 W
Consumer Electronics	
TV: >39-in. (active/standby)	142/3.5 W
TV: 25 to 27-in. color (active/standby)	90/4.9 W
TV: 19 to 20-in. color (active/standby)	68/5.1 W
Analog cable box (active/standby)	12/11 W
Satellite receiver (active/standby)	17/16 W
VCR (active/standby)	17/5.9 W
Component stereo (active/standby)	44/3 W
Compact stereo (active/standby)	22/9.8 W
Cordless phone	4 W
Clock radio (active/standby)	2.0/1.7 W
Computer, desktop (active/idle/standby)	125/80/2.2 W
Laptop computer	20 W

TABLE 9.10 (*continued*)

Ink-jet printer	35 W
Dot-matrix printer	200 W
Laser printer	900 W
Shop	
Circular saw, 7 1/4″	900 W
Table saw, 10-in.	1800 W
Hand drill, 1/4″	250 W
Water Pumping	
Centrifugal pump: 36 Vdc, 50-ft @ 10 gpm	450 W
Submersible pump: 24 Vdc, 100-ft @ 1.6 gpm	100 W
Submersible pump: 48 Vdc, 300-ft @ 1.5 gpm	180 W
DC pump (house pressure system), typical use 1–2 h/day	60 W

Source: Rosen and Meier (2000) and others.

consumer electronics category show power while they are being used (active) and power consumed the rest of the time (standby), both of which must be considered when determining energy consumption. Refrigerators are also unusual since they are always turned on, but their power demand varies throughout the day. The data given on refrigerator labels are average watt-hours per day based on measurements made with the refrigerator located in a 90°F room. This means that they are likely to overstate actual demand in someone's home—perhaps by as much as 20%. Tables like this one are useful for average values, but the best source of power and energy data are actual measurements that can easily be performed with readily available meters. Another source is the device nameplate itself, but those tend to overstate power since they are meant to describe maximum demand rather than the likely average. Some nameplates provide only amperage and voltage; and while it is tempting to multiply the two to get power, this can also be an overestimate since it ignores the phase angle, or power factor, between current and voltage.

Example 9.14 A Modest Household Demand. Estimate the monthly energy demand for a cabin with all ac appliances, consisting of a 19-cu. ft refrigerator, six 30-W compact fluorescents (CFLs) used 5 h/day, a 19-in. TV turned on 3 h/day and connected to a satellite, a cordless phone, a 1000-W microwave used 6 min/day, and a 100-ft deep well that supplies 120 gallons/day.

Solution. Using data from Table 9.10, we can put together the following table of power and energy demands. The total is just over 3.11 kWh/day.

Notice how little of the energy for TV/satellite is used during the 3 h that it is actually turned on (255 Wh out of a total of 698 Wh, or 36.5%). This household

really needs to provide a power strip to allow manual complete shut off of these electricity vampires.

Appliance	Power (W)	Hours	Watt-hours/day	Percentage
Refrigerator, 19 cu. ft	300		1140	37%
Lights (6 @ 30 W)	180	5	900	29%
TV, 19-in., active mode	68	3	204	7%
TV, 19-in., standby mode	5.1	21	107	3%
Satellite, active mode	17	3	51	2%
Satellite, standby mode	16	21	336	11%
Cordless phone	4	24	96	3%
Microwave	1000	0.1	100	3%
Washing machine	250	0.2	50	2%
Well pump, 100 ft, 1.6 gpm	100	1.25	125	4%
Total			3109	100%

9.5.2 The Inverter and the System Voltage

In Example 9.14, an example calculation was made for the average daily energy consumption for appliances and loads that were all assumed to run on ac power. To figure out how much power the batteries must supply, the calculation needs to be modified to account for losses in the dc-to-ac inverter. This can be tricky to do accurately since the inverter's efficiency is a function of the magnitude of the load it happens to be supplying at that particular instant. Most inverters now operate at around 90% efficiency over most of their range, as is suggested in Fig. 9.36. For calculations, an overall inverter efficiency of about 85% is considered to be a conservative default assumption.

When no load is present, a good inverter will power down to less than 1 watt of standby power while it waits for something to be turned on that needs ac. When it senses a load, the inverter powers up and while it runs uses on the order of 5–20 W of its own. That means those standby losses associated with so many of our electronic devices may keep the inverter running continuously, even though no real energy service is being delivered. In that case, that 5-to-20 watts of inverter loss adds to the other standby losses, which emphasizes the need for a manual shutdown of turned-off electronic equipment.

Example 9.15 Accounting for Inverter Losses. Suppose that a dc refrigerator that uses 800 Wh/day is being considered instead of the 1140 Wh/d ac one given in Example 9.14. Estimate the dc load that the batteries must provide if an 85% efficient inverter is used (a) with all loads running on ac and (b) with everything but the refrigerator running on ac.

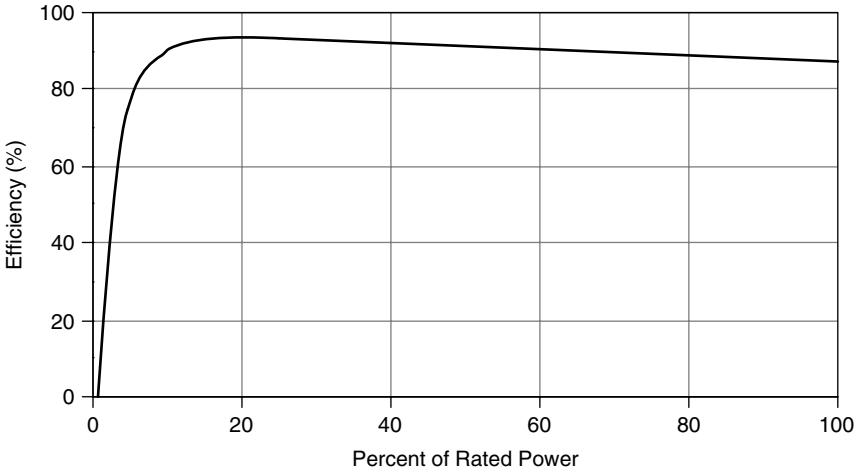


Figure 9.36 Typical efficiency of a stand-alone system inverter.

Solution

a. With all 3109 Wh/day running on an 85% efficient inverter, the dc load that the batteries must supply would be

$$\text{dc battery load} = \frac{3109 \text{ Wh/day}}{0.85} = 3658 \text{ Wh/day}$$

b. With the 1140-Wh/day refrigerator removed, the remaining ac load is

$$\text{ac load} = (3109 - 1140) \text{ Wh/day} = 1969 \text{ Wh/day}$$

accounting for the inverter efficiency, that ac load would be supplied by

$$\text{dc for ac loads} = \frac{1969 \text{ Wh/day}}{0.85} = 2316 \text{ Wh/day}$$

Adding in the 800-Wh/day dc refrigerator, the total dc load becomes

$$\text{dc battery load} = (2316 + 800) \text{ Wh/day} = 3116 \text{ Wh/day}$$

That's a 15% decrease in energy needed. Figure 9.37 shows these data.

Swapping out the ac refrigerator in the above example for a dc one reduces the total dc load that the batteries have to supply by about 15%. This can translate into a 15% reduction in the size and cost of the photovoltaic array as well as the batteries. Moreover, the inverter itself can be smaller and cheaper since it doesn't have to supply as much ac power. Offsetting those gains is the additional

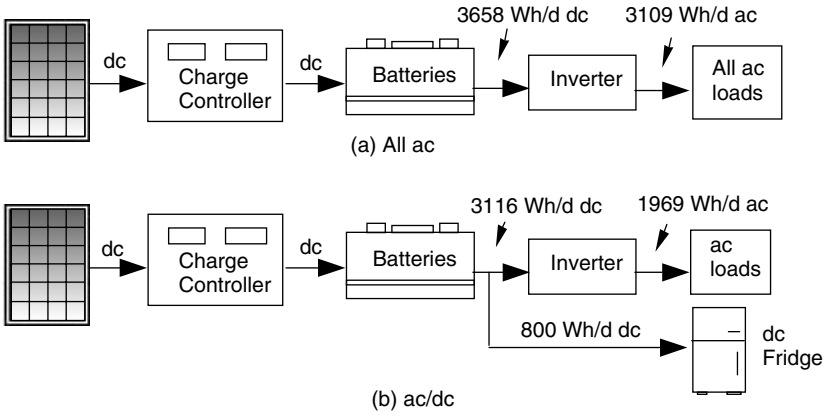


Figure 9.37 Switching out the ac refrigerator with a more efficient dc one. Numbers are based on Example 9.15.

cost of a dc refrigerator plus the added complexity and cost of wiring a house to provide for some ac and some dc loads. An economic analysis would be needed to determine the right decision.

System Voltage Inverters are specified by their dc input voltage as well as by their ac output voltage, continuous power handling capability, and the amount of surge power they can supply for brief periods of time. The inverter's dc input voltage, which is the same as the voltage of the battery bank and the PV array, is called the *system voltage*. The system voltage is usually 12 V, 24 V, or 48 V. Higher voltages need less current, making it easier to minimize wire losses. On the other hand, higher voltage means more batteries wired in series, which impacts the number of batteries that may be needed to supply the load.

One guideline that can be used to pick the system voltage is based on keeping the maximum steady-state current drawn below around 100 A so that readily available electrical hardware and wire sizes can be used. Using this guideline results in the system-voltage suggestions given in Table 9.11.

The maximum ac power that the inverter needs to deliver can be estimated by adding the power demand of all of the loads that will ever be anticipated to

TABLE 9.11 Suggested System Voltages Based on Limiting Current to 100 A

Maximum ac Power	System dc Voltage
<1200 W	12 V
1200–2400 W	24 V
2400–4800 W	48 V

TABLE 9.12 The Maximum Continuous Power Demand for The House in Example 9.14

Load	Watts
Refrigerator	300
Lights	180
TV/satellite, active mode	85
Cordless phone	4
Microwave	1000
Washing machine	250
Well pump	100
Maximum ac power demand	1919

be operating simultaneously. Table 9.10 provides representative values of steady-state power for a number of common household devices, which can be used as a starting point in the analysis. For the house described in Example 9.14, the total ac power demand with everything turned on at once would be 1919 W (Table 9.12), which would draw 160 A if the system voltage is only 12 V. A 24-V inverter would allow plenty of growth in demand without exceeding the 100-A guideline and so should be chosen for this system.

The most important specification for an inverter is the amount of ac power that it can supply on a continuous basis. But it is also critically important that the inverter be able to supply surges of current that occur when electric motors are started. Until the rotor starts spinning, there is no back emf to limit motor current and surges many times higher than steady state can occur. Table 9.13 provides estimates of steady-state and surge power requirements for typical household loads.

9.5.3 Batteries

Stand-alone systems obviously need some method to store energy gathered during good times to be able to use it during the bad. While various exotic technologies are possible—including flywheels, compressed air, or even hydrogen production—it is the lowly battery that makes the most sense today. And, among the many possible battery technologies, it is the familiar lead-acid battery that continues to be the workhorse of PV systems. In addition to energy storage, batteries provide several other important energy services for PV systems, including the ability to provide surges of current that are much higher than the instantaneous current available from the array, as well as the inherent and automatic property of controlling the output voltage of the array so that loads receive voltages that are within their own range of acceptability.

Competitors to conventional lead-acid batteries include nickel–cadmium, nickel–metal hydride, lithium–ion, lithium–polymer, and nickel–zinc technologies. Of these, only nickel–cadmium "Nicads" are even barely competitive with

TABLE 9.13 Steady-State and Surge Power Requirements for Example Loads

Load	Steady State (watts)	Surge (watts)
Refrigerator (ac)	300	1500
Refrigerator (dc)	58	700
Dishwasher	700	1400
Jet pump (1/3 hp) (ac)	750	1400
Submersible pump (ac)	1000	6000
Clothes washer (vertical axis)	650	1150
Clothes washer (horizontal axis)	250	750
Dryer (gas)	500	1800
Furnace fan 1/4 hp	600	1000
Furnace fan 1/3 hp	700	1400
Furnace fan 1/2 hp	875	2350
Air conditioner, window 10 kBtu	1200	1500
Worm drive 7 1/4″ saw	1800	3000
Table saw, 10″	1800	4500

Source: Real Goods (2002).

TABLE 9.14 Rough Comparison of Battery Characteristics[a]

Battery	Max Depth Discharge	Energy Density (Wh/kg)	Cycle Life (cycles)	Calendar Life (years)	Efficiencies Ah %	Efficiencies Wh %	Cost ($/kWh)
Lead-acid, SLI	20%	50	500	1–2	90	75	50
Lead-acid, golf cart	80%	45	1000	3–5	90	75	60
Lead-acid, deep-cycle	80%	35	2000	7–10	90	75	100
Nickel–cadmium	100%	20	1000–2000	10–15	70	60	1000
Nickel–metal hydride	100%	50	1000–2000	8–10	70	65	1200

[a] Actual performance depends greatly on how they are used.

Source: Linden (1995) and Patel (1999).

lead-acid batteries, but this may change in the near future due to the surge of interest and development in new battery technologies for electric and hybrid vehicles. Table 9.14 summarizes typical values of some of the important characteristics of these battery technologies. Lead-acid batteries are listed in three categories: conventional automobile batteries for engine starting, vehicle lighting, and engine ignition (SLI); low-cost, deep-cycle batteries typically used in golf carts; and longer-lifetime, true deep-cycle batteries. Two other battery types

are shown, nickel–cadmium (or Nicads) and nickel–metal hydride batteries, which are beginning to be used in some hybrid-electric vehicles. As can be seen, lead-acid batteries are by far the least expensive option, they have the highest efficiencies, and the more expensive ones, when used properly, can last nearly as long as their competitors. Nicads are much more expensive, but they last longer. Nicads also perform better in harsh climates; and since they can be discharged nearly 100% without damage, they are far more forgiving when abused.

9.5.4 Basics of Lead-Acid Batteries

Lead-acid batteries date back to the 1860s when inventor Raymond Gaston Planté fabricated the first practical cells made with corroded lead-foil electrodes and a dilute solution of sulfuric acid and water. Many advances since then have lead to a global market that now exceeds $30 billion in annual retail sales, with about three-fourths of that being starting, lighting, and ignition (SLI) automobile batteries. Lead-acid batteries are used in everything from small electronic devices such as cell phones, to car batteries, to enormous utility battery banks, the largest of which, in Chino, California, is capable of delivering 4 h of 10 MW power (5000 A at 2000 V) into the grid.

Automobile SLI batteries have been highly refined to perform their most important task, which is to start your engine. To do so, they have to provide short bursts of very high current (400–600 A!). Once the engine has started, its alternator quickly recharges the battery, which means that under normal circumstances the battery is almost always at or near full charge. SLI batteries are not designed to withstand deep discharges, and in fact they will fail after only a few complete-discharge cycles. This makes them inappropriate for most PV systems, in which slow, but deep, discharges are the norm. If they must be used, as is sometimes the case in developing countries where they may be the only batteries available, daily discharges of less than about 20% can yield approximately 500 cycles, or a year or two of operation.

In comparison with SLI batteries, deep discharge batteries have thicker plates, which are housed in bigger cases that provide greater space both above and beneath the plates. Greater space below allows more debris to accumulate without shorting out the plates, and greater space above lets there be more electrolyte in the cell to help keep water losses from exposing the plates. Thicker plates and larger cases mean that these batteries are big and heavy. A single 12-V deep-discharge battery can weigh several hundred pounds. They are designed to be discharged repeatedly by 80% of their capacity without harm, although such deep discharges result in a lower lifetime number of cycles. Figure 9.38 suggests that a typical deep-cycle, lead-acid battery can be cycled about 4000 times when discharged by 25% of its rated capacity, which would give it a lifetime of over 10 years. At a daily discharge of 80%, about 1800 cycles could be expected, which suggests a lifetime of around 5 years. While Fig. 9.38 provides a rough indication of battery life, other factors, including quality of battery, frequency of maintenance, charge rates, and final charging cut-off voltages, are also important.

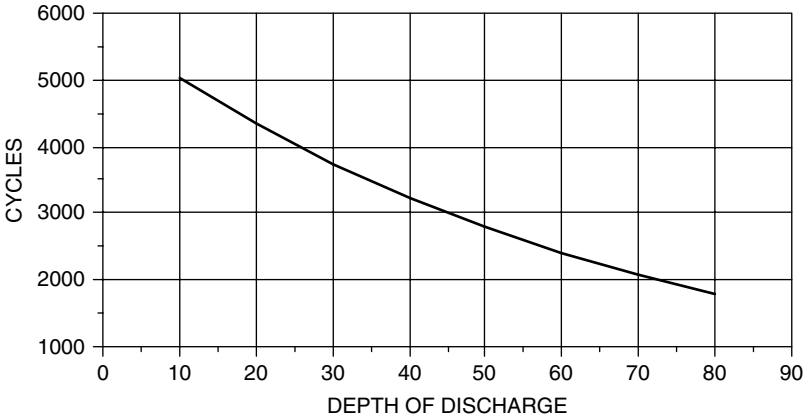


Figure 9.38 Impact of depth of discharge on the number of cycles a typical deep-cycle lead-acid battery might be able to provide. An automobile SLI battery delivers only around 500 cycles at 20% discharge.

To understand some of the subtleties in sizing battery systems, we need a basic understanding of their chemistry. Very simply, an individual 2-V cell in a lead-acid battery consists of a positive electrode made of lead dioxide (PbO_2) and an negative electrode made of a highly porous, metallic lead (Pb) structure, both of which are completely immersed in an electrolyte consisting of a dilute solution of sulfuric acid and water. Thin lead plates are structurally very weak and would not hold up well to physical abuse unless alloyed with a strengthening material. Automobile SLI batteries use calcium for strengthening, but calcium does not tolerate discharges of more than about 25 percent very well. Deep discharge batteries use antimony instead, and so are often referred to as lead-antimony batteries.

The chemical reactions taking place while the battery discharges are as follows:

$$\text{Positive plate}: \quad PbO_2 + 4H^+ + SO_4^{2-} + 2e^- \rightarrow PbSO_4 + 2H_2O \tag{9.21}$$

$$\text{Negative plate}: \quad Pb + SO_4^{2-} \rightarrow PbSO_4 + 2e^- \tag{9.22}$$

It is, by the way, simpler to refer to the terminals by their charge (positive or negative) rather than as the anode and cathode. Strictly speaking, the anode is the electrode at which oxidation occurs, which means that during discharge the anode is the negative terminal, but during charging the anode is the positive terminal.

As can be seen from (9.22), during discharge the electrons are released at the negative electrode, which then flow through the load to the positive plate where they enter into the reaction given by (9.21). The key feature of both reactions is that sulfate ions (SO_4^{2-}) that start out in the electrolyte when the battery is fully charged end up being deposited onto each of the two electrodes as lead sulfate ($PbSO_4$) during discharge. This lead sulfate, which is an electrical insulator, blankets the electrodes, leaving less and less active area for the reactions to take place.

As the battery approaches its fully discharged state, the cell voltage drops sharply while its internal resistance rises abruptly. Meanwhile, during discharge the specific gravity of the electrolyte drops as sulfate ions leave solution, providing an accurate indicator of the battery's state of charge. The battery is more vulnerable to freezing in its discharged state since the anti-freeze action of the sulfuric acid is diminished when there is less of it present. A fully discharged lead-acid battery will freeze at around −8°C (17°F), while a fully charged one won't freeze until the electrolyte drops below −57°C (−71°F). In very cold conditions, concern for freezing may limit the maximum allowable depth of discharge, as shown in Fig. 9.39.

The opposite reactions occur during charging. Battery voltage and specific gravity rise, while freeze temperature and internal resistance drop. Sulfate is removed from the plates and reenters the electrolyte as sulfate ions (Fig. 9.40). Unfortunately, not all of the lead sulfate returns to solution, and each battery charge/discharge cycle leaves a little more sulfate permanently attached to the

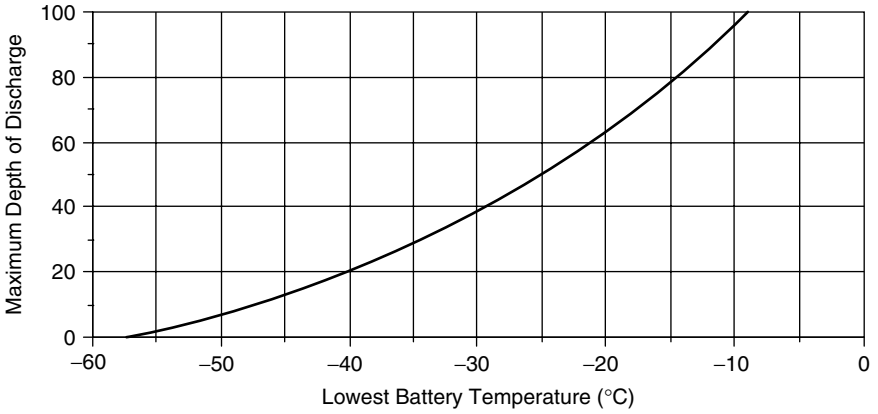


Figure 9.39 Concern for battery freezing may limit the allowable depth of discharge of a lead-acid battery.

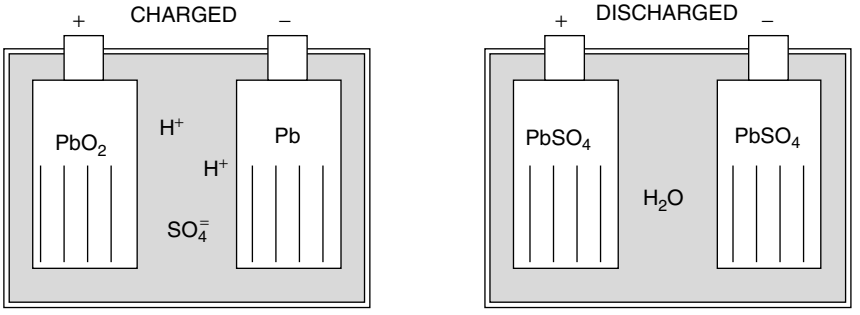


Figure 9.40 A lead-acid battery in its charged and discharged states.

plates. This sulfation is a primary cause of a battery's finite lifetime. The amount of lead sulfate that permanently bonds to the electrodes depends on the length of time that it is allowed to exist, which means that for good battery longevity it is important to keep batteries as fully charged as possible and to completely charge them on a regular basis. This suggests that a generator back-up system to top up batteries is an important consideration.

As batteries cycle between their charged and partially discharged states, the voltage as measured at the terminals and the specific gravity of the electrolyte changes. While either may be used as an indication of the state of charge (SOC) of the battery, both are tricky to measure correctly. To make an accurate voltage reading, the battery must be at rest, which means that at least several hours must elapse after any charging or discharging. Specific gravity is also difficult to measure since stratification of the electrolyte means that a sample taken from the liquid above the plates may not be an accurate average value. Bearing those complications in mind, Fig. 9.41 shows voltage for a nominal lead-acid 12-V battery at rest along with the specific gravity for a well-mixed electrolyte, as a function of the state of charge. It is interesting to note that a 12-V battery is only about 20% charged when its terminal voltage is 12 V.

9.5.5 Battery Storage Capacity

Energy storage in a battery is typically given in units of amp-hours (Ah) at some nominal voltage and at some specified discharge rate. A lead-acid battery, for example, has a nominal voltage of 2 V per cell (e.g., 6 cells for a 12-V battery), and manufacturers typically specify the amp-hour capacity at a discharge rate that would drain the battery down to 1.75 V over a specified period of time at a temperature of 25°C. For example, a fully charged 12-V battery that is specified to have a 10-h, 200-Ah capacity could deliver 20 A for 10 h, at which point the

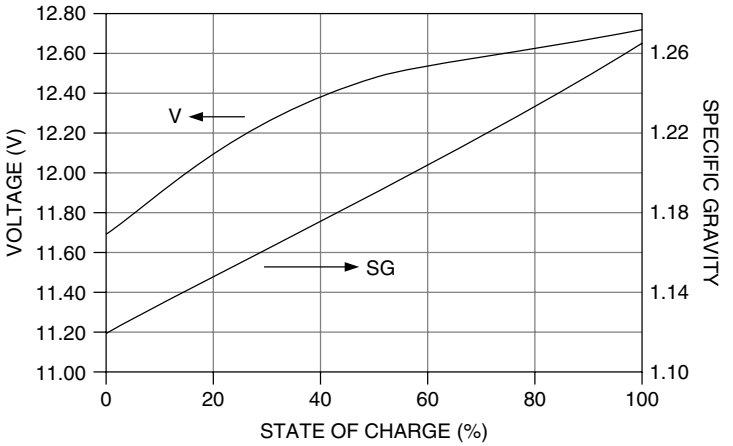


Figure 9.41 Voltage and specific gravity for a typical deep-cycle lead-acid 12-V battery. Data from Sandia National Laboratories (1991)..

battery would have a voltage of 10.5 V ($6 \times 1.75 = 10.5$) and be considered to be fully discharged. Notice how tricky it would be to specify how much energy the battery delivered during its discharge. Energy is volts × amps × hours, but since voltage varies throughout the discharge period, we can't just say 12 V × 20 A × 10 h = 2400 Wh. To avoid that ambiguity, almost everything having to do with battery storage capacity is specified in amp-hours rather than watt-hours.

A 200-Ah battery that is delivering 20 A is said to be discharging at a *C*/10 rate, where the *C* refers to Ah of capacity and the 10 is hours it would take to deplete (*C*/10 = 200 Ah/10 h = 20 A). That same 200-Ah battery won't be able to deliver 50 A for a full 4 h (*C*/4), however, and it will actually deliver 10 A for more than 20 h (*C*/20). In other words, the amp-hour capacity depends on the rate at which current is withdrawn. Rapid draw-down of a battery results in lower Ah capacity, while long discharge times result in higher Ah capacity. Deep-cycle batteries intended for photovoltaic systems are often specified in terms of their 20-h discharge rate (*C*/20), which is more or less of a standard, as well as in terms of the much longer *C*/100 rate that is more representative of how they are actually used. Table 9.15 provides some examples of such batteries, including their *C*/20 and *C*/100 rates as well as their voltage and weight.

The amp-hour capacity of a battery is not only rate-dependent, but also depends on temperature. Figure 9.42 captures both of these phenomena by comparing capacity under varying temperature and discharge rates to a reference condition of *C*/20 and 25°C. These curves are approximate for typical deep-cycle lead-acid batteries, so specific data available from the battery manufacturer should be used whenever possible. As shown in Fig. 9.42, battery capacity decreases dramatically in colder conditions. At −30°C (−22°F), for example, a battery that is discharged at the *C*/20 rate will have only half of its rated capacity. The combination of cold temperature effects on battery performance—decreased capacity, decreased output voltage, and increased vulnerability to freezing when discharged—mean that lead-acid batteries need to be well protected in cold climates. Nicad batteries, by the way, don't suffer from these cold weather effects, which is the main reason they are sometimes used instead of lead-acid batteries in cold climates. Also, the apparent improvement in battery capacity at high temperatures does not mean that heat is good for a battery. In fact, a rule-of-thumb estimate is that battery life is shortened by 50% for every 10°C above the optimum 25°C operating temperature.

TABLE 9.15 Example Deep-Cycle Lead-Acid Battery Characteristics

BATTERY	Voltage	Weight (lbs)	Ah @ *C*/20	Ah @ *C*/100
Concorde PVX 5040T	2	57	495	580
Trojan T-105	6	62	225	250
Trojan L16	6	121	360	400
Concorde PVX 1080	12	70	105	124
Surette 12CS11PS	12	272	357	503

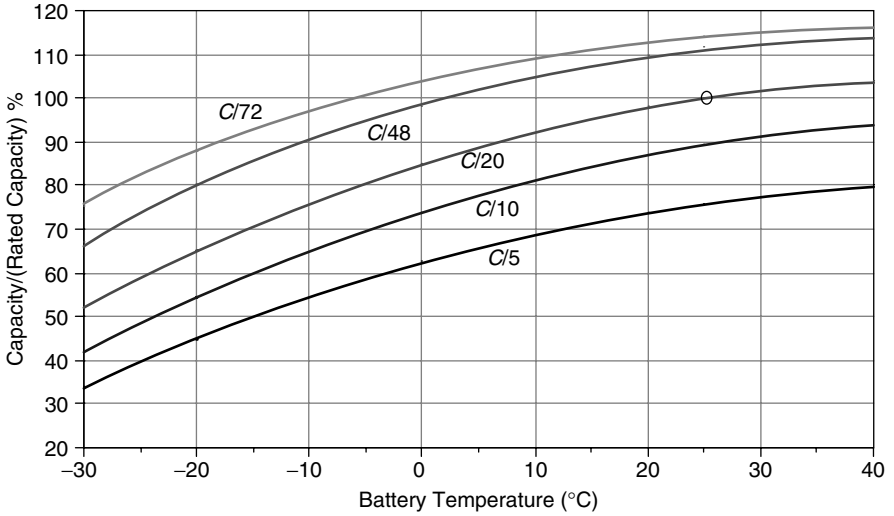


Figure 9.42 Lead-acid battery capacity depends on discharge rate and temperature. Ratio is based on a rated capacity at $C/20$ and 25°C.

Example 9.16 Battery Storage Calculation in a Cold Climate. Suppose that batteries located at a remote telecommunications site may drop to −20°C. If they must provide 2 days of storage for a load that needs 500 Ah/day at 12 V, how many amp-hours of storage should be specified for the battery bank?

Solution. From Fig. 9.39, to avoid freezing, the maximum depth of discharge at −20°C is about 60%. For 2 days of storage, with a discharge of no more than 60%, the batteries need to store

$$\text{Battery storage} = \frac{500 \text{ Ah/day} \times 2 \text{ days}}{0.60} = 1667 \text{ Ah}$$

Since the rated capacity of batteries is likely to be specified at an assumed temperature of 25°C at a $C/20$ rate, we need to adjust the battery capacity to account for our different temperature and discharge period. From Fig. 9.42, the actual capacity of batteries at −20°C discharged over a 48-h period is about 80% of their rated capacity. This means that we need to specify batteries with rated capacity

$$\text{Battery storage (25°C, 10-hour rate)} = \frac{1667 \text{ Ah}}{0.8} = 2083 \text{ Ah}$$

Most PV–battery systems are based on 6-V or 12-V batteries, which may be wired in series and parallel combinations to achieve the needed Ah capacity and

voltage rating. For batteries wired in series, the voltages add, but since the same current flows through each battery, the amp-hour rating of the string is the same as it is for each battery. For batteries wired in parallel the voltage across each battery is the same, but since currents add, the amp-hour capacity is additive. Figure 9.43 illustrates these notions.

Since there is no difference in energy stored in the two-battery series and parallel example shown in Fig. 9.43, the question arises as to which is better. The key difference between the two is the amount of current that flows to deliver a given amount of power. Batteries in series have higher voltage and lower current, which means more manageable wire sizes without excessive voltage and power losses, along with smaller fuses and switches, and slightly easier connections between batteries. On the other hand, a storage system consisting of batteries in parallel is easy to expand, one battery at a time. In series, a whole new string of batteries must be added to increase storage.

9.5.6 Coulomb Efficiency Instead of Energy Efficiency

As mentioned earlier, almost everything having to do with batteries is described in terms of currents rather than voltage or energy. Battery capacity C is given in amp-hours rather than watt-hours; charging and discharging are expressed in C/T rates, which are amps. And, as we shall see, even battery efficiency is more easily expressed in terms of current efficiency than in terms of energy efficiency. The reason, of course, is that battery voltage is so ambiguous without specifying whether it is a "rest" voltage measured some time after charging or discharging, a voltage during charging, or a voltage during discharge. And even those charging and discharging voltages depend on the rate at which current is entering or leaving the battery as well as the state of charge of the battery, its temperature, age, and general condition.

Imagine charging a battery with a constant current I_C over a period of time ΔT_C during which time applied voltage is V_C. The energy input to the battery

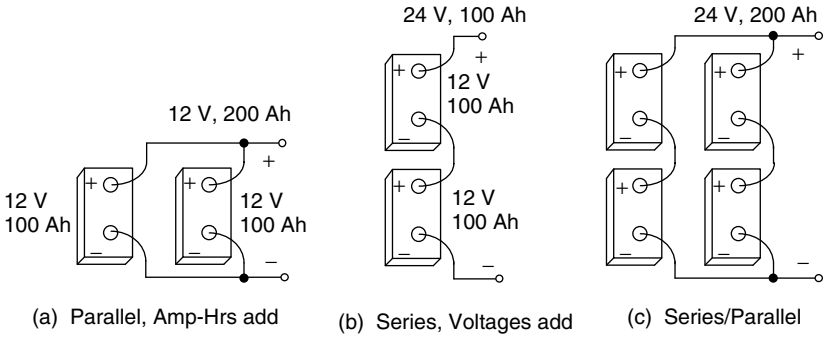


Figure 9.43 For batteries wired in parallel, amp-hours add (a). For batteries in series, voltages add (b). For series/parallel combinations, both add.

is thus

$$E_{\text{in}} = V_C I_C \ \Delta T_C \tag{9.23}$$

Suppose that the battery is discharged at current I_D and voltage V_D over a period of time ΔT_D, delivering energy

$$E_{\text{out}} = V_D I_D \ \Delta T_D \tag{9.24}$$

The energy efficiency of the battery would be

$$\text{Energy efficiency} = \frac{E_{\text{out}}}{E_{\text{in}}} = \frac{V_D I_D \ \Delta T_D}{V_C I_C \ \Delta T_C} \tag{9.25}$$

If we recognize that current (A) times time (h) is Coulombs of charge expressed as Ah, then

$$\text{Energy efficiency} = \left(\frac{V_D}{V_C}\right)\left(\frac{I_D \ \Delta T_D}{I_C \ \Delta T_C}\right) = \left(\frac{V_D}{V_C}\right)\left(\frac{\text{coulombs out, Ah}_{\text{out}}}{\text{coulombs in, Ah}_{\text{in}}}\right) \tag{9.26}$$

The ratio of discharge voltage to charge voltage is called the *voltage efficiency* of the battery, and the ratio of Ah_{out} to Ah_{in} is called the *Coulomb efficiency*.

$$\text{Energy efficiency} = (\text{Voltage efficiency}) \times (\text{Coulomb efficiency}) \tag{9.27}$$

A typical 12-V lead-acid battery might be charged at a voltage of around 14 V, and its discharge voltage might be around 12 V. Its voltage efficiency would therefore be

$$\text{Voltage efficiency} = \frac{12 \text{ V}}{14 \text{ V}} = 0.86 = 86\% \tag{9.28}$$

The Coulomb efficiency is the ratio of coulombs of charge out of the battery to coulombs that went in. If they don't all come back out, where did they go? When a battery approaches full charge, its cell voltage gets high enough to electrolyze water, creating hydrogen and oxygen gases that may be released. Among the negative effects of this *gassing* is loss of some of those charging electrons along with the escaping gases. While the battery state of charge (SOC) is low, little gassing occurs and the Coulomb efficiency is nearly 100%, but it can drop below 90% during the final stages of charging. Over a full charge cycle, it is typically between 90% and 95%. As we shall see later, when it comes to sizing batteries, the Coulomb efficiency will be the measure that is most appropriate.

Assuming a 90% Coulomb efficiency, the overall energy efficiency of a lead-acid battery with 86% voltage efficiency would be about

$$\text{Energy efficiency} = 0.86 \times 0.90 = 0.77 = 77\% \tag{9.29}$$

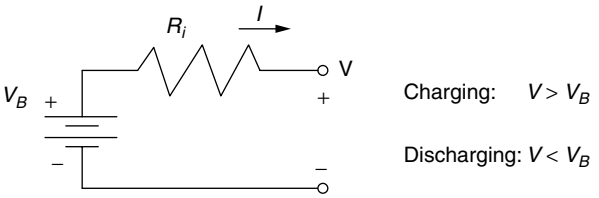


Figure 9.44 Thevenin equivalent circuit for a battery.

which is close to the commonly quoted estimate of 75% for lead-acid battery energy efficiency.

To help understand where that energy loss occurs, consider the simple Thevenin equivalent for a battery consisting of an ideal battery of voltage V_B in series with an internal resistance, R_i (Fig. 9.44). Voltage V_B can be considered to be the open-circuit "rest" voltage of the battery as measured some hours after either charging or discharging has occurred. To charge the battery, the voltage applied to the terminals must be greater than V_B; when the battery is discharging, the output voltage will be less than V_B. During those charge and discharge times, there are I^2R power losses in the internal resistance. Since those losses go as the square of current, faster charge or discharge times result in much higher losses.

Example 9.17 Losses at High and Low Charging Rates. A 100-Ah, 12-V battery with a rest voltage of 12.5 V (at its current SOC) is charged at a $C/5$ rate, during which time the applied voltage is 13.2 V. Using a simple Thevenin equivalent:

a. Estimate the internal resistance of the battery.
b. What fraction of the input power is lost in the internal resistance of the battery?
c. If the charging is done at a $C/20$ rate, what fraction of the input power would be lost due to the internal resistance?

Solution

a. At $C/5$, the current is 100 Ah/5 h = 20 A. The internal resistance must have been

$$R_i = \frac{V_{\text{in}} - V_B}{I} = \frac{13.2 - 12.5}{20} = 0.035\ \Omega$$

b. So the I^2R losses as a fraction of the input power would be

$$\frac{\text{Power lost in } R_i}{\text{Input power}} = \frac{I^2R}{V_{\text{in}}I} = \frac{(20)^2 \times 0.035}{13.2 \times 20} = 0.053 = 5.3\%$$

c. At $C/20$, the current is 100 Ah/20 h = 5 A. The input voltage needed to drive 5 A through the 0.035-Ω resistance is

$$V_{\text{in}} = V_B + IR = 12.5 + 5 \times 0.035 = 12.68 \text{ V}$$

So at the $C/20 = 5$-A rate, the losses are now only

$$\frac{\text{Power lost in } R_i}{\text{Input power}} = \frac{I^2R}{V_{\text{in}}I} = \frac{(5)^2 \times 0.035}{12.68 \times 5} = 0.014 = 1.4\%$$

The simple Thevenin model of a battery is complicated by the fact that both V_B and R_i depend on the battery's state of charge (SOC), its temperature, and its history. But since the model is so simple, it does help provide an intuitive understanding of the actual charging and discharging curves for a battery. Figure 9.45 shows representative values of battery voltage for differing charging and discharging currents as a function of SOC.

During charging, notice the sudden rise in cell voltage around the 14-V level as the battery nears full charge. This is when charging is most inefficient and when gassing occurs. The release of generated hydrogen and oxygen gases removes water from the battery, which has to be replaced or the plates may be damaged. It also creates a potentially dangerous situation since hydrogen gas is so explosive. In addition, very small quantities of the poisonous gases arsine (AsH_3) and stibine (SbH_3) can be released when hydrogen comes in contact with the lead alloys arsenic and antimony. Proper ventilation is clearly an important consideration in the design of a safe battery storage system.

To reduce the need for water replacement, some lead-acid batteries now use pressure-relief valves that allow most of the oxygen formed during overcharging to recombine with lead rather than being released. Hydrogen gas releases are also suppressed. While they do open as necessary to relieve the pressure built up by hydrogen and oxygen gases, these valve-regulated lead-acid batteries (VRLA) can reduce gas emissions by over 95% (Linden, 1995).

Gassing and charging losses can be minimized, by using a charge controller that has been designed to slow the charging rate as the battery approaches its fully charged condition. Charge controllers also protect batteries from overcharging by completely disconnecting the PV array at some predetermined battery voltage, usually around 14 V (for a 12-V battery). They also keep the batteries from being overly discharged by disconnecting the load when battery voltage drops below another set point, usually around 11.5 V.

9.5.7 Battery Sizing

If good weather could be counted on, battery sizing might mean simply providing enough storage to carry the load through the night and into the next day until

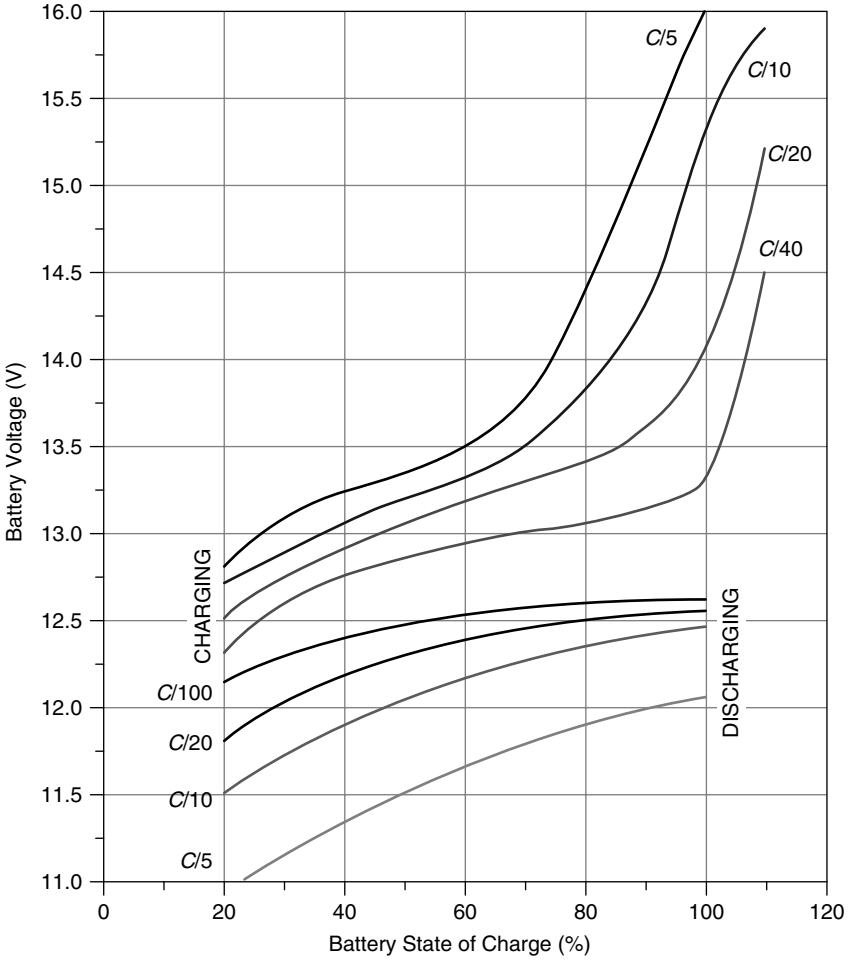


Figure 9.45 Terminal voltage and state of charge for 12-V lead-acid batteries for various rates of charging and discharging. Based on Sandia National Laboratories (1991).

the sun picks up the load once again. The usual case, of course, is one in which there are periods of time when little or no sunlight is available and the batteries might have to be relied on to carry the load for some number of days. During those periods, there may be some flexibility in the strategy to be taken. Some noncritical loads, for example, might be reduced or eliminated; and if a generator is part of the system, a trade-off between battery storage and generator run times will be part of the design.

Given the statistical nature of weather and the variability of responses to inclement conditions, there are no set rules about how best to size battery storage. The key trade-off will be cost. Sizing a storage system to meet the demand 99% of the time can easily cost triple that of one that meets demand only 95%